

Analysis Note:

Unbinned measurement of the charged-current inclusive cross section in νN collisions with public MINERvA data

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Code and file references: github.com/josephbailey/MINERvA-OmniFold

Abstract

We measure medium-energy charged-current neutrino–nucleus cross sections in multiple dimensions using the unbinned machine-learning method OmniFold. We first reproduce the published double-differential cross section in muon transverse and longitudinal momentum with an independent unfolding method, recovering both the spectrum and the uncertainty budget. We then perform *unbinned* simultaneous three-, four-, and five-observable unfolds of MINERvA data by adding available energy (E_{avail}), three-momentum transfer (q_3), and hadronic invariant mass (W); each reproduces its lower-dimensional predecessor under marginalization. Binned MINERvA multi-differential measurements, up to triple-differential [1], exist already; the methodological feature here is the unbinned, simultaneously-unfolded formulation. These unfolds recover the established MINERvA deficit of low-momentum-transfer multinucleon interactions and localize a residual high- E_{avail} excess to the deep-inelastic region, where all four tested generators underpredict the data. A recoil-only point-cloud transformer provides a low-level-representation cross-check, and a scalar study examines the effect of removing the truth-level muon kinematic cuts.

Executive summary

We apply *unbinned machine-learning unfolding* (OmniFold [2, 3]) to MINERvA’s medium-energy neutrino data. The analysis begins with a reproduction of a published measurement and extends it to simultaneous three-, four-, and five-observable scalar cross sections. We also study a recoil-only point-cloud representation and a preliminary extended fiducial domain with relaxed truth-muon cuts. The specific scalar extensions studied here do not have direct binned MINERvA counterparts.

Traditional unfolding corrects detector effects with a migration matrix between histogram bins, so each added observable multiplies the number of bins. OmniFold instead trains a classifier to learn, event by event, a weight that morphs the simulation into the data, and pulls that weight back to

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the true (pre-detector) level through the simulation’s truth–reco pairing. No reporting binning is imposed until the end, so a recorded observable can be added without building a new explicit response tensor; training, support and uncertainty propagation still become harder with dimension. This analysis uses gradient-boosted decision trees for the fixed-dimension measurements and validates the choice with closure, iteration and pull diagnostics, classifier-calibration checks, and neural-network cross-checks. The existing 2D toy ensemble does not provide a valid independent-truth coverage test, which remains open.

Results overview:

- **Reproduction.** Our unbinned measurement of the muon transverse and longitudinal momentum, $d^2\sigma/(dp_T dp_{\parallel})$, matches the published measurement [4]: total cross-section ratio 1.011, 94.1 % of bins within 10 %, per-bin standardized differences sub- 1σ , and a separately reconstructed uncertainty budget whose median per-bin uncertainty (6.87 %) matches the published 6.86 % (§5, §4, Fig. 7).
- **Unbinned 3D/4D/5D MINERvA unfolds.** Adding available energy (E_{avail}), momentum transfer (q_3), and hadronic mass (W) produces *unbinned* simultaneous three-, four- and five-observable unfolds of MINERvA data; every result reproduces its lower-dimensional predecessor when marginalized — a built-in cross-check (§7). Prior MINERvA multi-differential cross sections, including a binned triple-differential quasielastic-like measurement [1], used explicit binned unfolding; the methodological distinction here is the unbinned, simultaneously-unfolded formulation rather than multi-dimensionality itself.
- **Known physics recovered independently.** All four truth-level generators (GENIE [5], MINERvA Tune [6], NuWro [7], GiBUU [8]) under-predict the data at low E_{avail} ; pion final-state-interaction dials move the spectrum by less than 1 % while adding Valencia 2p2h [9, 10] fills the quasielastic- Δ dip — the established MINERvA low-recoil/2p2h deficit [6, 11], re-found by an independent method (§7.7, Fig. 23).
- **A new localization of the remaining excess.** The high- E_{avail} excess that 2p2h cannot explain is localized in the (E_{avail}, W) plane: its positive data-minus-generator difference concentrates in the deep-inelastic corner ($W \geq 1.8 \text{ GeV}$), and all four generators underpredict that region. The comparison is reported at central-value level because a corrected covariance for this projection has not been adopted (§8, Fig. 31).
- **A methodological finding.** The standard covariance adds matrices built for one systematic source at a time, which omits cross-source response terms. In MINERvA’s `GetTotalErrorMatrix`, each named source contributes its own covariance, built with all other sources held nominal; the sum retains within-source bin–bin correlations but contains no cross-source terms. The corrected unified-throw test perturbs all reweightable sources simultaneously using actual asymmetric endpoints, one fixed estimator seed, and mean-centered throws; it reports the joint mean shift separately. The resulting 5D products remain candidates until the selection-complete lateral replacement lands (§4.4).
- **Recoil-point-cloud representation cross-check.** A point-cloud transformer operates on non-muon reconstructed recoil clusters at step 1 and truth final-state hadrons at step 2. Muon kinematics enter selection and downstream binning but not either classifier, so the learned weight is a function of recoil information only. Its absolutely normalized central value and ordinary closure test this recoil-only estimator. Its historical five-component uncertainty assembly is quarantined because nuisance and retraining shifts were not combined with their

cross term and detector membership was CV-support-limited. A full-event estimator, meaning one trained jointly on the muon and recoil, requires a new nominal fit, an omitted-muon stress closure and a separate uncertainty budget (§9, §9.3).

- **A preliminary extended-fiducial study.** Re-running the scalar 2D $(p_T, p_{\parallel})$ unfold with the truth-muon kinematic cuts removed enlarges the signal definition while retaining the tracker fiducial interaction volume and the unchanged MINOS-based reconstructed selection. About 6 % of the rate lies in a newly acceptance-supported extension (tier 1), while about 28 % lies in low- or zero-efficiency cells and is reported separately as prior-dominated extrapolation (tier 2). The result includes a restricted-region anchor, closure tests and a three-prior envelope, but not a corrected extended-fiducial covariance (§10).

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1 Introduction and motivation

MINERvA (Main Injector Experiment for ν -A) is a dedicated neutrino–nucleus scattering experiment at Fermilab, situated in the NuMI beamline immediately upstream of the MINOS near detector [12]. Over roughly a decade of operation, ending in 2019, MINERvA recorded high-statistics samples in both the low-energy and medium-energy (ME) NuMI beam configurations [13] and in neutrino-dominated (forward horn current, FHC) and antineutrino-dominated beam modes. Its physics goal is the precision measurement of neutrino–nucleus interaction cross sections on a range of nuclear targets in the few-GeV region most relevant to accelerator-based oscillation experiments: these cross sections, and the nuclear effects that shape them, are a leading systematic uncertainty for experiments such as T2K [14], NOvA [15], Hyper-Kamiokande [16], and DUNE [17], and the few-GeV regime mixes quasistatic, resonant, and deep-inelastic processes that no single model describes cleanly. This note uses the public ME FHC CC-inclusive data product; the detector itself is described in §2.

The MINERvA ME FHC charged-current (CC) inclusive double-differential cross section in muon transverse and longitudinal momentum, $d^2\sigma/(dp_T dp_{\parallel})$, was published by Ruterbories *et al.* [4] using D’Agostini iterative Bayesian unfolding (IBU) [18] with a binned response matrix. This note reproduces that measurement with *unbinned* OmniFold [2, 3], an unfolding technique based on iterated reweighting with machine-learned likelihood ratios that operates on per-event observables rather than on a fixed histogram binning. Recent advances in machine learning have enabled unbinned unfolding techniques; see Refs. [19, 20] for overviews and Ref. [21] for cross-experiment practical guidance. These methods fall into two broad families: those that iteratively reweight a starting simulation, of which OmniFold [2, 3, 22] is the most widely adopted, and those that generate new samples directly with a neural network [23–26]. OmniFold was first demonstrated on collider data by the H1 collaboration [27, 28] and has since been adopted across a number of experiments; the application closest to this work is the OmniFold study on simulated T2K near-detector data [29]. These techniques lift the binned-response limitations of widely-used classical unfolding methods [30] like IBU, the SVD method [31], and TUnfold [32].

Why unbinned:

- **No binning at unfold time.** OmniFold learns event weights in the full observable space; the result is histogrammed only at the end, so the binning can be chosen (or refined) after unfolding without re-running.
- **Natural path to higher dimensions.** Adding observables is adding input features, not multiplying response-matrix bins — this is the enabling property for the available-energy extension (§7) that has no practical IBU analog.
- **Different regularization** OmniFold’s regularizer is the machine learning model capacity in addition to the iteration count in common with IBU. Comparing the two approaches on the same data quantifies the method-dependence of the result (§5).

Analysis strategy. The published 2D cross section provides a quantitative validation of unbinned OmniFold on MINERvA data, including its uncertainty budget. The validated pipeline is then extended by adding E_{avail} , q_3 and W as classifier inputs. Marginalization to each lower-dimensional result, closure, coverage and classifier-family comparisons test the extensions. A recoil-only point-cloud cross-check and a scalar study with relaxed truth-muon cuts probe alternative representations and signal definitions.

2 The MINERvA experiment, data and simulation samples

2.1 Detector and beam

MINERvA [12] was a fine-grained scintillator-tracker neutrino experiment on the NuMI beamline at Fermilab. The central tracker is built from hexagonal planes of triangular polystyrene scintillator strips in three stereo orientations, surrounded by electromagnetic and hadronic calorimetry; the magnetized MINOS near detector, 2 m downstream, serves as the muon spectrometer, providing the muon momentum and charge for MINOS-matched tracks. This analysis uses the *medium-energy* (ME) neutrino-mode (FHC) exposure with $\langle E_\nu \rangle \sim 6$ GeV. The neutrino flux is the *a posteriori* constrained NuMI ME prediction of Ref. [13] (PPFX hadron-production weights plus the neutrino-electron-scattering constraint), identical to the flux used by the published measurement [4].

2.2 Data and simulation samples

The data are the twelve ME FHC playlists (ME1A–ME1G, ME1L–ME1P), corresponding to 1.057×10^{21} protons on target — the same exposure as Ref. [4]. After the selection below the data sample contains 4 119 797 reconstructed events.

The reference simulation is GENIE 2.12.6 [5] with the MINERvA Tune v1 applied as per-event weights through the MAT (MINERvA Analysis Toolkit) reweighters: the flux-and-CV weight, the GENIE dial weights, the low-recoil 2p2h enhancement fit, and the non-resonant-pion reduction. Detector response is the full GEANT4-based [33] MINERvA simulation underlying the production AnaTuples. The simulated sample comprises 32 849 103 true signal events (POT-normalized to data exposure event-by-event), processed through the identical reconstruction and selection as the data.

2.3 Signal definition and event selection

The signal definition and phase space follow Ref. [4] exactly. The truth-level signal is a charged-current ν_μ interaction with vertex in the central-tracker fiducial volume, with muon phase space

$$\theta_\mu < 20^\circ, \quad 1.5 < p_\parallel < 60 \text{ GeV}/c, \quad p_T < 4.5 \text{ GeV}/c.$$

The fiducial volume is the tracker region $5980 < z < 8422$ mm within an 850 mm apothem, containing 3.2353×10^{30} nucleons (the fixed tracker-geometry constant).

Reconstructed events are selected by requiring: a reconstructed vertex in the same tracker fiducial volume and apothem; a muon track with $\theta_\mu < 20^\circ$ matched to a track in MINOS with a passing MINOS fit; no dead-time overlap; and a neutrino (rather than antineutrino) candidate. The MINOS-match requirement is implemented as `isMinosMatchTrack==1 && minos_trk_is_ok==1` (the current MINERvA-101 tutorial enforces the same `isMinosMatchTrack` match; we additionally require a passing MINOS fit). This yields a predicted genuine-background rate — charged-current events of other flavors or wrong sign, neutral-current interactions, and out-of-fiducial-vertex events; out-of-phase-space signal (*fakes*) is treated separately — of 0.35 % of the selected sample (evaluated on playlist 1A). This is the same sub-percent scale as the 8655 (0.2 %) predicted background events of Ref. [4], whose definition counts only the wrong-flavor, wrong-sign, and neutral-current components; exact agreement is not expected given the definitional and selection differences, and neither prediction carries an uncertainty. Backgrounds (reconstructed events whose true interaction fails the signal definition) and *fakes* (reconstructed in phase space, true muon outside it) are POT-scaled and subtracted from the reconstructed-data target before unfolding (§3).

2.4 Per-event analysis trees

Unbinned unfolding needs per-event records rather than histograms. A dedicated event loop (MINERvA101/MINERvA-101-Cross-Section/runEventLoopOmniFold.cpp), built on the MAT systematic-universe machinery, writes four flat trees per playlist: `mc_truth_denom` (every true signal event — the efficiency denominator), `mc_signal_reco` (reconstructed, truth-matched simulated events), `mc_background`, and `data`. Each entry carries the truth and reconstructed observables (p_T , $p_{\parallel}$, and in the extended runs E_{avail} , q_3 , W), the MnvTune central-value weight, and — in the `_universes_full` variants — the full set of systematic-universe weights and laterally-shifted observables (§4). Two invariants make the result truth-tree authoritative:

- **Truth-tree-authoritative reco gate:** a reconstructed signal event is filled only when its (run, subrun, event) key is present in the truth denominator; truth-only events enter as native OmniFold “miss” entries.
- **Completeness** $c = (\text{events in the OmniFold input})/(\text{all truth events})$ is 1.000000 by construction at ME FHC (32 849 103 entries), so the cross-section completeness division is a no-op self-check.

Because the event loop selects one global playlist flux/calibration from the first event, it is always run per playlist and the twelve outputs merged afterwards (with a `TTree::SetMaxTreeSize`-aware merger for the more-than-100 GB universe dumps). The flux histogram (`pTmu_reweightedflux_integrated`) comes from the corresponding baseline histogram event loop and is POT-weighted across playlists; playlist-to-playlist flux differences are a $\sim 0.2\%$ effect.

2.5 Reco-level control distributions

Figure 1 compares the selected data to the POT-scaled simulation (signal + background) at reconstruction level in all five observables. The selected event count exceeds the MnvTune-v1 prediction by $\sim 12\%$ in overall normalization, with a clear trend in p_T (the data/MC ratio rises from 0.88 in the lowest bin to ~ 1.2 above 1 GeV/c) — the reco-level image of the normalization deficit quantified at truth level in §5.4 and §7.5. The simulation describes the *shapes* well, which is what the unfolding relies on; the rate offset is what the data pull corrects.

The shapes themselves are set by the beam and the acceptance. The $p_{\parallel}$ spectrum peaks at 4–6 GeV/c — the image of the $\langle E_{\nu} \rangle \approx 6$ GeV NuMI medium-energy peak — and the spectrometer-match requirement removes the lowest- $p_{\parallel}$ muons; p_T peaks near 0.7 GeV/c and falls by over an order of magnitude across the measured range. On the hadronic side the sample is dominated by low-recoil (quasi-elastic and $2p2h$) events: E_{avail} is concentrated below ~ 0.5 GeV with a long inelastic tail collected by the wide upper bins, q_3 tracks the momentum transfer of the same population, and reconstructed W rises through the smeared quasi-elastic and $\Delta(1232)$ region before the deep-inelastic continuum takes over above ~ 2 GeV. The absolute background is at the few-percent level under the muon projections and is concentrated at high recoil (highest E_{avail} , q_3 and W bins). Beyond the p_T trend noted above, the ratio panels also show a mild enhancement in the lowest E_{avail} bins — the reco-level trace of the low-recoil excess examined at truth level in §5.4.

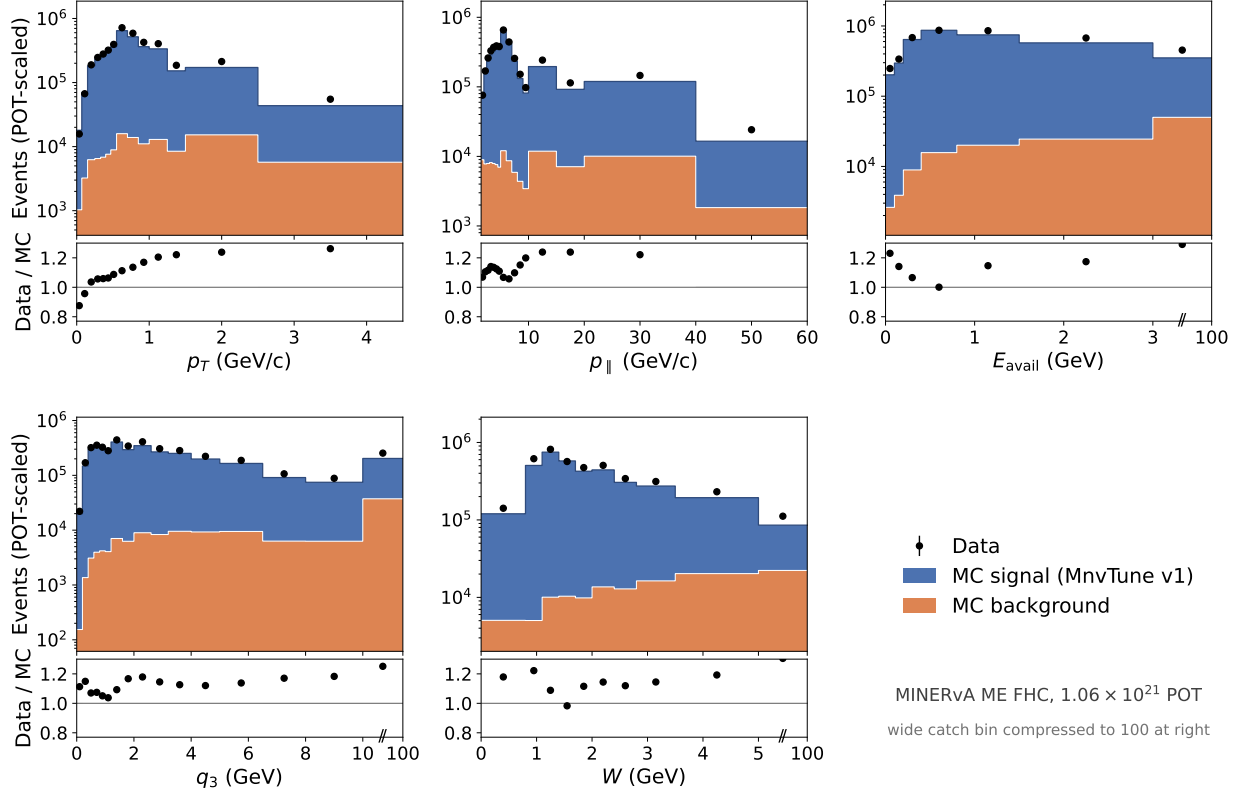


Figure 1: Reco-level control distributions for the analysis selection: data (points) versus POT-scaled simulated signal (blue) stacked on background (orange), with the data/MC ratio below, for p_T , $p_{||}$, E_{avail} , q_3 and W . On the hadronic axes the wide catch bin is drawn compressed at the right of each panel, running out to 100 GeV. The gray band on the ratio is the MC statistical uncertainty, mostly narrower than the points. The $\sim 12\%$ normalization excess of the data over MnvTune v1 is the reco-level counterpart of the truth-level generator deficits of §7.5.

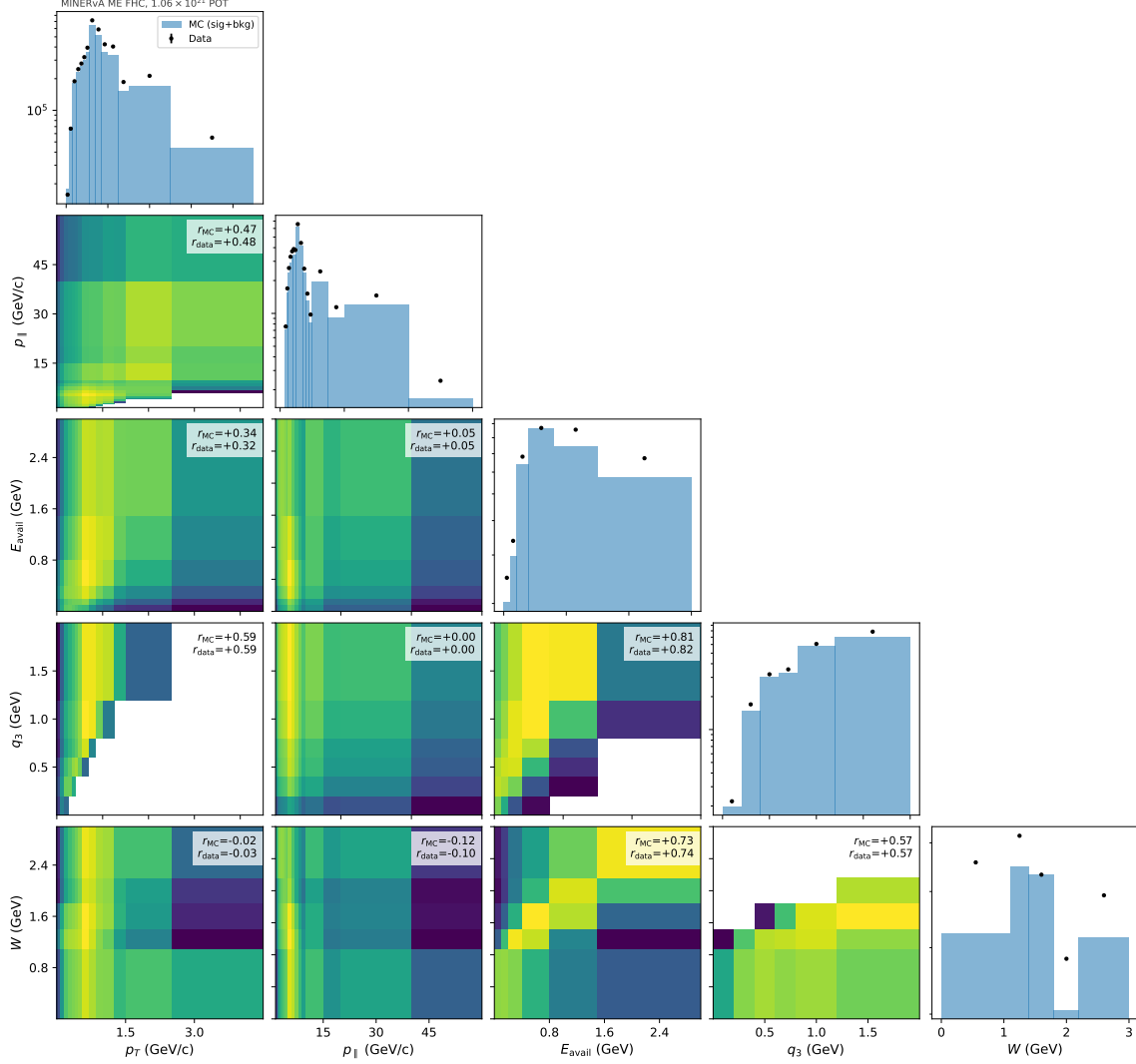


Figure 2: Reco-level corner plot of the five observables: pairwise POT-scaled MC (signal+background) densities (lower triangle, logarithmic color scale; wide catch bins dropped) with the weighted Pearson correlation of the simulation and the corresponding data value annotated per panel, and the 1D marginals (data points over MC) on the diagonal. The strongest correlations are kinematic by construction: q_3 with E_{avail} ($r \approx 0.81$) and W with E_{avail} ($r \approx 0.73$), which share the recoil energy, and q_3 with p_T ($r \approx 0.59$), which shares the muon transverse kick; E_{avail} and W are nearly uncorrelated with the muon variables. The data reproduce every simulated coefficient to within 0.02.

3 The unbinned OmniFold procedure

3.1 Algorithm

OmniFold [2, 3] unfolds by iterated reweighting with machine-learned likelihood ratios, operating on per-event observables rather than on a fixed histogram binning. Each iteration has two steps:

1. **Step 1 (detector space).** A classifier is trained to separate the measured data from the reconstructed simulation (the latter weighted by the current push-weights). Its calibrated

output p yields the likelihood-ratio reweight $w = p/(1 - p)$ on every reconstructed simulated event. The weight is *pulled back* to truth level through the event pairing: each simulated event is a (truth, reco) pair, so the reco-space weight attaches directly to the truth record. Truth events with no reconstructed partner (the “misses”) receive a step-1 weight from a regressor trained on the reconstructed subsample, $w(\text{truth features})$ — the native OmniFold acceptance treatment, with no separate efficiency correction.

2. **Step 2 (truth space).** A second classifier is trained to separate the original truth-level simulation from the same sample carrying the pulled-back step-1 weights, yielding a smooth truth-space reweight that becomes the next iteration’s push-weight.

After n iterations the push-weights define an efficiency-corrected, truth-level reweighted simulation ensemble; histogramming it (in any binning, chosen after the fact) gives the unfolded spectrum. The regularizer is the iteration count together with the model capacity of the learners.

Background subtraction (the one binned step). The measured sample entering step 1 is the full set of selected data events — none is discarded and no negative-weight events are injected. Backgrounds are instead removed by a per-reco-bin *purity* weight, applied event by event: the POT-scaled background prediction (genuine backgrounds plus out-of-phase-space signal, §2.3) is binned on the reconstructed analysis grid, and each data event in reco bin b is weighted by

$$w_{\text{pur}}(b) = \max(0, N_{\text{data}}(b) - N_{\text{bkg}}(b))/N_{\text{data}}(b),$$

the predicted signal purity of that bin (floored at zero in the rare bins where the prediction exceeds the data). The step-1 classifier still sees the full unbinned event kinematics; the analysis binning enters only through this multiplicative correction, an $\sim 3\%$ effect overall that is largest where the background concentrates (§2.3).

This correction is, deliberately, the one place the analysis binning enters before the final histogramming. Published unbinned measurements have so far avoided the problem rather than solved it: H1 cut backgrounds to a negligible level, LHCb corrected per track upstream of the unfold, and ATLAS left its the less-than-0.25% background in place and assigned it as an uncertainty, while identifying negatively-weighted simulated events as the natural in-method subtraction [3, 34]; Ref. [21] recommends that route for nontrivial backgrounds but records no analysis that has yet used it. The explicit per-reco-bin weight used here is a conservative middle ground at our sub-percent genuine-background scale. Its fully unbinned realization — injecting the background simulation onto the measured side with negative weight, which targets the identical background-subtracted density $(D - B)/S$ and reproduces the binned correction to -0.13% on the total 2D cross section, and its systematic and statistical covariances to within 2% (App. B) — removes the residual binning dependence and is the natural treatment for the extended-fiducial analysis, where backgrounds are larger.

OmniFold has moved from method paper to production measurement over the last few years. H1 published the first real-data applications — jet substructure in high- Q^2 deep-inelastic scattering, followed by a multi-differential eight-observable jet measurement [27, 28]; LHCb unfolded identified-hadron distributions in Z -tagged jets [35]; and ATLAS performed a simultaneous unbinned measurement of twenty-four Z +jets observables [34]. Ref. [21] collects these and other applications and distills cross-experiment practice; several of the practices audited there appear in our uncertainty budget (the ensemble-mean cross-check of the central value, the bootstrap statistical block, and the unified-throw covariance check of §4.4).

Closest to this analysis is the study of Huang *et al.* [29], the first application of OmniFold to a neutrino cross section: a mock-data unfolding of muon kinematics on the public T2K near-detector

simulated dataset, benchmarked against traditional binned approaches. Two of its findings shape expectations for neutrino applications. First, iteration count: Huang *et al.* select 45 iterations for the OmniFold-family methods (20 for their binned benchmark) by a binned χ^2 against the known truth of their mock data — a criterion they note is not feasible on real data. We do not adopt that recommendation, for two reasons. Their mock data injects a large prior-data *shape* discrepancy (an RPA-modified quasielastic model) that many iterations must overcome, whereas our prior already describes the data shapes well (§2.5) and the data pull is dominantly a normalization; and cross-experiment production practice is few iterations (typically ~ 5 [21]). Our $n = 5$ is instead justified empirically: doubling to $n = 10$ moves the total cross section by 0.026 % (§3.2), and the unfold reproduces an independently published binned measurement of the same data (§5). Second, an ensemble of reweighting trials with different train/test splits controls the ML stochastic component — which we budget explicitly via the seedscan band (§4). The present work differs in unfolding *real* MINERvA data to an absolutely normalized cross section with a complete finalized 2D uncertainty budget, in using GBDT learners suited to a low-dimensional tabular feature set, and in validating the method directly against a published binned measurement of the same data (§5).

3.2 Estimators and configuration

We use gradient-boosted decision trees (GBDTs) rather than the neural networks of the original OmniFold papers. The choice follows the character of the inputs: the scalar unfolds of this note use at most five high-level features per event, a tabular regime in which tree ensembles typically match or outperform neural networks [36] while training deterministically and far faster on CPU. Ref. [37] demonstrates this regime specifically for unbinned unfolding — boosted decision trees are effective with a small number of high-level features, complementing deep models that ingest low-level event representations — and our own measurements follow the same division of labor: GBDTs for the scalar unfolds and a point-cloud transformer as a recoil-cluster representation cross-check (§9). The PET cross-check omits the muon and therefore is not a full-event unfold. §6 tests the implementation with calibration tests and MLP/transformer cross-checks. The engine (`unbinned_unfolding/python/omnifold.py`; component terminology in App. D) supports interchangeable backends; the production configuration is **LightGBM** with 100 trees, 8 leaves, learning rate 0.1 (identical settings for the step-1 classifier, step-2 classifier, and miss regressor). Two alternative backends — **exact** (sklearn `GradientBoosting`) and **hist** (sklearn `HistGradientBoosting`, matched 8-leaf configuration) — are retained for method-dependence studies (§5.3). The production iteration count is $n = 5$; doubling to 10 moves the total cross section by 0.026 % (§6). The central value is a single-run unfold with pinned seeds; an ensemble-mean central value (averaging independent reweighting trials) agrees with it to 0.28 %, and ML stochasticity is budgeted explicitly via the seedscan band (§4).

Hyperparameter choice. The learner settings were fixed *a priori* rather than tuned on data: 100 trees, learning rate 0.1, and depth-3 capacity are the sklearn `GradientBoosting` defaults, and the production LightGBM backend is configured to the matched capacity (`num_leaves = 8 = 2^3`) so that every backend shares the same, deliberately modest, model capacity — capacity being one of OmniFold’s regularizers (§3.1). No hyperparameter search was performed on data, avoiding any tuning-on-result loop. The configuration is instead validated after the fact: closure and classifier-calibration tests (§6), insensitivity to doubling the iteration count (quoted above), backend agreement in the method-dependence studies (§5.3), and an explicit seedscan band for residual ML stochasticity (§4).

3.3 MINERvA driver and pipeline conventions

The MINERvA driver is `2d-unfolding/unfold_2d_omnifold_unbinned.py`; the higher-dimensional driver (`nd-unfolding/unfold_nd_omnifold_unbinned.py`) generalizes it to an axis list whose first two entries are always $(p_T, p_{\parallel})$ (carrying the fiducial gate and the per- p_T flux) and `imports` the same 2D helpers. The unfolded observables are the muon $(p_T, p_{\parallel})$, extended by E_{avail} , q_3 , W in the higher-dimensional runs; the phase space is that of §2.3. The following conventions hold in every production, universe-sweep, and closure run:

- **Reco-space masking.** The measured events entering step-1 training are exactly those passing the reconstructed-level selection; nothing outside the reconstructed phase space reaches the classifier.
- **Fake-free step-1 training.** The step-1 target is the background-subtracted data of the purity-weight construction of §3.1: genuine backgrounds *and* out-of-phase-space signal fakes are removed from the data side, and the simulated reco sample contains no fakes by construction — so neither side of the step-1 training contains events whose truth lies outside the measurement phase space.
- **No double efficiency correction.** When the unfolded sample is histogrammed, each simulated truth event carries the product of its final OmniFold push-weight and its central-value simulation weight (the MnvTune reweighting rides along). No additional efficiency-histogram division is applied: the step-1 miss regression already returns an efficiency-corrected truth spectrum, and dividing again would double-correct — a failure mode that occurred once in development and was caught by the marginal self-validation gate (§6).
- **Closure mirrors production.** In closure tests the pseudo-data are simulated events passing both the reconstructed and truth selections, weighted identically to the production data path, so closure exercises the same code path as the measurement rather than an idealized shortcut.

3.4 Cross-section extraction

Unfolded truth-level counts are converted to the differential cross section via

$$\frac{d^n \sigma}{dx_1 \cdots dx_n} = \frac{U}{c \Phi N \text{ POT } \Delta x_1 \cdots \Delta x_n} \times 10^4,$$

where U is the unfolded yield; c is the per-bin *completeness*, the fraction of the truth-level sample visible to the unfolding input. Because the event loop appends the truth-only misses explicitly, every truth event enters the unfolding and $c \equiv 1$ bin-by-bin; the factor survives in the formula as a monitored self-check (a configuration that dropped the misses would surface as $c < 1$). Φ is the per- p_T flux integral (broadcast over the other axes), N the fiducial nucleon count (3.2353×10^{30}), and POT the data exposure (1.057×10^{21}). The factor 10^4 converts the flux tabulation per square meter to the reported cm^2 ($1 \text{ m}^2 = 10^4 \text{ cm}^2$), so the result carries the conventional units of cm^2 per axis unit per nucleon. The implementation is `extract_cross_section_2d` and its N-dimensional generalization `xsec_nd.py` (self-tested to reproduce the frozen lower-dimensional extractions to $< 10^{-12}$). The flux normalization enters as $\sigma \propto 1/\Phi$, so Φ is itself varied per flux universe when the systematic covariance is built; this variation dominates the Flux band (§4).

3.5 Binning

The reported 2D binning ($14 p_T \times 16 p_{\parallel}$ bins; 205 of 224 reported) follows `minerva_paper_anc/bin_mapping.txt`; the 19 unreported diagonal cells have $p_T^{\text{hi}}/p_{\parallel}^{\text{lo}} > \tan 20^\circ$. The extension

axes use E_{avail} : $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV; q_3 : $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV; W : $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV — each ending in a wide catch bin so the marginals capture the full CC-inclusive tails (a truncated top edge would break the marginalization anchors of §7.3). Because the binning is applied only after unfolding, these choices can be revisited without re-running the unfold. The full edge table and reported-bin mapping are given in App. A (§A.3.1).

Neither underflow nor overflow appears in the reported product, by construction rather than by truncation. On the truth side, the muon-axis edges coincide with the measurement phase space itself (§2.3) — beyond them there is no signal definition to measure — and the hadronic axes span $[0, 100]$ GeV, the catch-bin edge sitting far beyond the kinematic reach at $\langle E_\nu \rangle \sim 6$ GeV, so the truth grids are exhaustive. On the reco side, a selected event whose reconstructed kinematics fall outside the grid is treated as unaccepted: it is excluded from the measured sample by the masking convention of §3.3, and its in-phase-space truth content is recovered by the miss treatment exactly as for any other unreconstructed event. Because the unfold is unbinned, there is no response-matrix under/overflow to track — the binning is applied once, at reporting, to events that all lie inside the grids.

Although OmniFold uses no response matrix, the detector smearing the unfold must undo is conveniently summarized by the truth→reco migration on the analysis binning (Fig. 3): the diagonal purity is ~ 0.6 (median) on every axis, i.e. bin-to-bin migrations are substantial and a bin-by-bin correction would be badly biased — the regime requiring unfolding.

The structure differs qualitatively between the muon and hadron sides. The spectrometer muon measurement makes the p_T and $p_{\parallel}$ maps nearly tridiagonal — migrations rarely exceed one bin, and the purity dips (to ~ 0.45 – 0.5 for p_T and ~ 0.35 for $p_{\parallel}$) occur where the binning is finest, recovering in the wide edge bins. The calorimetric axes are coarser-binned but relatively broader and visibly asymmetric: E_{avail} feeds preferentially *downward* (energy carried by neutrons and other unresponsive particles is not collected), with the diagonal purity falling from 0.61 in the lowest bins to 0.43 in the highest; q_3 , built from E_{avail} and the muon, inherits a milder version of the same pattern; and W , which compounds the E_{avail} and muon smearing, is the broadest, with two-bin migrations common in the resonance region while the lowest- W (quasi-elastic) bin stays the purest at 0.71.

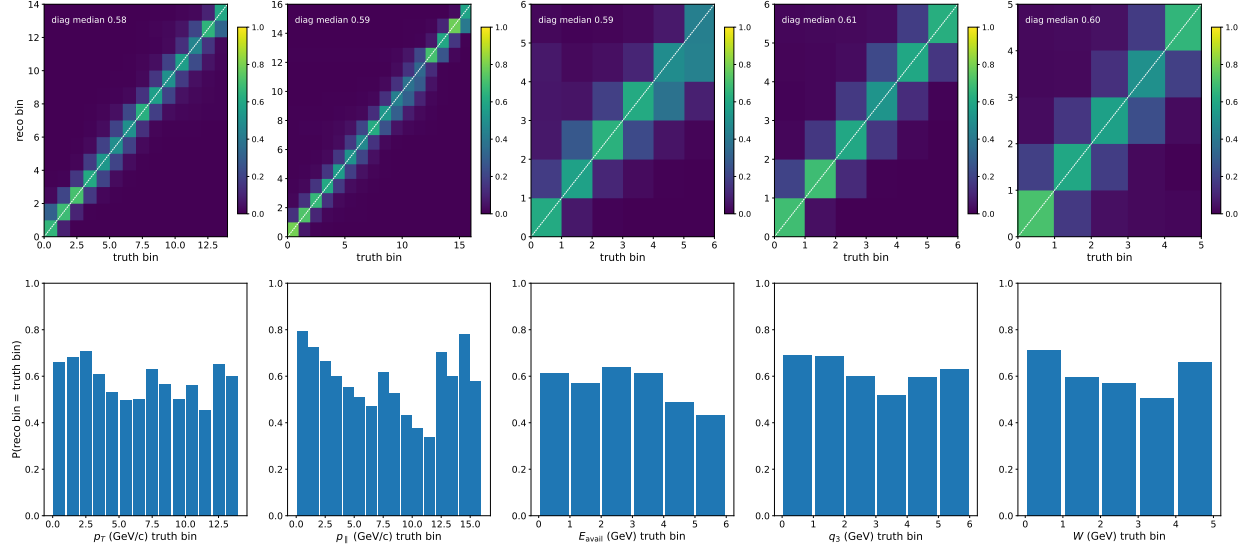


Figure 3: Truth→reco migration of the simulated signal on the analysis binning (top: column-normalized maps; bottom: diagonal purity per truth bin) for the five observables. Median diagonal purity is 0.58–0.61 per axis. Shown for illustration only — the unfold itself is unbinned.

4 Uncertainty quantification

The uncertainty budget combines three independent blocks — systematic (universe), statistical (bootstrap), and ML (seedscan) — summed assuming independence (distinct RNGs / physics sources). This section describes each block and their combination for the 2D measurement; the 3D and 4D budgets (§7.6, §7.8) repeat the identical procedure on the higher-dimensional grids. Full mathematical procedures, with worked examples on the reported numbers, are in App. A. Our *standalone* combined budget matches the published total.

Component	N	$\sqrt{\text{Tr } C}$	per-bin median rel. σ
ML noise (lgbm seedscan)	10	5.06×10^{-41}	0.166 %
Statistical (Poisson bootstrap)	300	1.82×10^{-40}	0.549 %
Systematic (universe sweep + 1.4 % norm)	187+1	3.21×10^{-39}	6.830 %
Combined (block sum)	—	3.22×10^{-39}	6.865 %
Paper TotalCov (reference)	—	2.68×10^{-39}	6.86 %

Table 1: Uncertainty budget for the 2D cross section over the 205 reported bins. For each component, C denotes its 205×205 covariance matrix; $\sqrt{\text{Tr } C}$ is the quadrature sum of its absolute per-bin uncertainties. The standalone combined median relative uncertainty (6.865 %) matches the published total (6.86 %).

4.1 Systematic universes

The detector and model systematics use the MAT-MINERvA universe machinery: 44 error bands and 187 universes in total, identical to the published analysis’s `GetStandardSystematics` set. Each universe is re-unfolded with the central estimator seed. All per-band covariances use the MAT convention: universe-mean centering and biased $1/N$. For a $\pm 1\sigma$ pair this is the outer product of the

half-difference; a common displacement of both endpoints from CV is excluded from the variance. No per-band displacement artifact is stored; only the joint-throw mean shift described below is stored separately. The 100 PPFX universes use the same mean-centered $1/N$ formula.

Dump-all shifted branches are limited to CV event-loop support. For the five genuinely kinematic bands (BeamAngleX/Y, MuonResolution, and Muon_Energy_MINERvA/MINOS), selection-complete active-universe event loops are the production migration gate and are still in progress; until their replacement covariance lands, higher-dimensional uncertainty products that use the dump-all lateral branches remain candidates. MinosEfficiency and GEANT are weight-only. Top 2D bands: Flux 4.99 %, Muon_Energy_MINOS 2.31 %, MinosEfficiency 1.47 %, Muon_Energy_MINERvA 1.29 %. Figure 4 shows the grouped fractional-uncertainty projections in the published Fig. 6/7 style (the full 205×205 covariance projected exactly, $C_{1D} = PC_{2D}P^\top$, before dividing by the 1D central value).

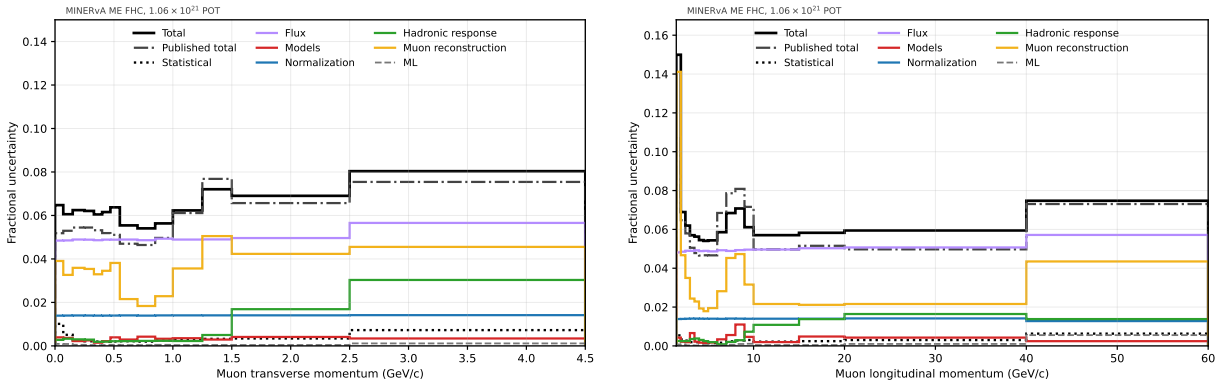


Figure 4: Fractional uncertainty on the 2D cross section projected onto p_T (left) and $p_{||}$ (right), grouped by source in the style of Figs. 6–7 of Ref. [4]: flux, interaction models, normalization, hadronic response, muon reconstruction, and the statistical band.

Flux $1/\Phi$ normalization. The dominant flux term enters as $\sigma \propto 1/\Phi$; varying Φ per flux universe, rather than dividing every universe by the CV Φ , gives a Flux band of 4.99 % (versus ~ 1 % under the CV convention) and a standalone combined budget of 6.87 %. The 1.4 % target-nucleon normalization is a fully-correlated rank-1 band. The 100-universe flux covariance has a participation-ratio effective rank of ≈ 1 (one coherent normalization mode carries 99.6 % of the variance), and its leading mode correlates at $\rho = 0.900$ with the rank-1 muon-energy-scale band: the rederived Bashyal [38] joint block, whose cross term is rank-2, with two independent estimators giving $\rho = 0.87$ – 0.90 . Adding the block leaves the systematic rank unchanged — its directions already lie within the span of the universe covariance (App. A.4).

4.2 Statistical uncertainty: Poisson bootstrap

The statistical block is a 300-replica per-event Poisson(1) weight bootstrap applied jointly to data and simulation, with two independent sub-RNGs (data and MC are independent variance sources), the MC reco and truth weights riding a *single* per-event draw (preserving the within-event migration correlation), and the GBDT seed pinned per replica so the bootstrap and seedscan variances are separable. Each replica is a full re-unfold; the covariance is the sample covariance over replicas (Fig. 5). Our bootstrap block is genuinely smaller than the published statistical block ($\approx 2.5\times$ by trace). We report this as an *observed* difference in the statistical covariance rather than a

demonstrated causal efficiency advantage; it is consistent with an unbinned statistical-efficiency gain, and the data-only stream of the data/MC split bootstrap ($C_{\text{data}}/\text{StatOnly} = 0.356$) confirms it is not a missing term (App. A.4).

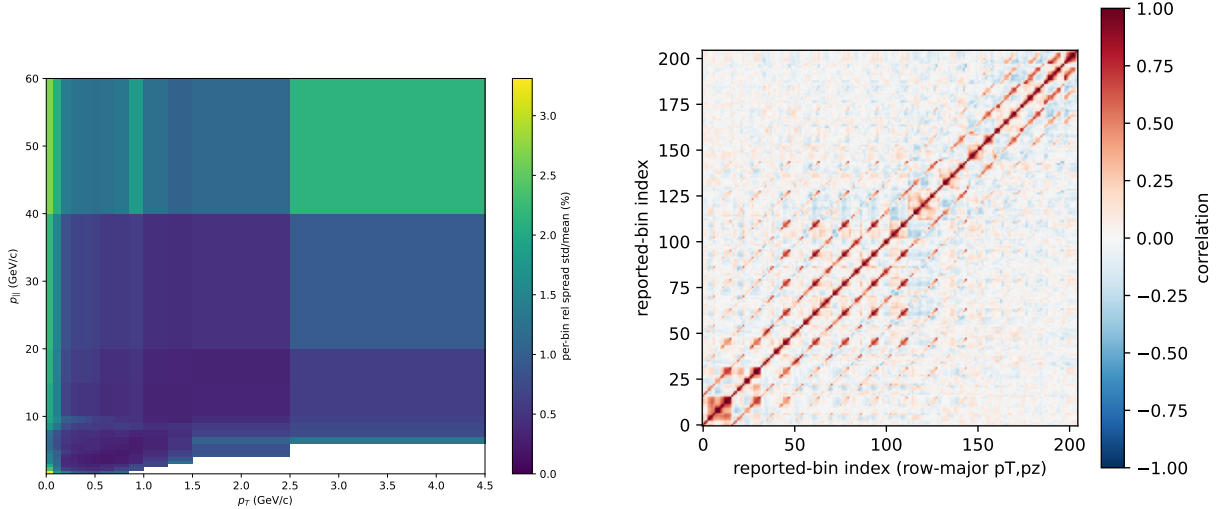


Figure 5: 300-replica Poisson bootstrap: per-bin relative spread of the unfolded 2D cross section (left) and the bootstrap correlation matrix on the 205 reported bins (right).

4.3 ML/optimization uncertainty: seedscan

ML stochasticity is budgeted by re-unfolding with 10 independent estimator seeds at fixed data and fixed universe (Fig. 6); the sample covariance over seeds is the ML block. It is the smallest block (median 0.166 % per bin). Two refinements harden it: (i) a *train/test-split* seedscan, which also re-draws the internal training split, gives a band $1.24\times$ the pure-seed band, moving the combined 3D budget by only $+0.04\%$ in $\sqrt{\text{Tr}}$ — the published budgets use the split-inclusive band; (ii) the ensemble-mean central value (averaging the seed ensemble) agrees with the production single-run CV to 0.28 %. For the corrected 5D candidate campaign, the ML ensemble fixes the estimator seed at 42 and varies the train/test split seed. It therefore measures split response, not the full estimator stochasticity. A dedicated fixed-data estimator-seed scan (twelve estimator seeds at fixed data and fixed split) closes this scope: its covariance has $\sqrt{\text{Tr}C} = 1.306 \times 10^{-39} \text{ cm}^2/\text{nucleon}$, 87.5 % of the candidate split-varied band ($1.493 \times 10^{-39} \text{ cm}^2/\text{nucleon}$), so the adopted band is dominated by genuine estimator stochasticity rather than by the particular train/test partition. The scan remains an auxiliary robustness check rather than an independent matrix — split and estimator responses are not assumed uncorrelated — and both bands are minor components, about three percent of the combined budget in $\sqrt{\text{Tr}}$.

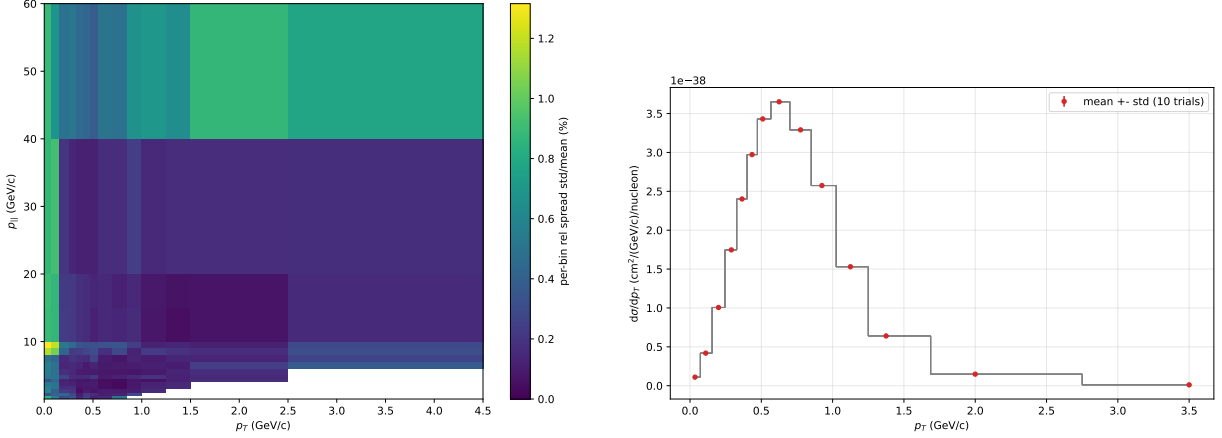


Figure 6: ML seedscan: per-bin relative spread over 10 estimator seeds (left) and the resulting fractional band projected onto p_T (right).

4.4 Combination and the unified-throw test

The nominal covariance decomposition is the block sum $C = C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$, which assumes the blocks are independent *and* that the unfolding responds linearly to simultaneous perturbations (so that per-band covariances add). The second assumption is tested directly by the *unified throw* (§7.8): joint re-unfolds with all reweightable knob bands and the flux multisim perturbed simultaneously per event. The corrected construction uses the actual -1σ and $+1\sigma$ event-weight endpoints, interpolates asymmetrically between them, holds the unfolding-estimator seed fixed for every throw, and forms the joint covariance about the throw mean. The joint displacement of that mean from the CV is stored separately as a per-bin vector and its norm; per-band endpoint displacements are not stored. The matched block comparator uses the MAT convention: universe-mean centering and the biased $1/N$ normalization, with both endpoints re-unfolded.

For the 5D candidate, the background-aware block sum has median per-bin uncertainty 13.36 %; including cross-source nonlinear response raises the candidate mean-centered covariance to $\sqrt{\text{Tr } C} = 5.81 \times 10^{-38} \text{ cm}^2/\text{nucleon}$. The joint mean shift, norm $1.65 \times 10^{-38} \text{ cm}^2/\text{nucleon}$, is reported separately rather than folded into that covariance. A CV-centered construction gives the larger $6.24 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ and is retained as a conservative variant. Both are positive semidefinite, but neither is adopted for publication until the selection-complete lateral replacement lands. Repeating all 188 universe unfolds with per-universe background subtraction changes the combined block sqrt-trace by only 0.30 %, closing the background-CV concern as numerically negligible. The 4D and extended-fiducial results do not yet have covariances constructed with this procedure; the 5D covariance is not transferred to them.

More generally, this reframes the block sum as an approximation — valid only where the unfolding responds linearly — rather than as a default construction. We retain the per-source block decomposition for attribution, but use it only after joint throws and matched block endpoints pass the stated validation tests.

χ^2 caveat. The combined paper+ours $\chi^2/\text{ndf} = 1.481$ double-counts systematics shared with the paper (flux/GENIE/MINOS); the ours-only inverse-covariance χ^2 is evaluated only with a truncated-spectral convention on the well-constrained subspace. We therefore quote per-bin pulls, the regularization scan, and the truncated result together rather than treating an unconstrained

pseudo-inverse as a standalone goodness-of-fit (App. A.4).

5 Results

Throughout this section, error bars on the unfolded points are the total uncertainty — the combined systematic (universe), statistical (bootstrap), and ML (seedscan) budget constructed in §4; the published points carry the published total covariance.

5.1 Double-differential cross section

Figure 7 shows the unfolded $d^2\sigma/(dp_T dp_{\parallel})$ and its 1D projections. The total cross section is $3.073 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ (published: $3.039 \times 10^{-38} \text{ cm}^2/\text{nucleon}$).

The shapes are those expected for CC-inclusive scattering at $\langle E_{\nu} \rangle \sim 6 \text{ GeV}$. The p_T projection vanishes at $p_T = 0$ (phase space), rises to a broad maximum at $p_T \approx 0.55\text{--}0.70 \text{ GeV}/c$, and falls by more than two orders of magnitude toward the $4.5 \text{ GeV}/c$ edge. The peak position is set by the momentum-transfer scale of few-GeV neutrino scattering: for the forward muons selected here $p_T^2 \simeq Q^2 E_{\mu}/E_{\nu}$, and the cross section concentrates at $Q^2 \lesssim 1 \text{ GeV}^2$ — nucleon form-factor suppression in the quasi-elastic and resonance channels, structure-function kinematics in the deep-inelastic contribution — which with $E_{\mu}/E_{\nu} \sim 0.6$ gives the observed $p_T \sim 0.6 \text{ GeV}/c$ scale. The $p_{\parallel}$ projection rises from the $1.5 \text{ GeV}/c$ threshold to a broad maximum at $p_{\parallel} \approx 3\text{--}4.5 \text{ GeV}/c$, the image of the NuMI medium-energy flux peak at $E_{\nu} \approx 6 \text{ GeV}$, and falls steadily thereafter (two orders of magnitude by $p_{\parallel} \approx 20 \text{ GeV}/c$). In the 2D map the $\theta_{\mu} < 20^\circ$ acceptance empties the high- p_T , low- $p_{\parallel}$ corner (the 19 unreported diagonal cells of §3.5).

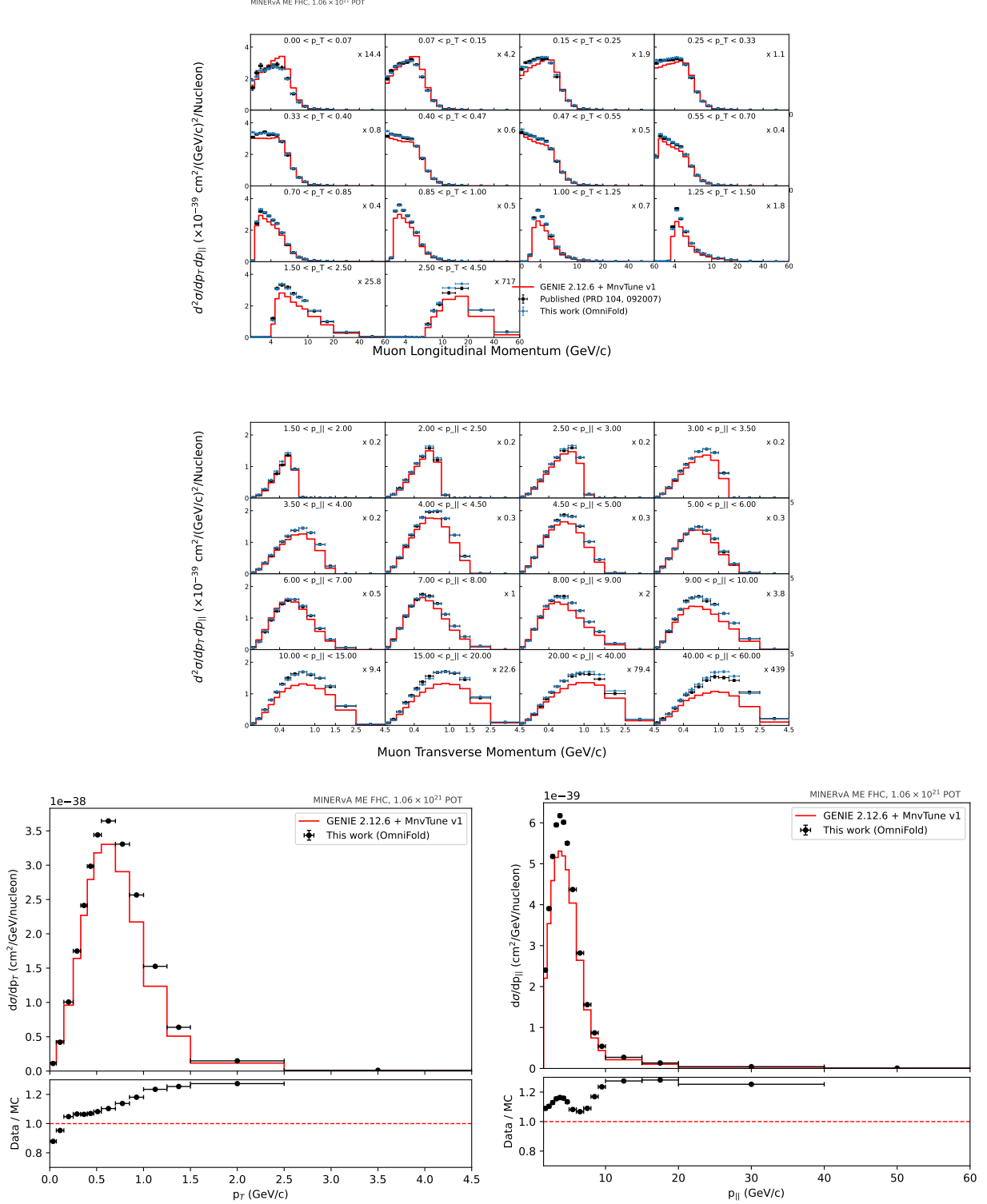


Figure 7: Unfolded 2D cross section (top, Fig. 13 style) and its $p_T/p_{||}$ projections (bottom). The p_T peak at $\sim 0.6 \text{ GeV}/c$ reflects the $Q^2 \lesssim 1 \text{ GeV}^2$ momentum-transfer scale and the $p_{||}$ peak the medium-energy flux maximum (see text).

5.2 Comparison to the published measurement

Table 2 quantifies the agreement with Ref. [4] over the 205 reported bins. The total cross section reproduces the published one to 1.1 % (ratio 1.011), with a median per-bin ratio of 1.006; 77.6% of bins agree within 5 % and 94.1% within 10 %. Two indicative covariance distances per reported bin are quoted: against the published covariance alone (3.66) and against the combined covariance that also credits this measurement’s own uncertainty (1.481); neither is a calibrated goodness-of-fit because the two results share data and their cross-covariance is unavailable. The anatomy of the former is examined in §5.3. The per-bin standardized difference using the published uncertainty as a scale (Fig. 8) is centered (mean 0.089) with RMS 0.60, well inside the published uncertainties, and Figs. 9–10 show the same comparison bin-by-bin in p_T and $p_{\parallel}$ slices.

Quantity (205 reported bins)	Value
$\sigma_{\text{tot}}(\text{ours})/\sigma_{\text{tot}}(\text{paper})$	1.011
Median bin ratio (ours/paper)	1.006
Bins within 5 %/10 %/20 %	77.6/94.1/98.5 %
Indicative covariance distance/bin, paper covariance	3.66
Indicative covariance distance/bin, paper + ours	1.481
Published- σ standardized-difference mean / RMS	0.089/0.598

Table 2: Agreement with arXiv:2106.16210. Per-bin standardized differences using the published uncertainties (Fig. 8) are sub- 1σ ; shared-data cross-covariance is not available.

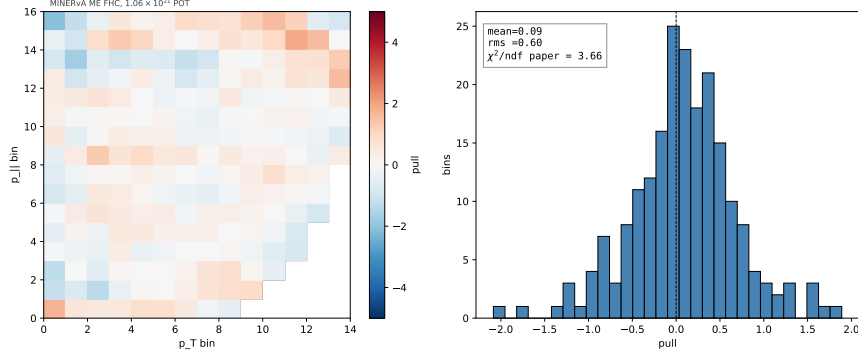


Figure 8: Per-bin standardized difference (ours – paper)/ σ_{paper} on the 14×16 grid and its distribution. It is a descriptive scale, not an independent-result pull.

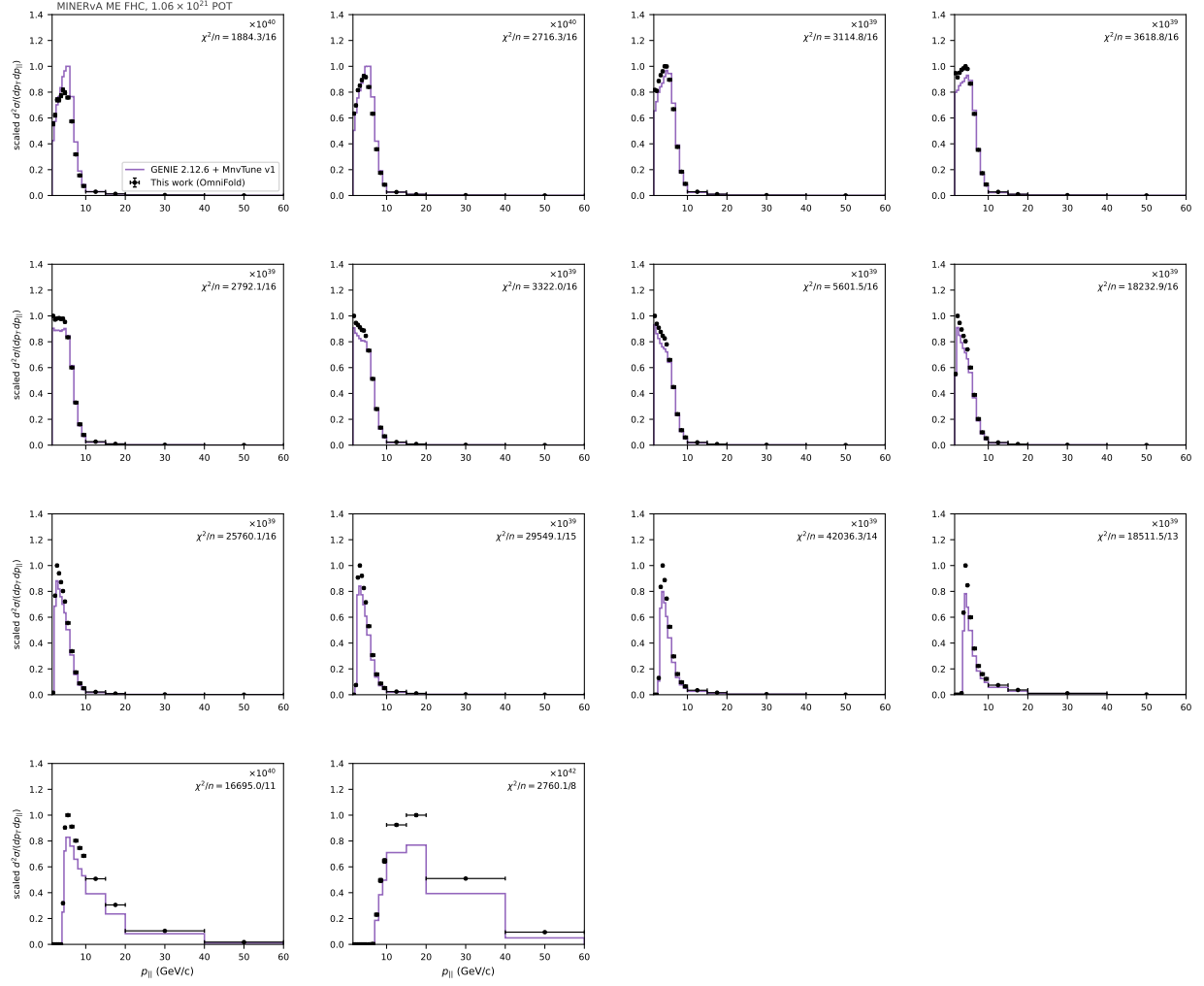


Figure 9: Bin-by-bin comparison to the published measurement: the unfolded cross section (points) overlaid on the published data (histogram) in slices of p_T versus $p_{||}$.

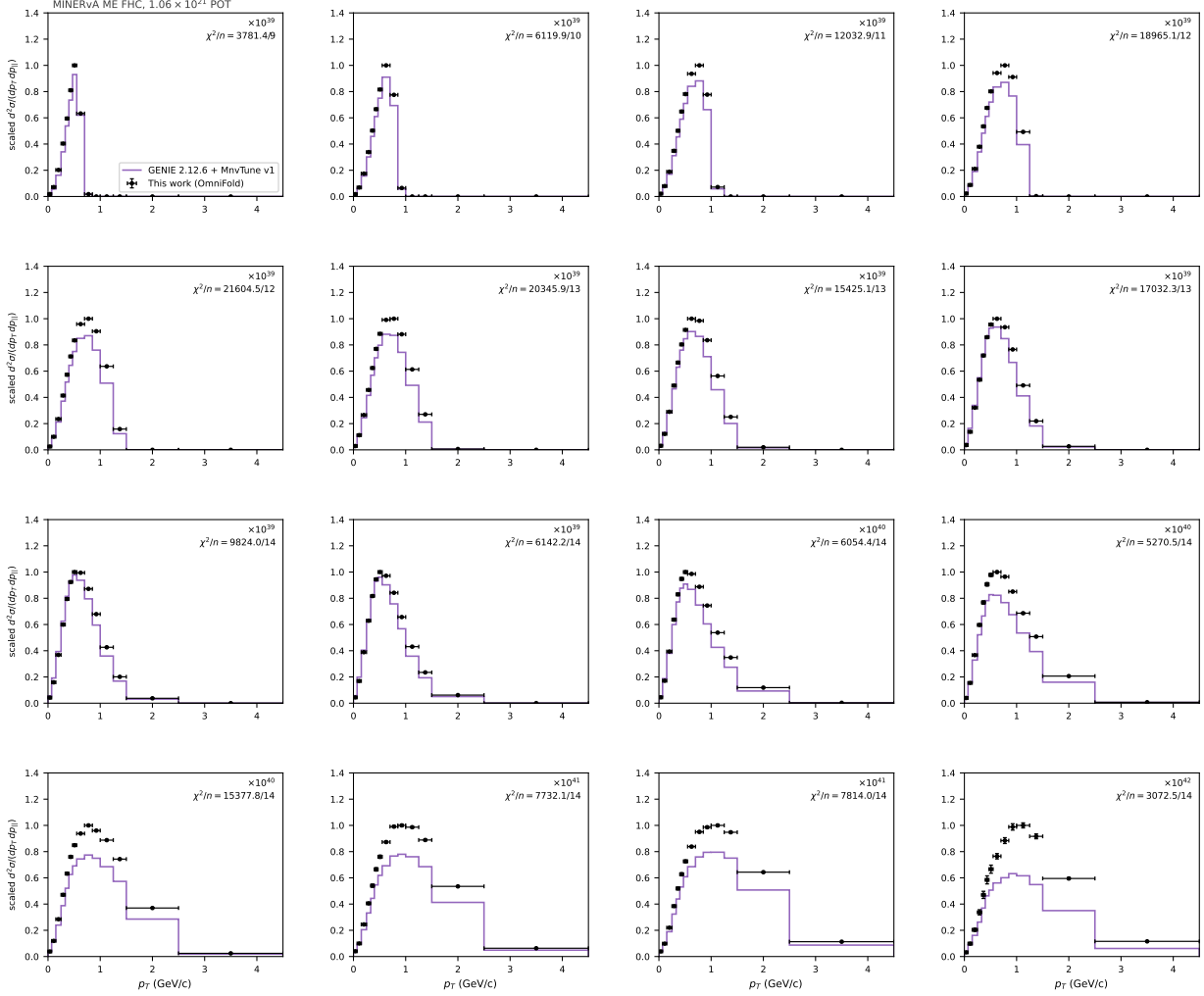


Figure 10: Same comparison as Fig. 9, sliced in $p_{||}$ versus p_T .

5.3 Anatomy of the $\chi^2 = 3.66$ tension

Using the published uncertainties as a descriptive scale, the central values have pull RMS 0.598 (a diagonal-only χ^2/nfd would be 0.37), while the full paper-covariance distance is 3.66 per bin: a $10\times$ inflation living entirely in the off-diagonal structure of the published covariance. The eigenmode decomposition (Fig. 11) shows the indicative distance is carried by moderate- λ modes whose paper-covariance-standardized components have magnitude 5–6 (descriptive eigen-coordinates, not calibrated significances) (the 10 smallest- λ modes carry only 3%; keeping the 73 best-measured directions still gives 1.42), so it is *not* a small- λ inversion artifact. It is *shape*, not normalization ($\chi^2_{\text{norm}} = 0.10$ vs $\chi^2_{\text{shape}} = 750$), localized to the low- p_T cross-section peak ridge where the flux and muon-energy bands dominate. A ~ 1 -unit methodological band comes from the GBDT estimator (exact-GBT 3.66 vs HistGBT 2.70 ± 0.04 vs LightGBM 2.65; iteration count is negligible, $5 \rightarrow 10$ iters move it by +0.01). Full treatment: App. A.5. The eigenmode-localization maps and the additional model-comparison ratio maps are shown in Figs. 11–12. Because the two results share data and systematic sources and their cross-covariance is unavailable, none of these coordinates is a calibrated independent-result goodness-of-fit statistic.

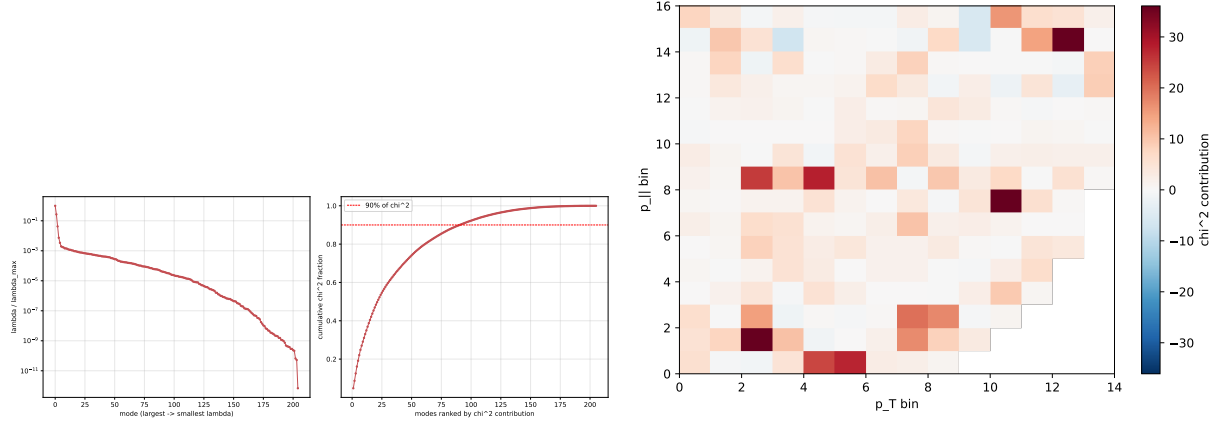


Figure 11: Tension anatomy: eigenvalue spectrum with cumulative χ^2 (left) and the per-bin χ^2 -contribution map (right).

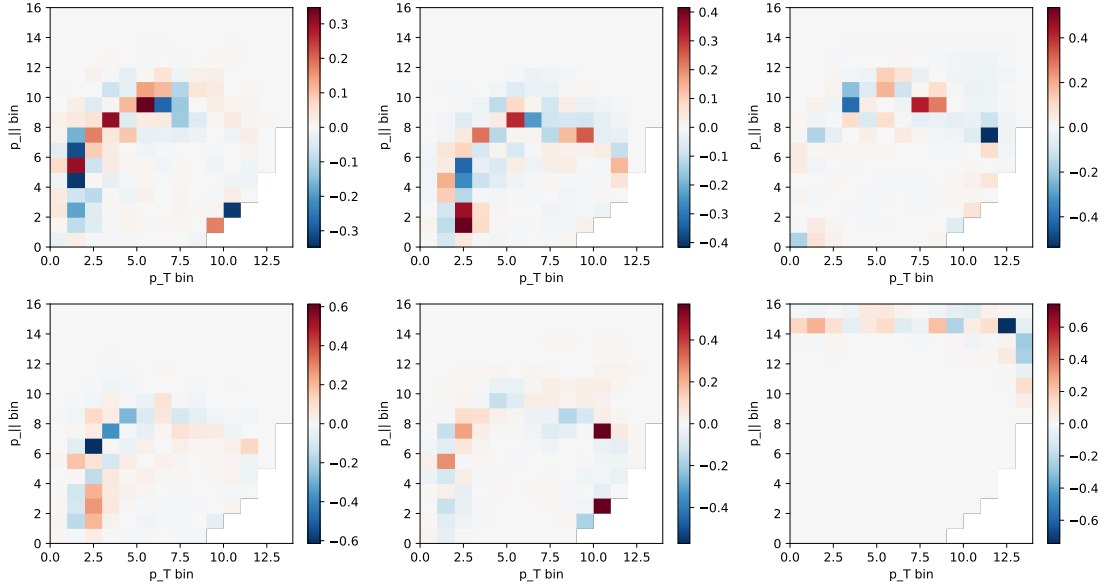


Figure 12: Bin-space maps of the leading tension-carrying eigenmodes of the published covariance: the magnitude-5–6 standardized components live on the low- p_T cross-section-peak ridge where the flux and muon-energy bands dominate.

5.4 Comparison to GENIE MINERvA Tune v1

Both the published data and our unfolded result are compared to the GENIE 2.12.6 [5] MINERvA Tune v1 prediction shipped with the ancillary (Fig. 13). The tune under-predicts the total by $\sim 9\%$ ($\sigma_{\text{tune}}/\sigma_{\text{data}} = 0.91$), concentrated in the low- p_T peak, and is strongly disfavored by the data covariance (χ^2/ndf : data vs tune 33.0, ours vs tune 26.5). Our unfolded result tracks the data far more closely than the model does. The published analysis [4] additionally compares to NuWro [7], GiBUU [8] and NEUT [39]; because OmniFold delivers a per-event reweighted truth ensemble, any such generator prediction can be compared to our result in *any* projection or dimension without

re-unfolding — a capability we exploit for the 3D extension (§7).

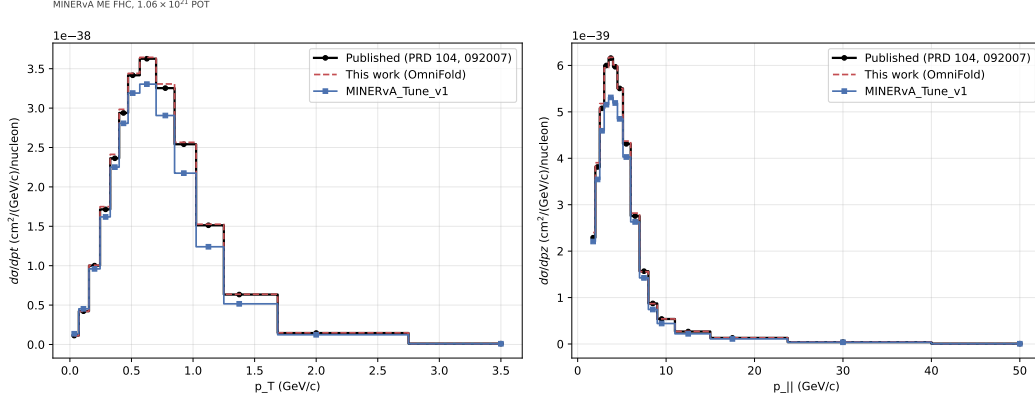


Figure 13: Our result (dashed) on top of the published data (black); MINERvA Tune v1 (orange) dips below in the peak.

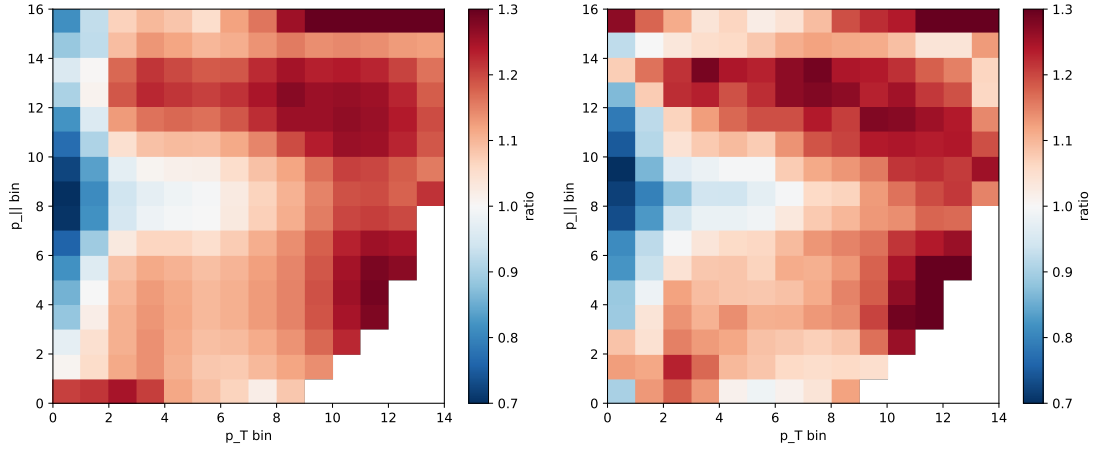


Figure 14: Bin-by-bin ratio maps on the $(p_T, p_{||})$ grid: ours/paper (left) and Tune v1/paper (right). The tune’s deficit is concentrated in the low- p_T peak; our result is flat near unity.

6 Validation

Closure. Three closure families, all passing (1A lgbm 5-iter; thresholds in parentheses):

- **Truth-reweight** (multiply truth and reco weights by the same factor): gaussian- p_T residual 0.046 %, tilt- $p_{||}$ 0.013 % (thr 1.5 %) — injected truth recovered.
- **Hidden-variable** (bias on $\Delta p_T = p_T^{\text{reco}} - p_T^{\text{truth}}$, a resolution variable not in the feature set): leakage linear in amplitude, $A = 0.05/0.10/0.30 \rightarrow 0.57/1.14/3.43$ % (thr 3.0 %).
- **Alt-model** (train on CV, pseudo-data from an alternative universe): MaCCQE 0.068 %, Flux 0.376 % (thr 2.0 %).

Bottom-line test. Following the unbinned-unfolding guide [21], the unfolded result must recover the data’s deviation from the prior with a residual far smaller than the feature itself (the unfolding cannot manufacture discriminating power). Using the closure runs, in the bins carrying the injected feature the unfolded result tracks the injected truth to $\approx 10\times$ better than the feature size: 2D (gaussian-bump truth reweight) residual 1.69 % vs injected 17.2 % (ratio 0.098); 3D (+30 % E_{avail} bump, E_{avail} projection) 1.84 % vs 18.1 % (ratio 0.102). Both pass. (The naive data-vs-prior form—reco vs truth χ^2 —instead has truth>reco, as expected for a high-resolution measurement where detector smearing lowers the reco baseline and a stat-only diagonal omits the unfolding covariance; the proper full-covariance goodness-of-fit is the χ^2 -vs-paper of §5.3 and the generator comparisons.)

These closure tests validate the estimator but are not used to assign a non-closure uncertainty. Such an uncertainty would require a predeclared suite of injection shapes and amplitudes rather than calibration to a single Gaussian bump.

Unbinned diagnostic (C2ST). As a binning-free check that the unfold removes the data $\leftrightarrow$ MC mismatch, we use a classifier two-sample test (C2ST): a classifier is trained to separate the data from the (reweighted) simulation in the full unbinned feature space, and its held-out AUC/accuracy is compared to the chance value 0.5. The discriminator AUC drops from ≈ 0.535 before unfolding to ≈ 0.501 after, indicating that the reweighting removes the reconstructed data–MC mismatch. We report this strictly as a *descriptive* diagnostic and do *not* quote a calibrated p -value: the C2ST as run trains and evaluates on the same sample without cross-fitting, and a valid null would require re-running the full (cross-fitted) OmniFold pipeline per pseudoexperiment (hundreds of unfolds), which is not computationally justified here. Calibrated unbinned goodness-of-fit remains an open problem [21].

Toy residuals and coverage status. The existing 200 2D closure toys fluctuate both the unfolded result and the stored `hTruthXSec2D` through the MC bootstrap. The stored truth varies in all 205 reported bins, so its all-toy mean is not an independent fixed truth for a split-sample coverage claim. We therefore quote no 2D coverage number from this ensemble. Its same-ensemble figures (68.71 %, $\langle|r|\rangle = 0.794$ vs $\sqrt{2/\pi} = 0.798$, and 97.6 % of bins above 65 %) are retained only as standardized-pull/Gaussianity diagnostics. A coverage test requires an independently stored unfluctuated truth reference (or a redesigned ensemble) and remains open.

Stability. 5-iter total σ within 0.026 % of 10-iter (per-bin shape RMS 1.54 %); HistGBT and exact-GBT agree on total σ to 0.04 %. Efficiency (Fig. 19) reproduces the published Fig. 5.

Classifier choice (GBDT vs. NN). Published OmniFold analyses use neural-network classifiers [2, 21]; we use gradient-boosted decision trees, whose output must still be a calibrated posterior for $w = p/(1 - p)$ to be valid. On the step-1 reco classifier (data vs. MC reco) the reliability curve is near-diagonal (Brier ≈ 0.25 at AUC ≈ 0.537 : data and reco MC are nearly indistinguishable, so the reweight is a modest correction). A full scalar Keras-MLP OmniFold cross-check on the same weighted 3D inputs and five-iteration loop gives total ratio 1.0078 versus the GBDT reference and median projection differences of 0.7 % to 1.4 % (Fig. 15). The agreement in the populated bulk establishes that the scalar result is not specific to the production classifier family. As a stronger family cross-check, a vendored PET (point-cloud transformer [40]) unfold was run on the actual set-valued recoil representations — the non-muon reconstructed clusters (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack`) read by `CVUniverse::GetRecoClusters`, where `cluster_pos` is the measured coordinate in one X/U/V scintillator view. Area-normalized against the frozen 4D

GBDT result, the PET unfold reproduces the scalar shape to a median 2.3% to 3.9% per bin (p_T 3.9%, $p_{\parallel}$ 2.4%, E_{avail} 2.6%, q_3 2.3%; Fig. 16). In its role here this is a recoil-representation sensitivity check: it compares classifier families and engineered-scalar versus recoil-cloud inputs, but does not test a complete joint muon–hadron representation. Figure 17 visualizes representative events on a common energy–polar-angle plane and shows the reconstructed and truth muon together with the recoil activity. The PET classifier receives a variable-length set of single-view energy deposits at reconstruction level and particles with full four-momenta at truth level. The muon kinematics exist at both levels and are stored as event scalars for downstream binning, but are not part of the recoil-PET classifier tensor. Figure 18 shows the per-event cardinality of each cloud. Section 9 elevates the PET track to an absolutely-normalized recoil-estimator cross-check with its own ordinary closure and systematic diagnostics.

Figure 18 uses the coverage-complete point-cloud input: only 174 of the 32 849 103 truth events have an empty cloud. Including the GENIE residual-nucleus bookkeeping particles, as the PET input does, gives a mean truth cardinality of 4.57. The depleted $k = 1$ bin is a generator-record feature: a nominally single-hadron event, most often quasielastic, usually also contains a residual-nucleus pseudo-particle and therefore enters at $k = 2$. Removing those pseudo-particles gives a mean physical-hadron multiplicity of 4.43 and a distribution that rises smoothly to its $k = 2$ maximum. The twelve-constituent cap is a compute/memory choice: PET cost grows with cardinality, while only 2.31% of truth events reach the cap and the twelfth constituent carries only 0.09% of the retained truth energy on average.

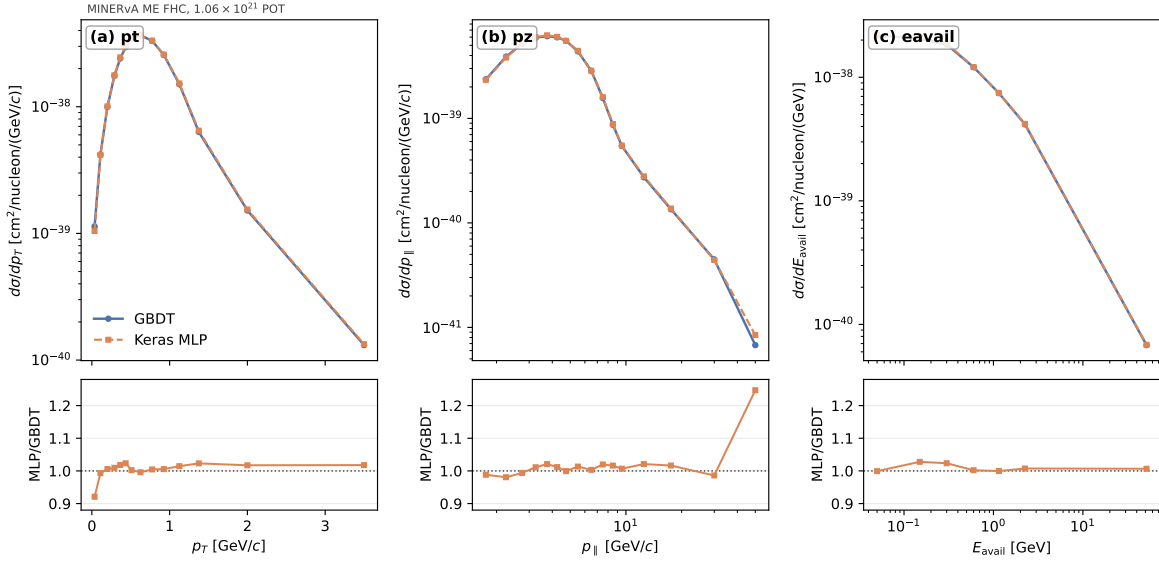


Figure 15: Full five-iteration scalar OmniFold cross-check using identical weighted 3D inputs and extraction, with only the classifier family changed. Top: absolutely normalized GBDT and Keras-MLP projections. Bottom: their binwise ratio. The MLP/GBDT total-cross-section ratio is 1.0078; median absolute projection differences are 1.20% in p_T , 1.36% in $p_{\parallel}$, and 0.66% in E_{avail} . The larger excursions are confined to the sparsely populated outermost p_T and $p_{\parallel}$ bins.

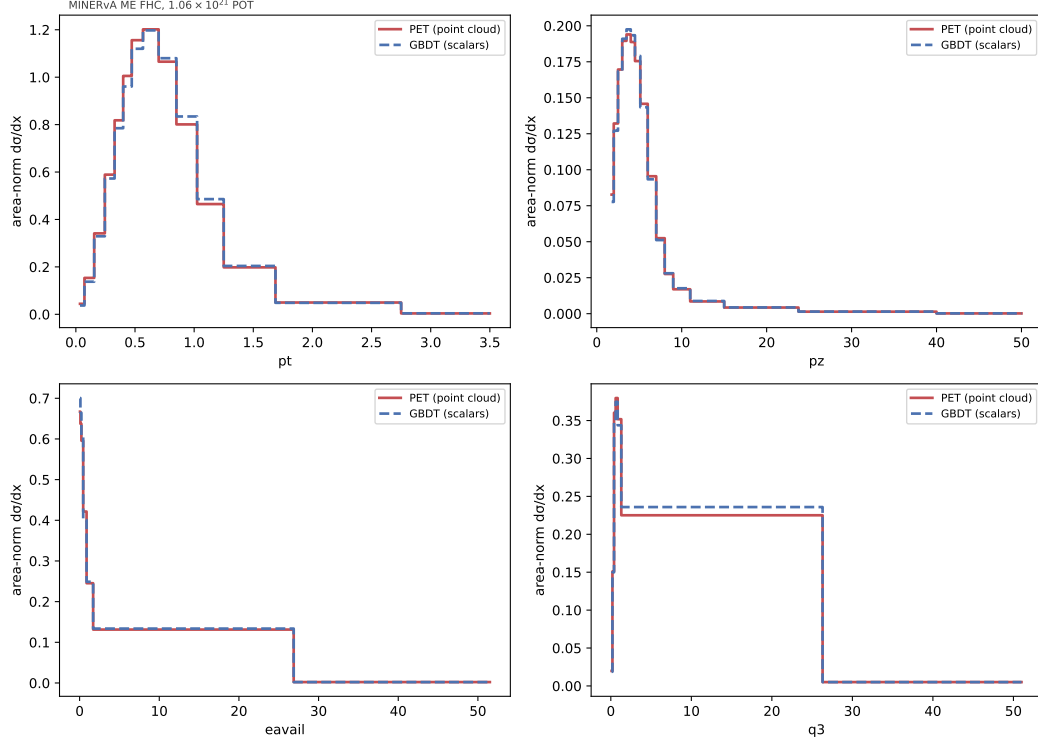


Figure 16: Recoil-representation cross-check: area-normalized 4D unfolded shapes from the recoil-only PET point-cloud transformer (solid) versus the production scalar GBDT (dashed), projected onto each axis. Median per-bin shape differences are 2.3 % to 3.9 %; the only visible deviation is the wide q_3 catch bin, which collects a large, structured high- q_3 tail into one wide bin — a binning artifact of that catch bin rather than a genuine shape disagreement. Shape-only comparison (PET on a 2M-event subsample), not an independent physics result.

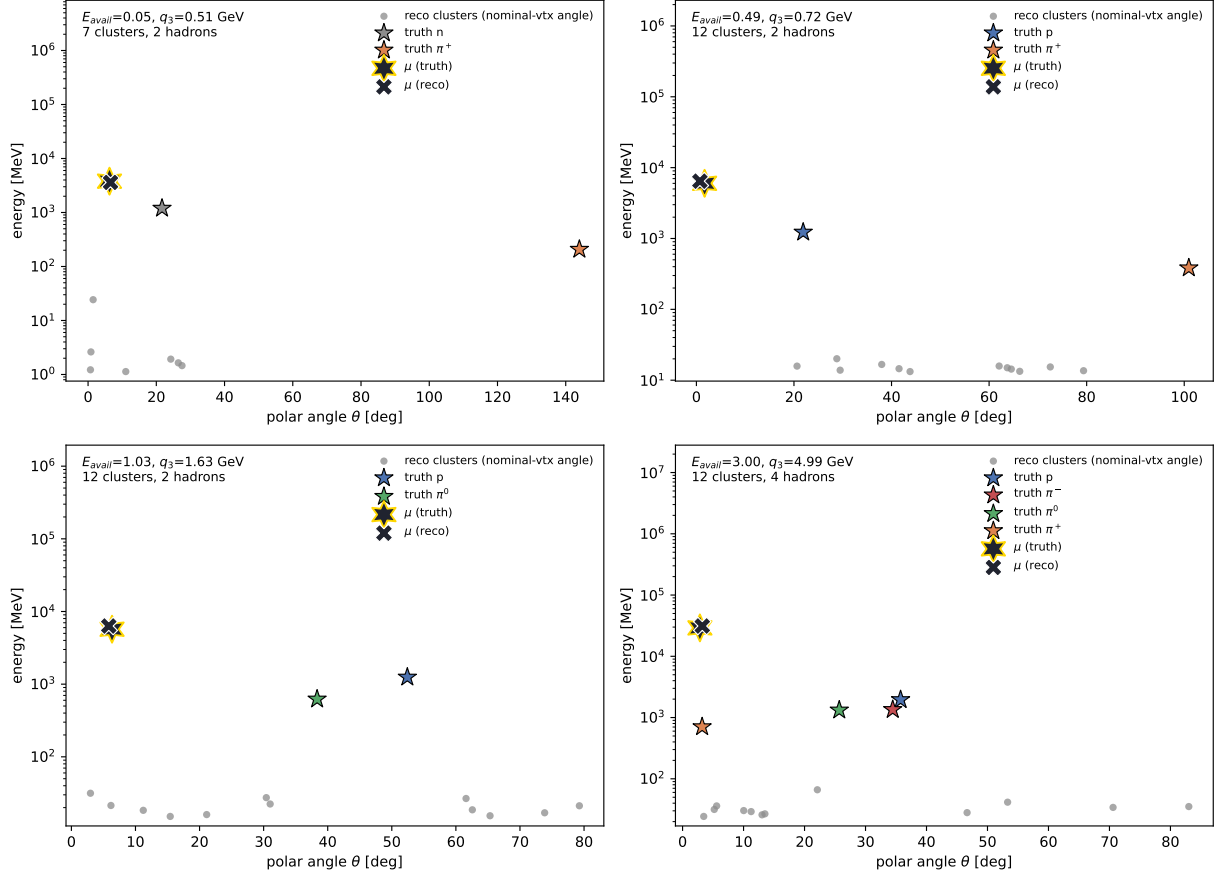


Figure 17: Four representative PET events on a common energy–polar-angle plane. Gray points are reconstructed recoil clusters; colored stars are truth final-state hadrons, and the reconstructed (X) and truth (six-point star) muons nearly overlap. The muon momentum and direction are measured from the MINERvA–MINOS track and stored as event-level $(p_T, p_{||})$ scalars. The cluster angle is illustrative, formed relative to a nominal upstream vertex because the current reco-cloud dump stores $[E_{\text{dep}}, \text{pos}_{\text{view}}, z]$ but not the reconstructed vertex. Thus this common projection is a visual comparison, not the PET input tensor: the classifier consumes the raw single-view cluster set at reco level and hadron four-vectors at truth level, with the muon currently omitted from both classifier inputs.

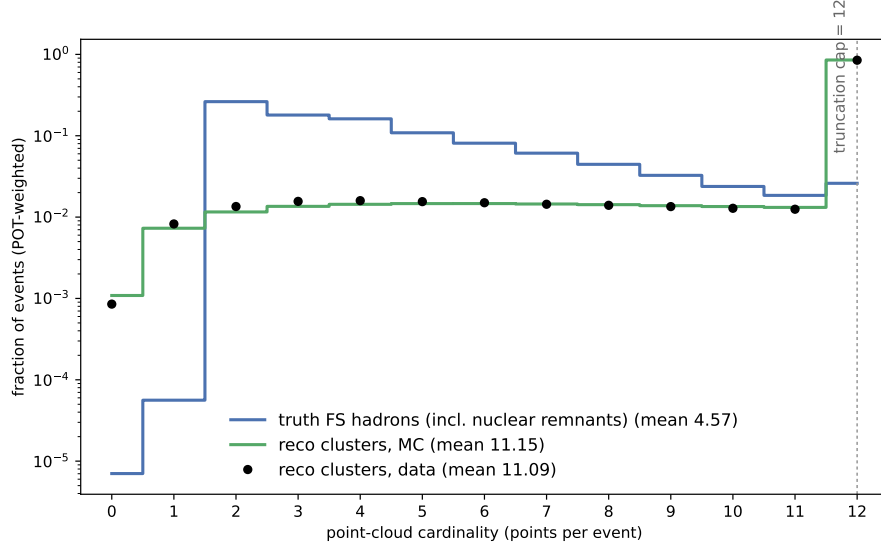


Figure 18: Per-event point-cloud cardinality (number of constituents) for the truth final-state hadrons, the reconstructed clusters in MC, and the reconstructed clusters in data, POT-weighted. The reco-cluster multiplicity greatly exceeds the truth hadron multiplicity — a single hadron deposits energy in many detector clusters — while data and MC reco agree closely (means 11.09 vs. 11.15). Clouds are truncated to the twelve highest-energy constituents per event, so the reco distributions pile up at the cap; the truth clouds are sparser (mean 4.57, including GENIE nuclear-remnant bookkeeping particles) and rarely reach it. The depleted $k = 1$ truth bin and the truncation choice are explained in the text.

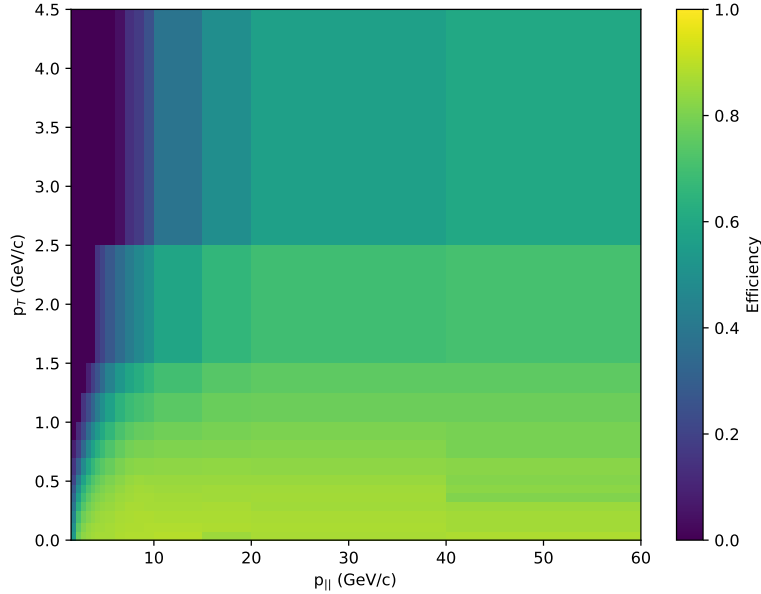


Figure 19: Selection efficiency over the $(p_T, p_{||})$ truth grid. Efficiency is diagnostic only — OmniFold’s miss treatment performs the acceptance correction natively (§3.1).

7 Higher-dimensional extension: available energy

The enabling property of unbinned OmniFold is that adding an observable is adding an input feature: the step-1/step-2 classifiers simply gain a column, with no explicit migration-matrix or per-bin response tensor. This removes the combinatorial binning penalty of matrix unfolding, but it is not free of a dimensional cost — classifier training, acceptance/support coverage, sample sparsity, and uncertainty quantification all become more demanding as observables are added. There is no practical D’Agostini IBU analog, which is the primary methodological motivation for the technique. We exercise it by extending the measurement to $d^3\sigma/(dp_T dp_{\parallel} dE_{\text{avail}})$ with available energy as a third axis.

Prior MINERvA measurements include binned triple-differential results (Fig. 20, Table 4). The distinction here is the unbinned, simultaneous CC-inclusive scalar formulation and its extension to three, four and five recorded observables, not multidimensionality by itself.



Figure 20: Dimensionality of the full set of MINERvA differential cross-section measurements over time (number of kinematic variables unfolded *simultaneously*; a double-differential $d^2\sigma/dx dy$ counts as two), colored by channel. Prior MINERvA results use explicit binned unfolding and include a triple-differential quasielastic-like measurement; this work extends an unbinned CC-inclusive scalar unfold to five recorded observables. Source: the MINERvA data-release page and the cited publications; full list in Table 4. Dimensionality is classified from each analysis’s headline differential result; flux, reconstruction, generator-tuning and total-only entries are excluded.

7.1 Definitions and pipeline

Available energy follows the MINERvA low-recoil convention (Ref. [41], Eq. 4): truth $E_{\text{avail}} = \sum_p \text{KE} + \sum_{\pi^\pm} \text{KE} + \sum_{\pi^0, \gamma} E$ (excluding neutrons and the muon), via `CVUniverse::GetEAvailableTrue()`; reco E_{avail} is the calorimetric tracker+ECAL estimate scaled by 1.17, `CVUniverse::NewEavail()`. The event loop evaluates both accessors over all twelve ME FHC playlists, carrying the E_{avail} branches alongside the $p_T/p_{\parallel}$ branches. The unfolding driver is the 2D pipeline with a three-

column feature stack $(p_T, p_{\parallel}, E_{\text{avail}})$; the cross-section extraction generalizes to a 3D Jacobian $\Delta p_T \Delta p_{\parallel} \Delta E_{\text{avail}}$ with the flux Φ still applied per p_T (broadcast over $p_{\parallel}$ and E_{avail}). The E_{avail} binning $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV ends in a wide catch bin so that the E_{avail} -marginal captures the full CC-inclusive recoil tail; the global OmniFold completeness is $c = 1.0000$, as in 2D.

7.2 The available-energy spectrum

Figure 21 shows the unfolded $d\sigma/dE_{\text{avail}}$ (the 1D projection of the 3D result). It falls monotonically from $2.19 \times 10^{-38} \text{ cm}^2/\text{GeV}/\text{nucleon}$ at the low-recoil peak to 6.8×10^{-41} in the catch bin — the expected CC-inclusive shape, with most of the rate at $E_{\text{avail}} \lesssim 0.8 \text{ GeV}$.

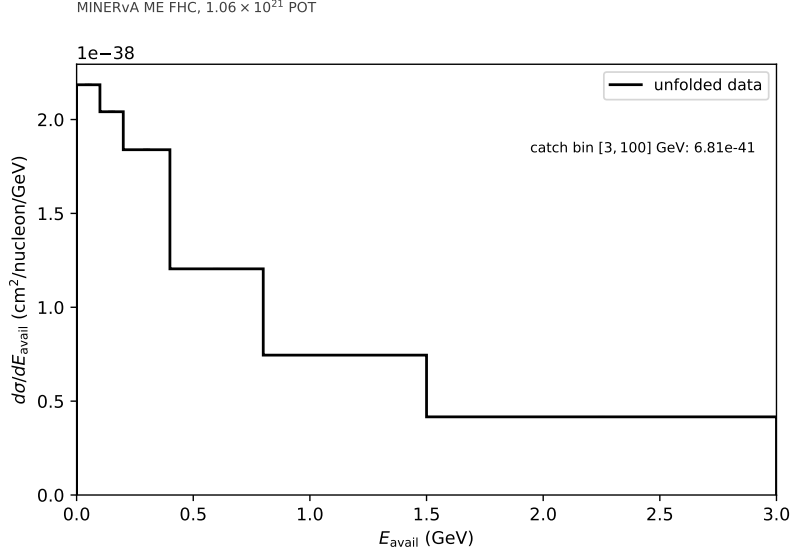


Figure 21: Unfolded available-energy spectrum $d\sigma/dE_{\text{avail}}$, the 1D projection of the 3D measurement. The wide $[3, 100]$ GeV catch bin (value annotated) is excluded from the axis.

7.3 Validation against the 2D measurement

With no published 3D reference, the validation anchor is that marginalizing the 3D cross section over E_{avail} must reproduce the 2D result of §5. The marginal recovers the 2D *normalization* closely: total σ agrees to $+0.95\%$ and the per-bin marginal/2D ratio has median 1.0016. The *shape* agreement is looser — an indicative distance using the published covariance gives $\chi^2/\text{ndf} = 4.98$, above the frozen 2D values (3.66 default, 2.65 LightGBM-CV), driven by a $\sim 4.4\%$ per-bin scatter that the third axis injects into the $(p_T, p_{\parallel})$ marginal (Fig. 22). The two results share the same data and their cross-covariance is unavailable, so this number is not a calibrated goodness-of-fit statistic.

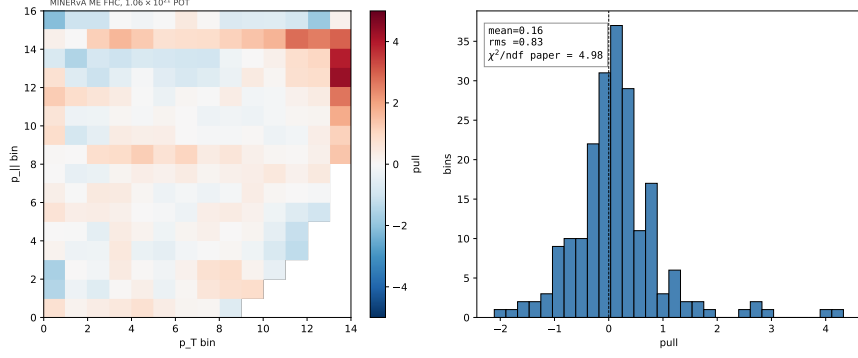


Figure 22: Per-bin standardized differences of the E_{avail} -marginal using the published uncertainties as a scale, on the 14×16 grid and in distribution. Because both results use the same data and their cross-covariance is unavailable, these are descriptive rather than formal independent-result pulls.

7.4 Closure: the scatter is physical, not a method bias

A reweight closure isolates whether that 4.4 % scatter is method bias or real information. We inject a known +30 % Gaussian bump in *truth* E_{avail} (center 0.3 GeV, width 0.15 GeV) into the pseudo-data and require the 3D unfold to recover it. It does: the recovered step-2 mean weight is 1.048 against the injected 1.0485, and the residual of the unfold against the reweighted-truth reference is, on the E_{avail} -marginal, median 0.9999 with standard deviation 0.0006 (0.06 %); the per-3D-bin residual standard deviation is 3.2 % (MC statistics in sparse high-recoil corners). The closure marginal scatter is thus $\sim 70\times$ tighter than the 4.4 % data-versus-2D scatter. No nonclosure is observed for this tested deformation (a single injected +30% Gaussian bump), and we read the elevated marginal distance as consistent with *data* $\leftrightarrow$ *MC structure resolved by the available-energy dimension*, rather than nonclosure of the particular tested deformation (the total normalization matches to less than 1 %).

7.5 Generator comparison in 3D

Because the published ancillary ships only binned 2D model histograms, a quantitative comparison of the 3D result (or of arbitrary projections) requires generating truth-level events ourselves. No detector simulation is needed — the unfolded result already lives at truth level — so a generator prediction is obtained by histogramming its truth $d^3\sigma$ in the same fiducial phase space and the same truth E_{avail} definition. We compare four predictions. The GENIE, NuWro, and GiBUU samples were generated and processed in this analysis, while the MINERvA Tune v1 prediction was extracted from the analysis MC truth with its full tune weight; none of the spectra were digitized or externally supplied.

- **GENIE 2.12.10 central value** — generated with `gevgen` on a CH target with the MINERvA ME FHC flux and the DefaultPlusValenciaMEC configuration [5].
- **MINERvA Tune v1** — the analysis tune (base GENIE + RPA, low-recoil 2p2h, and non-resonant-pion reweights), extracted directly from the event-loop MC truth (the `w_truth` weights), which reproduces the shipped ancillary curve to 0.01 % in normalization.
- **NuWro 21.09** [7] — an independent generator (C target), the same flux and phase space.

- **GiBUU 2019** [8] — a transport-theory generator (C target), built and run *natively* on the compute system (no container), same flux and phase space.

NEUT [39] is not included. Each prediction is normalized by its own flux-averaged total CC cross section per nucleon ($\langle\sigma\rangle \approx 3.6\text{--}4.0 \times 10^{-38} \text{ cm}^2$).

Figure 23 overlays the four on the unfolded result, with the combined-covariance systematic band of §7.6 on the data. The flux-averaged totals in the measurement phase space order as GiBUU (2.22×10^{-38}) < NuWro (2.34×10^{-38}) < GENIE CV (2.52×10^{-38}) < Tune v1 (2.71×10^{-38}) < data ($3.08 \times 10^{-38} \text{ cm}^2/\text{nucleon}$): all four models under-predict the rate, by 12% to 28%. On $(p_T, p_{\parallel})$ the four track the data shape closely; the discriminating axis is E_{avail} , where NuWro, Tune v1 and GiBUU *suppress* the low-recoil quasielastic peak below bare GENIE (RPA and nuclear effects) — yet the data sit *above* all four in the lowest- E_{avail} bins. This low-available-energy excess is invisible to the 2D measurement and is a direct statement of the unbinned method’s high-dimensional reach.

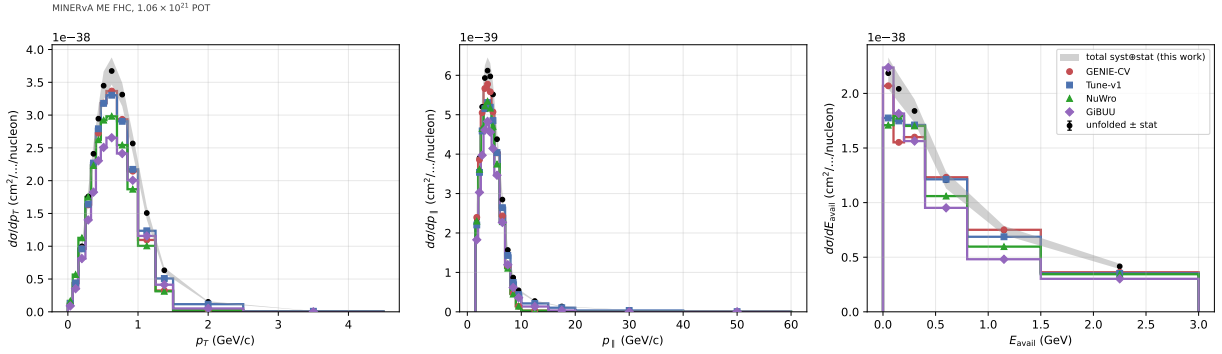


Figure 23: 1D projections of the unfolded 3D central cross section (black; the historical covariance band is shown for orientation only and is not final) versus four generator predictions produced at truth level: GENIE 2.12 CV, MINERvA Tune v1, NuWro 21.09, and GiBUU 2019. On p_T and $p_{\parallel}$ the models track the data shape; on E_{avail} they split, and all under-predict the data at low available energy. The wide E_{avail} catch bin is omitted from the axis.

At central-value level, the integrated E_{avail} rate (catch bin dropped) places GENIE CV, Tune v1, NuWro and GiBUU respectively 7.2%, 9.5%, 15.3% and 21.9% below the data. The historical truncated-spectral scan ranked Tune v1 closest and GiBUU farthest, but its covariance is superseded by the corrected dependency contract: the final 3D covariance must be projected from the adopted, selection-complete 5D trunk. Consequently no 3D generator χ^2 , p -value or significance is quoted here. The old scan is retained only as a quarantined numerical diagnostic in Fig. 24.

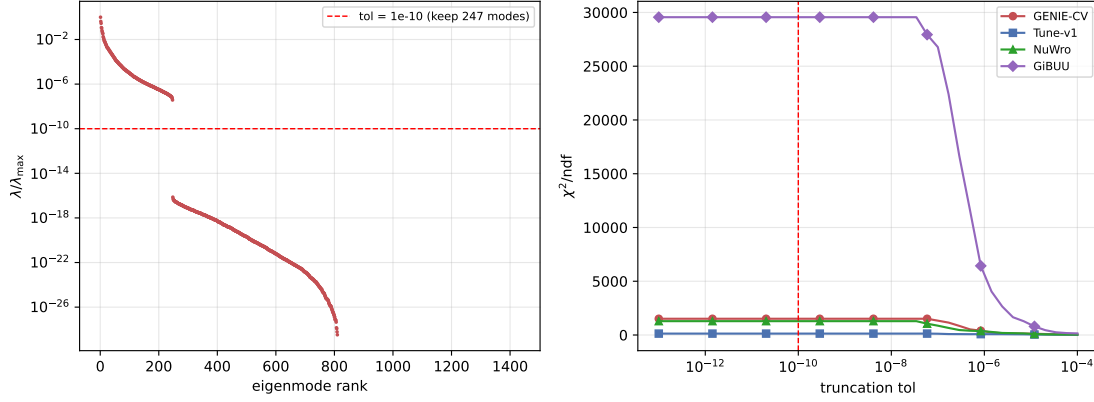


Figure 24: Quarantined historical full-3D truncated-spectral scan. It shows the old per-generator χ^2/ndf versus eigenvalue tolerance and is retained only to document numerical behavior; no value is interpreted until the selection-complete 5D covariance is adopted and projected to 3D.

7.6 Historical 3D uncertainty budget and replacement gate

The historical band on Fig. 23 was built as a full 3D analog of the 2D uncertainty budget of §4. The systematic universes are dumped on the three-axis $(p_T, p_{\parallel}, E_{\text{avail}})$ grid (187 universes), the bootstrap statistical and ML-seedscan covariances propagated, and the three summed into a single historical combined covariance $C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$ over all 1431 reported bins. It has $\sqrt{\text{tr}} = 5.72 \times 10^{-39}$, median per-bin relative uncertainty 10.4 %, numerical rank 247 (of 1431), and the same flux-dominated band ordering as 2D (Fig. 25). These are audit descriptors, not publication uncertainties. Under the corrected contract, the quotable 3D covariance is the exact projection of the final adopted 5D covariance after the selection-complete lateral replacement; all covariance-dependent 3D comparisons are gated on that product.

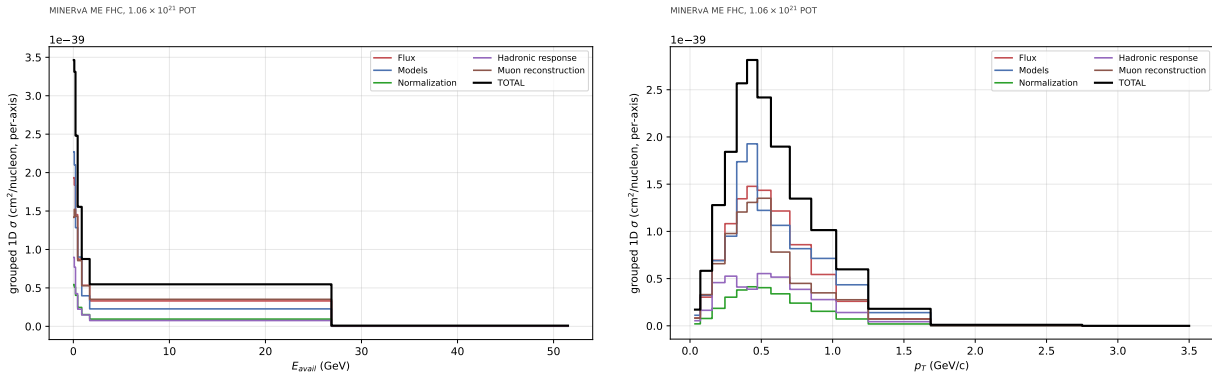


Figure 25: Grouped fractional systematic bands of the historical 3D covariance projected onto E_{avail} (left) and p_T (right): flux-dominated, with the same band ordering as the 2D budget. The magnitudes are quarantined pending the final selection-complete 5D-to-3D projection.

7.7 The low-available-energy excess recovers the known low-recoil 2p2h deficit

The low- E_{avail} excess is the most physically interesting feature the third axis exposes, and the 3D framework lets us identify its origin directly by generating the relevant physics ourselves rather than

only reweighting. It sits in a well-established MINERvA low-recoil program: the enhancement of the event rate between the quasielastic and Δ peaks, with multi-nucleon final states, was first isolated in the low-three-momentum-transfer subsample [6] and is the effect the empirical MINERvA tune was built to absorb — our measurement re-exposes it on the muon-kinematics $\times E_{\text{avail}}$ axes.

Final-state interactions do not explain it. A `grwght1p` reweight pass on the 2M central-value GENIE events varying the pion-FSI dials shows both are sub-percent on $d\sigma/dE_{\text{avail}}$: `FrInel_pi` shifts the spectrum by $\leq 0.74\%$ (peak in 0.10–0.20 GeV) and `FrAbs_pi` by $\leq 0.82\%$ (peak in the lowest bin), with the integrated rate moving only ± 0.02 – 0.03% at $\pm 1\sigma$. A less-than-1 % knob cannot bridge a 7–15 % excess, so the excess points to the initial-state/nuclear model, not rescattering.

The shape and size match 2p2h. A subtlety drives the diagnosis: our base GENIE central value was generated with the default event-generator list, which contains *no* meson-exchange-current (2p2h/MEC) events (`mec = 0` for all 1.48×10^6 CC events; the modes are QE 11 %, RES 22 %, DIS 66 %, COH 0.9 %). Any 2p2h is therefore *added*, not reweighted. Decomposing the central-value $d\sigma/dE_{\text{avail}}$ by interaction mode (per-bin mode count-fractions $\times$ the bin value) and overlaying the data localizes the largest positive gaps to the quasielastic– Δ dip, [0.10, 0.20) and [0.20, 0.40), while the QE-dominated [0, 0.10) and [0.40, 1.50) regions are much closer at central-value level. 57 % of the integrated deficit sits at $E_{\text{avail}} \leq 0.4$ GeV, exactly where 2p2h lives, and closing it requires a 2p2h component $\approx 43\%$ of the QE rate (62 % locally) — the standard empirical/Valencia-MEC magnitude, in contrast to the sub-percent FSI dials (Fig. 26).

Enabling Valencia 2p2h confirms it. We regenerated the GENIE central value with the empirical 2p2h turned on (`-event-generator-list Default+CCMEC`, 2M events). Because the Nieves–Simo–Vacas MEC channel, though present in the splines, is absent from the `gsp12root` total-CC graph, the sample is normalized by the non-MEC CC count (anchoring QE+RES+DIS+COH to the known total and letting MEC add on top by $1/(1 - f_{\text{MEC}})$, with $f_{\text{MEC}} = 2.87\%$). The added 2p2h lands squarely in the dip and moves GENIE toward the data throughout the three low- E_{avail} bins (Fig. 27). It fills 46 % of the data–CV gap in the dip and 27 % of the integrated deficit (CV $-7.2\% \rightarrow -5.2\%$). Historical 3D χ^2 values are not quoted pending the corrected covariance projection. The prediction remains below Tune v1 at central-value level because Tune v1 layers MINERvA’s empirical low-recoil 2p2h *enhancement* and RPA on top of the stock Valencia MEC — i.e. stock Valencia 2p2h is real but under-strength for MINERvA low recoil, which is precisely the known MINERvA-tune story and is consistent with NuWro and GiBUU (native, un-enhanced 2p2h) sitting low as well. The one feature 2p2h does not touch is a separate central-value excess in the high- E_{avail} [1.50, 3.00) bin, a DIS-tail modeling issue resolved on the (E_{avail}, W) plane in §8.

This low-available-energy deficit is the same feature MINERvA’s dedicated low-recoil inclusive measurement reports: Ascencio *et al.* [11], which measures $d^2\sigma/(dq_3 dE_{\text{avail}})$ for $q_3 < 1.2$ GeV at the same $\langle E_\nu \rangle \sim 6$ GeV, finds GENIE and NuWro underpredicting the data in the low- E_{avail} region absent an enhanced 2p2h component. Our excess appears in the muon-kinematics extension (integrated over all q_3) rather than at fixed q_3 , so the comparison is of the E_{avail} *shape* and direction rather than absolute normalization; the consistency confirms that the OmniFold 3D extension recovers the established MINERvA low-recoil/2p2h feature as a by-product of the muon-kinematics unfolding. A quantitative shape comparison awaits the public [11] data release.

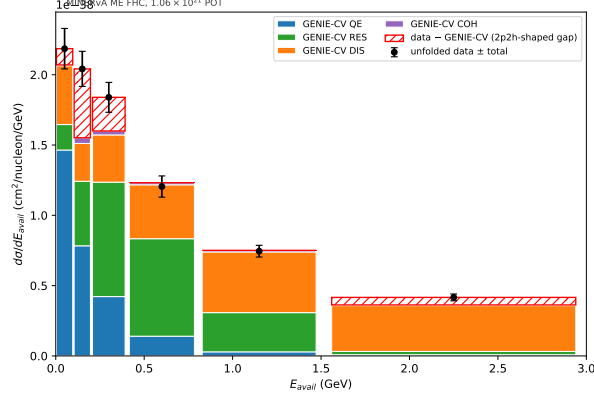


Figure 26: Central-value $d\sigma/dE_{\text{avail}}$ decomposed by interaction mode and overlaid with the unfolded data. The data excess is localized to the quasielastic- Δ dip ($E_{\text{avail}} \leq 0.4$ GeV), the shape and size of a missing 2p2h component.

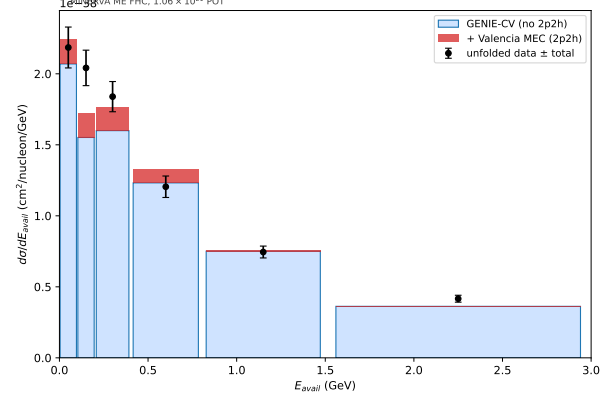


Figure 27: Re-generating GENIE with the empirical Valencia 2p2h enabled (red, stacked on the no-2p2h CV) fills part of the low- E_{avail} dip — about half the data-CV gap, the remainder being the MINERvA-tune enhancement.

7.8 Fourth axis: three-momentum transfer q_3

The same one-feature-per-observable property lets us add a fourth axis without constructing an explicit response tensor, although training, support and uncertainty propagation become more demanding. We choose the three-momentum transfer $q_3 = |\vec{q}|$ because (E_{avail}, q_3) is the plane in which the quasielastic, 2p2h and resonance processes separate, and because it makes the comparison to MINERvA's dedicated low-recoil measurement (Ascencio *et al.* [11], $d^2\sigma/dq_3 dE_{\text{avail}}$) bin-compatible. Truth q_3 is the generator momentum transfer (`CVUniverse::Getq3True()`); reco q_3 is the calorimetric estimator built from the muon Q^2 and the recoil system (`CVUniverse::RecoQ3()`). The event loop evaluates both accessors over all twelve playlists, exactly as for E_{avail} . Unlike E_{avail} , q_3 is *not* invariant under the lateral systematic shifts, so the shifted- q_3 universes are dumped per universe rather than reused from the central value. The driver feature stack becomes $(p_T, p_{\parallel}, E_{\text{avail}}, q_3)$; the q_3 binning $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV is deliberately coarse (seven bins) and ends in a catch bin, for $14 \times 16 \times 7 \times 7 = 10\,976$ four-dimensional bins (4830 populated).

Marginal closure. Marginalizing the 4D result over q_3 reproduces the 3D E_{avail} spectrum and its low-recoil excess (Fig. 29, right): the E_{avail} -marginal data/GENIE-CV ratio is 1.21 in the lowest bin, dips to 1.01 across the quasielastic- Δ region, and rises again to 1.16–1.23 in the high- E_{avail} DIS tail — the same shape established in 3D (§7.7), now recovered as a by-product of the higher-dimensional unfold (integrated data/GENIE-CV = 1.13). This is the 4D→3D analog of the 3D→2D Jacobian anchor of §7.3.

What q_3 adds. The new content is the resolved (E_{avail}, q_3) plane (Fig. 29, left). Integrated over E_{avail} the q_3 -marginal discrepancy is mild (1.03–1.19, center panel), but the joint map is structured: at fixed E_{avail} in the quasielastic- Δ dip the data/GENIE-CV ratio *rises* with q_3 , the central value over-predicting the low- q_3 , moderate- E_{avail} cells (ratio down to 0.66) while the data excess concentrates at low E_{avail} (all q_3) and toward high q_3 . This is precisely the correlation not resolved by the muon-only or fixed- q_3 projections considered here, and it is the natural input for the bin-compatible Ascencio comparison. We show it here at central-value level (the base GENIE

event-generator list contains no 2p2h, as in §7.7); no per-cell significances are reported without a corrected 4D covariance.

Bin-identical comparison to the published low-recoil measurement. The Ref. [11] supplemental data (44-cell $d^2\sigma/(dE_{\text{avail}} dq_3)$ with its full covariance, distributed inside the public arXiv source package) enables a direct quantitative cross-check. Merging both measurements onto their maximal common grid — edges present in *both* binnings, with our $(p_T, p_{\parallel})$ marginalization restricted to $p_{\parallel} < 20$ GeV/c to mirror the Ascencio muon gate ($\theta_{\mu} < 20^\circ$, $1.5 < p_{\mu} < 20$ GeV) — gives two super-cells, $E_{\text{avail}} < 0.4$ GeV in $q_3 \in [0.4, 0.6)$ and $[0.6, 1.2)$ GeV: the low-recoil region where the 2p2h physics lives. Our central result sits 9.2 % and 6.3 % above Ascencio. No pulls or full-covariance χ^2 are reported without a corrected 4D covariance (Fig. 28). Caveats: the two share MINERvA flux/detector systematics, treated here as independent; the p_{μ} -vs- $p_{\parallel}$ gate differs by $\lesssim 6\%$ kinematic smearing at the 20 GeV boundary. A finer 44-cell comparison would require re-unfolding onto the Ascencio edges (trivial for the unbinned method) *and* a systematic sweep on that binning; given the agreement above it is left to future work.

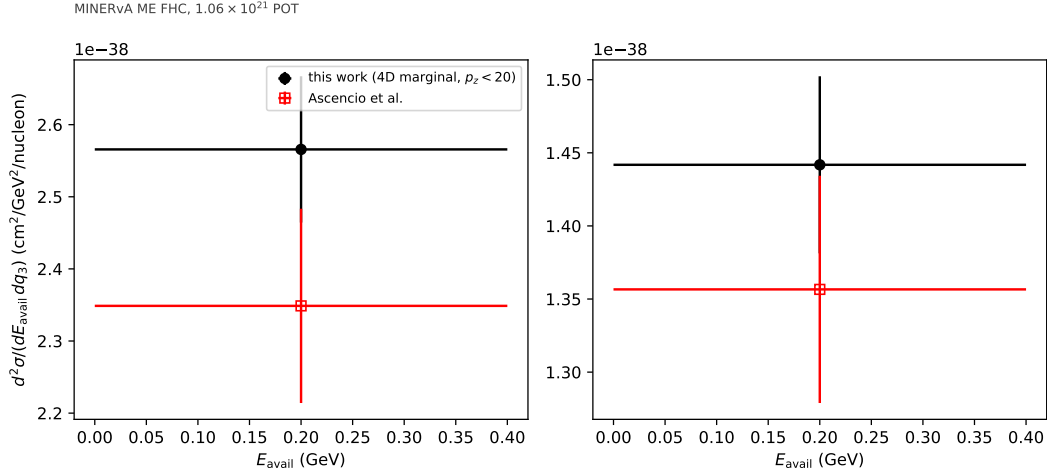


Figure 28: Bin-identical comparison to Ascencio *et al.* [11] on the maximal common (E_{avail}, q_3) grid. Only the central-value comparison is used; the uncertainty-derived annotations in the panel do not use the corrected 4D covariance.

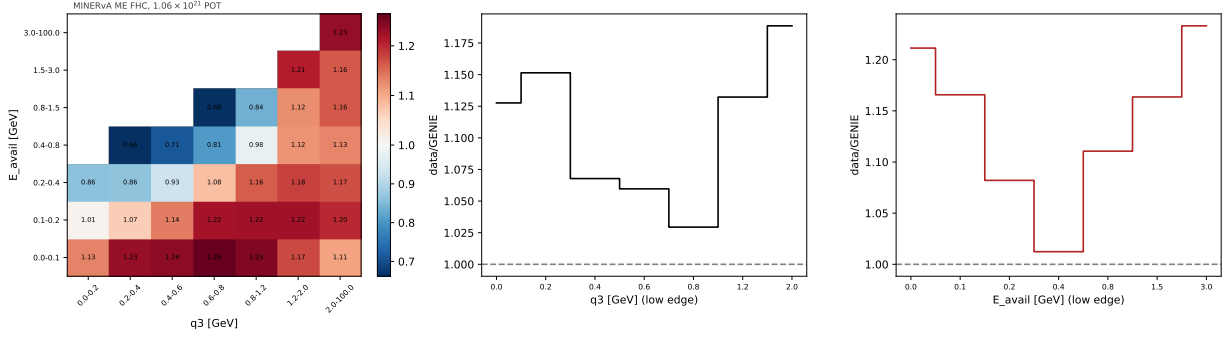


Figure 29: Data/GENIE-CV ratio of the 4D unfold projected onto the (E_{avail}, q_3) plane (left; blank cells are kinematically empty), with the q_3 (center) and E_{avail} (right) marginals. The E_{avail} marginal reproduces the 3D low-recoil excess; the joint plane shows the data – MC discrepancy to carry genuine q_3 structure (the ratio rising with q_3 across the quasielastic- Δ dip). GENIE central value, no 2p2h in the base event-generator list.

Systematic covariance. A corrected 4D covariance would use actual minus/plus endpoints, MAT mean-centered $1/N$ blocks, one fixed estimator seed, mean-centered joint throws with a separately reported mean shift, and no scalar jitter subtraction. Because dump-all lateral branches are CV-support-limited, selection-complete active-universe event loops are required to bound migration effects. Accordingly, no 4D uncertainty magnitude or unified/block ratio is reported.

8 Fifth axis: the (E_{avail}, W) deep-inelastic-tail study

Enabling Valencia 2p2h resolves the low-recoil excess (§7.7) but leaves a separate high- E_{avail} data excess where 2p2h does not contribute and the sub-percent pion-FSI dials cannot reach. The comparison is made at central-value level because no corrected covariance has been adopted for the (E_{avail}, W) projection. GiBUU spreads its deficit across the entire tail (data/generator 1.59 at $E_{\text{avail}} \in [0.8, 1.5)$, 1.36 at $[1.5, 3.0)$) rather than piling it in the deep catch bin like the GENIE variants — an independent, qualitatively distinct confirmation. To test the deep-inelastic hypothesis directly, we added the hadronic invariant mass W as a fifth OmniFold axis.

8.1 W as a fifth axis

Truth W is the experimenter’s invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$ with M the struck-nucleon mass (`CVUniverse::GetTrueExperimentersW()`); reco W uses the same calorimetric energy transfer q_0 and muon-kinematic Q^2 as the reco q_3 estimator, with the average nucleon mass (`CVUniverse::RecoW()`). Like q_3 — and unlike E_{avail} — W is not invariant under the lateral systematic shifts, so the event loop dumps shifted- W lateral universes per universe, mirroring the q_3 treatment (§7.8); the dump also carries truth hadronic diagnostics (proton and pion multiplicities, leading-hadron angle). The W binning $[0, 1.1, 1.4, 1.8, 2.2, 3.0, 100]$ GeV brackets the quasielastic peak, the Δ , the transition region, and the DIS tail. The five-axis unfold $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$ passes the same marginalization anchors as its predecessors: the W -marginal recovers the frozen 4D cross section to 0.11 % in normalization (5D/4D total = 1.0011), with per-axis median differences of 0.3 % to 1.5 %, and the $d\sigma/dW$ spectrum is finite and non-negative across all six bins.

8.2 Where the excess lives in the (E_{avail}, W) plane

Projecting the unfolded five-axis result onto (E_{avail}, W) and subtracting the GENIE central value localizes the excess (Fig. 30): it is *predominantly deep-inelastic*. The two highest- E_{avail} bands carry 57 % of the positive excess; of all positive excess, high- E_{avail} (≥ 0.8 GeV) holds 67 %, and 83 % of *that* sits at $W \geq 1.8$ GeV, with the single largest cell the deep-DIS corner ($E_{\text{avail}} > 3$ GeV, $W > 3$ GeV) at 22 %. A secondary low- W (below 1.1 GeV, QE-like) excess and a small Δ -region deficit are also present.

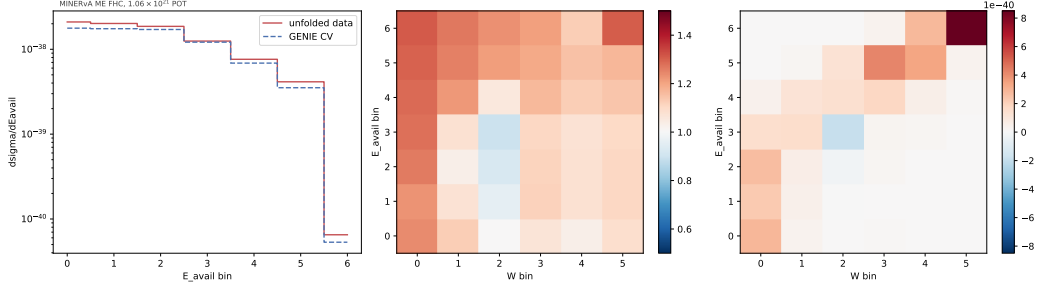


Figure 30: Unfolded-data minus GENIE-CV excess across the (E_{avail}, W) plane. The positive excess concentrates at high E_{avail} and high W — the deep-inelastic region — with a secondary low- W (QE-like) component and a small Δ -region deficit.

8.3 Generator band: the corner is generator- and tune-robust

We confirm this against independent spline-normalized (E_{avail}, W) predictions from GENIE CV, GENIE+Valencia 2p2h/MEC, NuWro, and GiBUU (Fig. 31). All four underpredict the high- $E_{\text{avail}} \times$ high- W corner by 54 % to 58 % (data/generator = 1.54, 1.58, 1.56). Crucially, enabling Valencia 2p2h does *not* close the corner — it slightly worsens it ($1.54 \rightarrow 1.58$), since 2p2h populates low W — and the independent NuWro and GiBUU generators miss it by the same or a larger margin (GiBUU is the most deficient overall, integrated $\sigma = 2.22 \times 10^{-38}$ cm²/nucleon against the data’s 3.07×10^{-38}); at $W \in [2.2, 3.0]$ GeV all four sit 23 % to 25 % below the data.

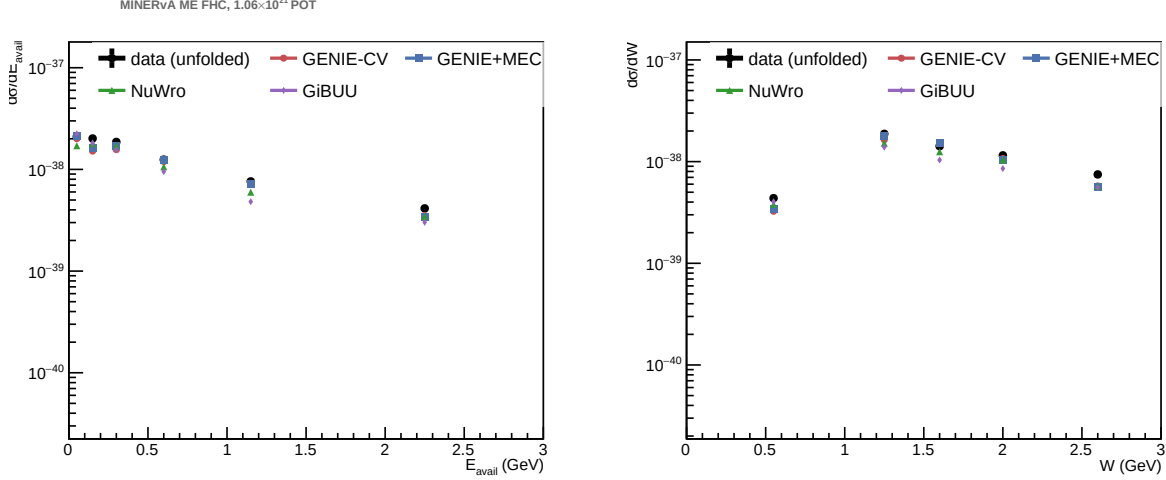


Figure 31: Unfolded (E_{avail}, W) cross section versus the four-generator band (GENIE CV, GENIE+MEC, NuWro, GiBUU). Every generator underpredicts the high- W , high- E_{avail} DIS corner; adding 2p2h does not help there because 2p2h populates low W .

8.4 The (E_{avail}, W) covariance replacement gate

To turn the band into a number we build a dedicated 42-bin (E_{avail}, W) covariance, $C = C_{\text{syst}} + C_{\text{stat}} + C_{\text{lateral}}$, by a *frozen-reweighter* block sum on the merged five-axis `_universes_full` sample: the central-value OmniFold push-weights are held fixed and only the per-event truth reweight ratios change per systematic universe (the same frozen-reweighter *technique* as the PET budget, §9.2 — not the PET covariance matrix, which is never applied to this GBDT central value; no per-universe re-inference). The covariance carries 13 knob bands $\pm 1\sigma$, the 100-universe flux multisim, the projected full five-axis bootstrap covariance $C_{EW} = MC_{5D}M^T$, and a W -resolved detector (lateral) block. A valid replacement must project the corrected five-axis statistical covariance as $MC_{5D}M^T$ and use the corrected systematic endpoint convention. The detector block uses five shifted-kinematic bands (`Muon_Energy_MINERvA/MINOS`, `MuonResolution`, and `BeamAngleX/Y`) plus weight-only `MinosEfficiency` and `GEANT` bands. Selection-complete active-universe event loops are also required to bound migrations omitted by the CV-support-limited dump-all sample.

All four generators underpredict the high- W , high- E_{avail} region at central-value level, and adding 2p2h does not close it; statistical compatibility is not evaluated without the corrected projected covariance. The complementary cross-check against MINERvA’s published low-recoil measurement is presented in the (E_{avail}, q_3) plane (§7.8): on the maximal common grid the central values agree, while no covariance-based consistency metric is reported.

9 Recoil-point-cloud representation cross-check

As a low-level-representation cross-check, a point-cloud transformer (PET, the architecture of OmniLearn [40], vendored with the upstream OmniFold implementation) unfolds the same ME FHC data using, on the reconstructed side, the actual calorimetric recoil clusters — per-cluster energy, one view-projected transverse coordinate and z (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack` via `CVUniverse::GetRecoClusters()`). At truth level it receives the final-state hadron cloud with the muon removed. Muon-derived event scalars are retained for selection and binning the extracted result, but they are not classifier features at either level. The learned

weight is therefore a function of the recoil representation, not of the complete joint muon–hadron event. We therefore compare its absolutely normalized central value, closure and uncertainty only as properties of the recoil-only estimator. A full-event estimator would instead train jointly on the declared muon and recoil inputs.

9.1 From shape check to absolute cross section

The PET network trains on a GPU on a 2M-event subsample (2 OmniFold iterations); the learned step-2 truth reweight is then applied to the *full* 32.8M-event generator sample (`MultiFold.reweight`), binned on the 4D grid, and pushed through the standard extraction (`extract_cross_section_nd` with the GBDT run’s completeness — completeness is reweight-independent — and the same flux/POT/nucleon constants). Three gates validate the absolute path:

1. **Full-statistics reweight sanity:** the pushed weights over the 32.8M generator events are finite with mean 1.028;
2. **Ordinary closure:** unfolding MC-reco-as-pseudo-data recovers MC truth to total 0.9884 with uniform per-axis median residuals of $\sim 1.1\%$ — the absolute machinery is internally consistent. Because pseudo-data and prior come from the same simulation, this test cannot expose omitted muon dependence at fixed recoil;
3. **Cross-check vs GBDT:** on data, the PET absolute total is $2.796 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ versus the GBDT 3.066×10^{-38} , ratio 0.912, with per-axis median differences of 6.5% to 9.9% (Fig. 32).

The $\sim 9\%$ data-side gap is not a normalization bug, but ordinary closure alone cannot assign it uniquely to training configuration. The PET run used a 2M-event, two-iteration training rather than the full-statistics five-iteration GBDT, and a higher-epoch retrain moved the total by -1.6% (ratio 0.897), demonstrating training sensitivity. The recoil-only feature contract is an additional structural difference: at fixed recoil, the conditional muon distribution remains inherited from the prior.

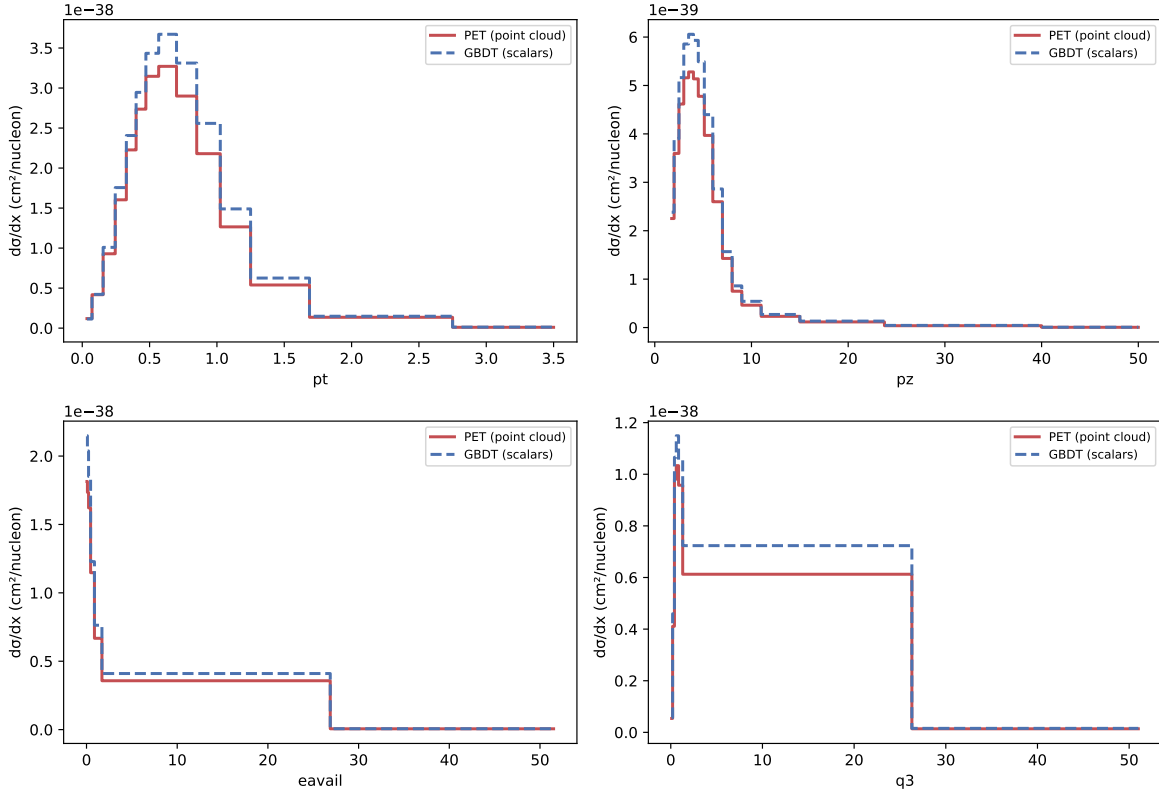


Figure 32: Recoil-only PET representation cross-check: absolutely-normalized 4D cross section from the point-cloud unfold (full-statistics reweight of the generator sample) versus the production scalar GBDT result, projected on each axis. The PET/GBDT total ratio is 0.912; ordinary closure validates the extraction machinery to $\sim 1\%$ but does not test omitted muon dependence.

9.2 Uncertainty of the recoil-only estimator

The PET statistical replicas independently Poisson-fluctuate data and MC, retrain PET at a fixed estimator seed, and apply the same coherent MC draw in the final truth binning and completeness.

The calculation uses the same background-subtracted five-dimensional measured target as the scalar GBDT. Event-by-event data alignment is exact, the MC arrays are byte-identical to the full-cloud input. The nominal PET extraction, an identical-seed GPU-floor repeat, a 20-member coherent-Poisson retraining ensemble, and a 12-member crossed subsample/estimator-seed ensemble have all passed their ordering, event-alignment, and finite-output gates. The GPU nondeterminism is negligible relative to both sampling and training variation. The C_{stat} estimate is based on 20 coherent replicas, so its finite-ensemble precision is more limited than the 100-replica target. These statements characterize the recoil-only estimator and do not supply an uncertainty for a joint muon–recoil estimator.

The historical recoil-PET campaign produced vertical, retraining, ML, statistical and detector blocks on a common nominal. That assembly is now quarantined rather than quoted as an uncertainty. For a nuisance endpoint u , the physical shift and the induced retraining response form one joint shift,

$$\delta_u = x_u^{\text{varied+retrained}} - x_{CV}. \quad (1)$$

Writing this as a frozen-map piece plus a retraining increment is an algebraic decomposition, but adding the two separate covariances omits $\text{Cov}(\delta_{\text{frozen}}, \delta_{\text{retrain}})$ and its transpose. Positive semidefiniteness of that sum does not restore the missing cross term.

The historical detector block is additionally CV-support-limited: it propagates shifted weights and observables through the nominal cloud membership instead of regenerating the selected cloud for every lateral endpoint. The replacement therefore uses end-to-end joint nuisance/retraining shifts and the selection-complete FPS lateral samples. It belongs to the full-event PET estimator and receives a fresh statistical and ML ensemble. Until that product lands, this section quotes only the recoil-PET central value and closure; no PET precision comparison is made.

9.3 Truth-cloud projection within the recoil cross-check

The truth side of the recoil-only PET unfold is *already* a point cloud: the generator-level final-state hadron set (`part_gen`, the top-12-by-energy truth hadrons with their PDG codes, built by `CVUniverse::GetTruthFSHadrons`) is carried per event alongside the learned step-2 push weight. Any truth observable that is a function of those 4-vectors can therefore be obtained after the unfold by re-binning the push-weighted truth events — no re-training and no re-unfolding. We demonstrate this *reweight-then-project* capability on three observables computed post-hoc from the stored cloud: available energy E_{avail} , the hadronic-system invariant mass $W_{\text{had}} = \sqrt{(\sum E)^2 - |\sum \vec{p}|^2}$, and the proton/charged-pion multiplicities. The exercise also quantifies the two limitations of the stored cloud.

Truncation is essentially lossless. Because E_{avail} recomputed from the cloud uses the same formula and inputs as the value stored at dump time, the cloud-vs-stored residual isolates the cost of keeping only the 12 highest-energy hadrons. Over the full 32.8 million truth events (every event now carries a cloud, below) the residual has median 0 MeV and mean -6.4 MeV, with 98.78 % of events within 10 MeV (Fig. 33). The entire deficit lives in the 2.31 % of events that saturate the 12-particle cap (median deficit -35 MeV there, exactly zero below it), and because the ranking is by energy the dropped constituents are the lowest-energy stored final-state hadrons — a small, one-sided, bounded effect. The 12-particle cap therefore does not limit E_{avail} -scale observables and does not need to be raised.

Coverage: closing the acceptance gap. Originally the cloud was filled only in the reco-signal event loop, so the truth-only “miss” events appended for the unfold (37.9 % of the truth sample) carried no cloud — only 72.6 % of truth events had one, and cloud projections were correspondingly acceptance-limited, landing exactly on the stored- E_{avail} projection restricted to the has-cloud subset but sitting $1.5\text{--}2\times$ below the full unfold per bin. This is a truth-side instance of the same dangling-miss-row dump issue tracked for the systematic bank. We fixed it at source by carrying the truth final-state hadrons through the truth-denominator loop and filling `part_gen` for the appended miss rows (the reco cloud `part_reco` correctly stays empty for a miss), then re-dumping, rebuilding the inputs, and re-training. Coverage is now complete: 99.999 % of truth events carry a cloud (only 174 residual empties out of 32.8 million), and the cloud-recomputed $d\sigma/dE_{\text{avail}}$ now reproduces the full unfold to $\lesssim 0.5$ % in every E_{avail} bin (Fig. 34, center, where the cloud, stored, and full-unfold curves coincide). The projection is no longer acceptance-limited.

Exploratory projection yields new hadronic diagnostics. W_{had} is not a lossy copy of the leptonic “experimenter’s” W stored at dump time — the latter is reconstructed from muon kinematics assuming a single struck nucleon at rest, while W_{had} is the true invariant mass of the hadron system.

The two differ by a median 0.96 GeV, and the 2D correlation (Fig. 34, left) makes the physics explicit: a sharp locus at $W_{\text{had}} \approx 0.94$ GeV (single-baryon final states, whose hadronic mass is just the nucleon rest mass irrespective of leptonic W) and diagonal bands stepped above it by $\sim$ one nucleon mass per extra final-state baryon (multi-nucleon knockout, 2p2h), Fermi-smeared. Projecting the cloud thus provides a physically interpretable hadron-cloud diagnostic from the existing recoil-only unfold; it is not promoted here as a separate full-event measurement.

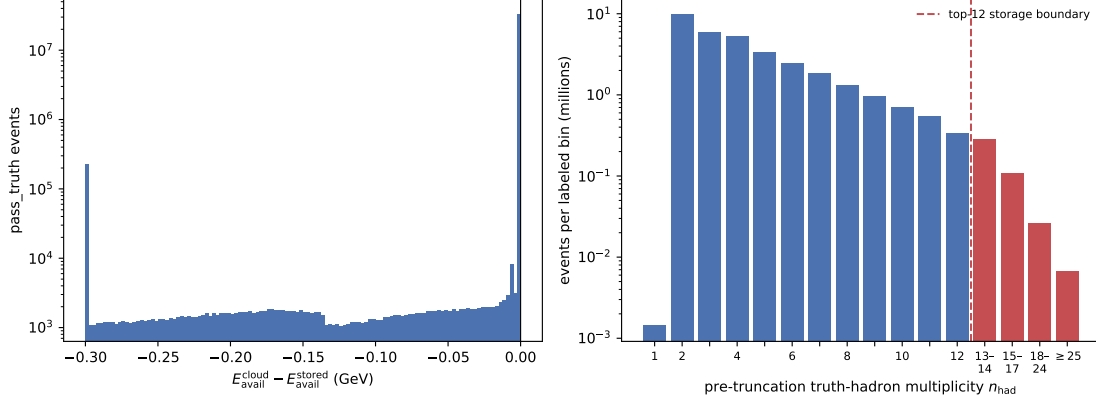


Figure 33: Recoil-cloud truncation validation: cloud-recomputed E_{avail} minus the value stored at dump time, over the full 32.8 million truth events. The residual is zero to 98.78 % within 10 MeV; the small negative tail is confined to the 2.31 % of events that reach the 12-hadron cap. The right panel shows ordinary event counts versus the pre-truncation multiplicity from the variable-length ROOT branch. Multiplicities 1–12 are shown exactly; the sparse tail is grouped into the labeled 13–14, 15–17, 18–24, and ≥ 25 categories. The dashed boundary separates fully stored events from categories for which only the 12 highest-energy hadrons are retained. The cap-reaching events have a median E_{avail} deficit of only -35 MeV. Keeping the 12 highest-energy hadrons is therefore essentially lossless for E_{avail} -scale observables.

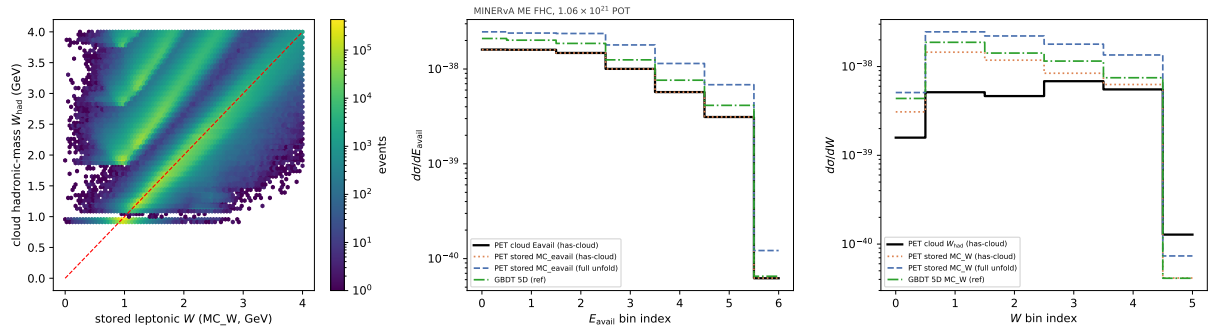


Figure 34: Exploratory recoil-only reweight-then-project demonstration. *Left*: hadronic-system invariant mass W_{had} versus the leptonic stored W — they differ by definition (median 0.96 GeV), with the QE locus at 0.94 GeV and multi-baryon bands above $y = x$. *Center*: projected $d\sigma/dE_{\text{avail}}$ from the push-weighted cloud now reproduces the full unfold to $\lesssim 0.5\%$ per bin after the miss-row coverage fix — the cloud, stored- E_{avail} , and full-unfold curves coincide; the GBDT 5D reference is shown for comparison. *Right*: projected $d\sigma/dW_{\text{had}}$, whose shape differs from the leptonic- W reference as expected.

The projection and extrapolation limits. Reweighting can only redistribute weight among truth configurations the generator already produced; it cannot populate phase space the prior never sampled — the same extrapolation limit flagged for the extended-fiducial study (§10). Within that limit, hadron-cloud diagnostics can be projected over the stored truth spectrum. They do not turn the recoil-derived weight into a full-event weight or remove the prior conditional for omitted muon information. Two extensions lie beyond re-projection from the stored cloud (they would require regenerating the event dump): adding the muon through separately declared reco/data and truth event-feature interfaces (it is excluded from the recoil cloud but retained as a stored scalar); and raising the 12-particle cap for the rare very-high-multiplicity tail (unnecessary at E_{avail} scale per the validation above). Full details and reproduction are in `nd-unfolding/pet/POINTCLOUD_-PROJECTION.md`.

9.4 Scope of the recoil-PET cross-check

The cross-check establishes that the point-cloud infrastructure works on real MINERvA recoil data: paired reco/truth clouds, GPU training, full-sample reweighting, absolute extraction, coherent estimator bootstraps and targeted response studies all operate end to end. It also provides a useful comparison against the scalar result and supports exploratory post-hoc hadron-cloud projections. It does *not* establish simultaneous muon-plus-hadron unfolding, because the muon was not a classifier input.

A full-event extension requires separate observable `event_reco/event_data` and truth-level `event_truth` interfaces containing a distinguished muon, explicit masks/types and validated neighborhood geometry. An omitted-variable stress closure that changes muon kinematics within narrow recoil strata would test whether the joint estimator recovers information that the recoil-only estimator necessarily misses. Any such estimator requires its own nominal fit and uncertainty budget.

10 Extended fiducial phase-space study

This section reports a preliminary scalar central value, a three-prior dependence envelope, and an extension region tested by a hidden-variable closure and 200 coverage toys. No corrected covariance is available for the extended fiducial domain, so the result is limited to central values and these validation tests. The signal remains ν_μ CC in the tracker interaction fiducial volume: truth-muon kinematic cuts are removed, but the reconstructed MINOS-based selection is unchanged. This is therefore an extended *fiducial* measurement domain, not unrestricted neutrino phase space.

The published phase space ($\theta_\mu < 20^\circ$, $1.5 < p_\parallel < 60$ GeV/c, $p_T < 4.5$ GeV/c) hides a third of the signal: weighting the true ν_μ CC rate in the tracker fiducial volume, 66.4 % lies inside the cuts, 22.3 % at $p_\parallel < 1.5$ GeV/c, 11.3 % at $\theta_\mu > 20^\circ$, and the high- $p_T/p_\parallel$ tails are negligible (Fig. 35). Because OmniFold’s miss treatment performs acceptance correction natively, removing the truth-level kinematic cuts is a configurational change (an event-loop switch enlarging the truth denominator from 32.8M to 49M events; the reconstruction-level selection, including the MINOS match, is untouched), and the unfold extrapolates into the unmeasured regions through the step-1 miss regressor. The question is not *whether* it can extrapolate but *at what prior cost*; this section quantifies that.

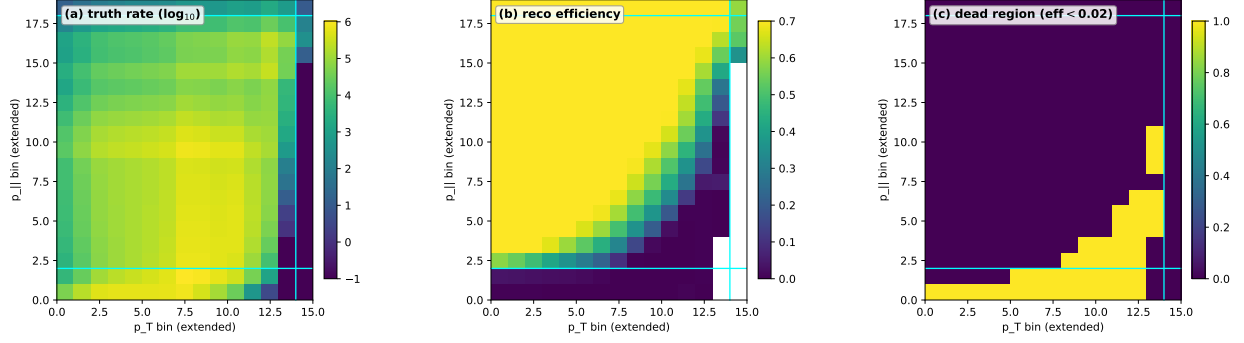


Figure 35: Extended-fiducial acceptance study (all twelve playlists): (a) weighted truth rate ($\log_{10}$), (b) reconstruction efficiency, and (c) the dead region with $\epsilon < 2\%$, on the extended $(p_T, p_{||})$ grid (bin number; the catch bins are too wide for a linear axis). Cyan lines mark the published phase-space boundary. Efficiency is ~ 0.65 and flat through the published region, degrades smoothly across the $\theta > 20^\circ$ wedge, and dies in the $p_{||} < 1.5 \text{ GeV}/c$ strip and the high- p_T corner; the dead region carries 27.7% of the total truth rate.

Validation battery. Three results establish the extension (Fig. 36):

1. **Anchor.** Restricted to the published-phase-space sub-block, the extended-fiducial unfold reproduces a control unfold (standard omnifile, paper grid, identical settings) to 0.9994 in integral, with a median per-cell ratio of 1.0013 and median deviation 0.57% over all 185 reported cells — opening the phase space does not damage the measurement where it was already made. The control itself reproduces the frozen production total ($3.073 \times 10^{-38} \text{ cm}^2/\text{nucleon}$) exactly.
2. **Closure.** A self-closure on the extended grid (simulated reconstruction as pseudo-data) returns recovered/truth = 1.0000 in every cell including the extension — no bookkeeping or normalization bias in the enlarged-denominator configuration. A *hidden-variable* closure goes further: injecting a Gaussian truth bump in true E_{avail} (an axis the $(p_T, p_{||})$ unfold is blind to) and unfolding, the recovered $(p_T, p_{||})$ cross section reproduces the bump-reweighted truth to median 0.17% (90th percentile 0.65%) in the published cells and median 0.77% (max 4.05%) in the extension — both well inside the tier-2 prior band.
3. **Three-prior envelope.** Repeating the unfold with three priors — MnvTune v1, bare GENIE, and a NuWro-shaped prior (built by reweighting the truth $(p_T, p_{||})$ shape to two million raw NuWro events, which carry no phase-space cut) — spans the total cross section by only $\pm 1.5\%$. Per cell, the half-spread/mean is median 2.9% inside the published region but median 7.9% (90th percentile 62%) in the extension, concentrated in the zero-acceptance cells (Fig. 37).
4. **Containment diagnostic.** With the estimator seed fixed, one 100/100 calibration–evaluation split, 68.67% of evaluation bin–toy cells contain the fixed truth using widths learned from the disjoint calibration half, compared with the 68.27% Gaussian nominal. Correlated bins and estimated calibration widths preclude a global binomial error or calibrated test; this probes the statistical closure-toy stream, not the full adopted band. Separately, the independent C_{stat} band contains 77.6% of closure bin–toy cells and is wider than the closure-toy spread. The same-ensemble 68.9% ($\langle |r| \rangle \approx 0.79$) is retained as a pull/Gaussianity diagnostic. The published and extension regions have descriptive split-sample fractions of 68.5% and 70.0%, respectively; neither is used as a covariance-certification test.

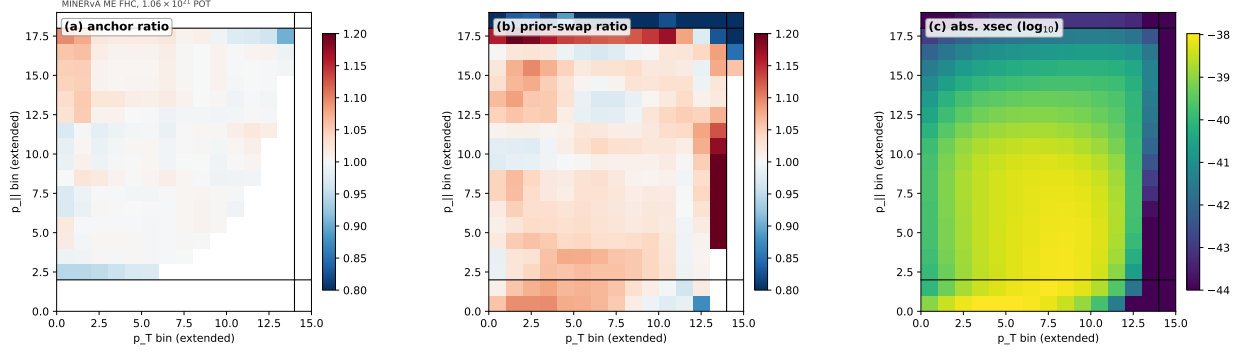


Figure 36: Extended-fiducial validation battery (all playlists): (a) anchor ratio in the published sub-block (median 1.0013), (b) the Mnvtune/bare-GENIE prior-swap ratio, and (c) the extended-fiducial cross section ($\log_{10}$). Panels (a) and (b) are *ratios* (diverging scale centered at unity); (c) is an *absolute* magnitude (sequential scale) — hence the two color schemes. Black lines mark the published boundary; axes are bin number on the extended grid.

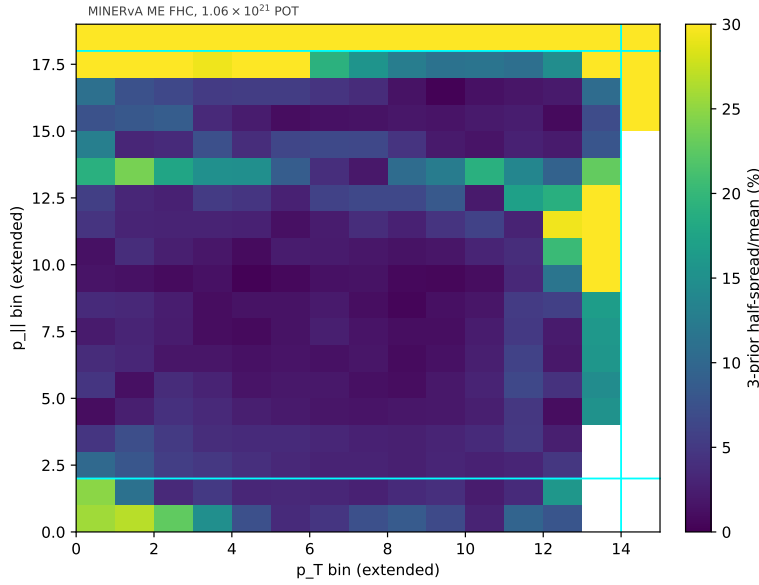


Figure 37: Three-prior dependence envelope (half-spread/mean per cell, in percent). Prior dependence is at the few-percent level wherever the detector has acceptance and reaches tens of percent only in the dead cells — the quantitative basis for the two-tier reporting.

Result and reporting convention. The preliminary extended-fiducial CC-inclusive central value on the tracker fiducial volume is

$$\sigma_{\text{ext}} = 4.502 \times 10^{-38} \text{ cm}^2/\text{nucleon} \quad (\text{central value}),$$

46 % above the restricted-phase-space $3.073 \times 10^{-38} \text{ cm}^2/\text{nucleon}$. We adopt **two-tier reporting**: *tier 1* (acceptance-supported reporting) covers cells with $\epsilon \gtrsim 2\%$ — the published region plus the measurable extension ($\sim 6\%$ of the rate, chiefly the $\theta > 20^\circ$ wedge and the $p_{\parallel} \in [0.75, 1.5] \text{ GeV}/c$ edge); *tier 2* (prior-extrapolated) covers the dead cells ($\sim 28\%$ of the rate), separately flagged and

carrying the three-prior envelope as an explicit additional band. The summed central value includes both tiers; only tier 1 is interpreted as an acceptance-supported measurement. No uncertainty magnitude is assigned beyond the three-prior envelope because the extended-domain covariance has not been constructed under the contract of §4.4 with selection-complete lateral support.

10.1 Recoil-PET feasibility study on the extended population

The PET feasibility study on the enlarged event population and reporting grid remained recoil-only: step 1 saw non-muon reconstructed clusters and step 2 saw truth hadrons with the muon removed. The muon coordinates used to define and bin the extension were not classifier inputs. Those products are therefore feasibility diagnostics for the recoil estimator; their populated-bin counts, prior-transfer envelope and retraining floor are not a final full-event measurement and are not used to argue that low-efficiency cells are data-constrained.

A full-event estimator must satisfy two independent gates. It must first carry a distinguished muon and validated event context through separate observable reco/data and truth interfaces, and pass an omitted-variable stress closure that shifts muon kinematics at fixed recoil. It must also consume the canonical extended-fiducial event loops and 15×19 muon grid while preserving the enlarged truth denominator, native misses and unchanged reconstructed selection.

Its validation would need to repeat the ordinary and stress closures, acceptance census, anchor, coverage and prior envelope before constructing statistical, ML, vertical, retraining-response and selection-complete-lateral uncertainties. Additional features can use more observed information, but cannot create detector information in zero-efficiency cells.

11 Summary and outlook

Summary. Unbinned OmniFold with gradient-boosted trees reproduces the published MINERvA ME FHC CC-inclusive $d^2\sigma/(dp_T dp_{\parallel})$ (σ_{tot} ratio 1.011; 94.1 % of bins within 10 %; published- σ standardized-difference RMS 0.60) with a standalone uncertainty budget that matches the published total (6.87 % vs 6.86 % median per bin) and passes a full validation suite: three closure families, a bottom-line test, iteration and pull diagnostics, a binned goodness-of-fit and an unbinned (C2ST) descriptive diagnostic, and classifier-family cross-checks. The existing 2D toys do not furnish an independent-truth coverage test. On that foundation we performed *unbinned* simultaneous three-, four- and five-observable unfolds of MINERvA data ($+E_{\text{avail}}$, $+q_3$, $+W$), each validated by recovering its lower-dimensional predecessor under marginalization. (Binned MINERvA multi-differential measurements, up to triple-differential [1], exist; the methodological distinction is the unbinned, simultaneously-unfolded formulation.) The physics output:

- the low- E_{avail} excess over all four truth-level generators recovers the established MINERvA low-recoil/2p2h deficit through an independent method, with the expected FSI insensitivity and Valencia 2p2h response (§7.7);
- the (E_{avail}, q_3) plane resolves the deficit’s momentum-transfer structure (§7.8); (the 4D systematic-budget composition is not quoted — no corrected 4D covariance is adopted, §7);
- the unified-throw machinery now tests the block sum with asymmetric endpoints, a fixed estimator seed, and mean-centered joint throws; the resulting 5D covariance remains a candidate until the selection-complete lateral replacement lands; no corrected 4D covariance is reported (§4.4);

- the (E_{avail}, W) study localizes the remaining high- E_{avail} excess to the deep-inelastic region, where all four generators underpredict the central result; no significance is quoted without a corrected projected covariance (§8);
- the recoil-only PET point-cloud track provides a low-level-input representation cross-check, yielding an absolutely-normalized central value that passes ordinary closure. Its historical five-component budget is quarantined because nuisance and retraining shifts require a joint covariance (or explicit cross term) and the detector sample was CV-support-limited (§9). Because the muon is omitted from both classifiers, these products do not establish a joint muon–hadron unfold and no covariance component transfers automatically to the full-event replacement.

Outlook. The scalar extended-fiducial study (§10) yields a preliminary central value $\sigma_{\text{ext}} = 4.502 \times 10^{-38} \text{ cm}^2/\text{nucleon}$ (46 % above the restricted phase space), with an anchor, closure tests and a three-prior envelope but without a corrected covariance (§4.4). About 6 % of the rate is a newly acceptance-supported extension, while about 28 % is a separately labeled prior-dominated extrapolation. A full-event PET estimator would add the distinguished muon and validated event context, require an omitted-muon stress closure, and repeat the nominal fit and complete UQ on the canonical extended grid. Natural additional refinements are NEUT as a fifth generator and the finer 44-cell Ascencio comparison (the maximal-common-grid comparison is presented in §7.8). Two paper-independent cross-checks would further strengthen the analysis — a flux $\leftrightarrow$ muon posterior from MAT-MINERvA for a fully self-derived ρ , and published precedent for the truncated-spectral “ours-only” goodness-of-fit convention (App. A.4).

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A Statistical methods

This appendix is the complete uncertainty-quantification methodology of the measurement — the seedscan, bootstrap and universe-covariance procedures, the MAT-conformant covariance construction, the $1/\Phi$ flux normalization, the comparison conventions (χ^2 , pulls, shape-only), the ill-conditioning/rank analysis, the tension anatomy, the coverage toys, and the closure tests. Each procedure gives the recipe, the math, the implementation pointer, and a worked example using a reported number. We document only the analysis-specific numerical conventions, not a general statistics primer; the χ^2 convention referenced throughout is Eq. (11) in §A.3.2.

A.1 Numerical conventions: inverting a covariance (Cholesky, ridge jitter, pseudo-inverse)

Equation (11) requires inverting $\mathbf{C}$. Three practical issues come up in this analysis.

Positive-definiteness check (Cholesky). A matrix $\mathbf{C}$ is positive-definite (PD) iff there is a lower-triangular $\mathbf{L}$ with $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$ (the Cholesky factorization). If Cholesky succeeds, $\mathbf{C}$ is invertible and Eq. (11) is well-posed.

Ridge jitter. Numerical noise can leave a PD matrix barely non-PD. The standard fix is to add an isotropic ridge $\varepsilon \mathbf{I}$. The `uq/analyze_uq.py` script does exactly this:

$$\varepsilon = 10^{-12} \frac{\text{Tr}\mathbf{C}}{n}, \quad (2)$$

where the trace sets the natural scale of the eigenvalues. Twelve orders of magnitude below the mean diagonal is far below any real signal in $\mathbf{C}$ and safely above floating-point round-off.

Rank deficiency at $N \leq n$. If you estimate $\mathbf{C}$ from N replicas in a space of n bins with $N \leq n$, the resulting sample covariance has rank at most $N - 1$: there are not enough independent “directions” to fill the full $n \times n$ matrix. This was the regime of the Stage-1 bootstrap ($N = 50$ replicas, $n = 205$ bins): Cholesky failed without jitter, and the required jitter, 1.6×10^{-94} , was a numerical artifact of the rank deficiency rather than a real near-singularity. The Stage-2 bootstrap ($N = 300 > 205$) is full-rank. The *systematic* covariance suffers a more stubborn version of this problem that survives at any universe count — see §A.4.

Pseudo-inverse. For the χ^2 comparison against the paper covariance, `compare_to_paper_full-cov.py` replaces $\mathbf{C}^{-1}$ with the Moore–Penrose pseudo-inverse $\mathbf{C}^+$, computed via SVD. Writing $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n)$,

$$\mathbf{C}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top, \quad \Sigma_{ii}^+ = \begin{cases} 1/\sigma_i & \sigma_i > \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

with a small threshold τ (NumPy’s default is $\tau = n \cdot \sigma_{\max} \cdot \varepsilon_{\text{machine}}$). This silently drops singular modes that are numerically indistinguishable from zero, replacing them with 0 in the inverse rather than ∞ . The result is a stable quadratic form even when $\mathbf{C}$ has zero or near-zero eigenvalues.

A.2 Variance-source procedures

Each subsection below follows the same template: *What it estimates · Procedure · Math · Code pointer · Worked example*.

A.2.1 ML stochasticity: the seedscan

What it estimates. The variance in the unfolded cross section due to random choices inside the GBDT (gradient-boosted decision tree) optimizer. OmniFold is deterministic given the input weights, the network architecture, and the random seed; flipping only the seed produces slightly different fits to the per-event reweighting functions, and that spread is the “ML noise floor”.

Procedure. Run $n = 10$ full 5-iteration MEFHC unfolds with everything fixed except the GBDT `random_state`. The driver `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` pins three seeds per trial via `--seed N`:

$$(\text{step-1 classifier}, \text{step-2 classifier}, \text{miss regressor}) = (N, N + 1, N + 2), \quad N \in \{1, \dots, 10\}. \quad (4)$$

The canonical seedscan uses the `lgbm` (LightGBM) backend, so it is backend-matched to the production unfold.

Math. For each reported bin i , build the n -vector $X_i^{(1)}, \dots, X_i^{(n)}$ across trials and compute the unbiased sample standard deviation

$$\sigma_i^{\text{ML}} = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (X_i^{(k)} - \bar{X}_i)^2}, \quad \bar{X}_i = \frac{1}{n} \sum_k X_i^{(k)}. \quad (5)$$

The headline diagnostic is the per-bin *relative* spread $\sigma_i^{\text{ML}}/\bar{X}_i$, summarized by its median over the 205 reported bins.

Code pointer. `seedscan/analyze_seedscan.py` (rollup); `sbatch_unfold_2d_MEFHC_5iter_seedscan.sh` (driver); covariance output `uq/seedscan_lgbm_ml/uq_covariance_ml.root`.

Worked example (UQ envelope). With $n = 10$ `lgbm` trials on the 205-bin set:

- Total- σ relative spread: 0.0055%.
- Per-bin median relative spread: 0.166%.
- Component trace $\sqrt{\text{Tr} \mathbf{C}^{\text{ML}}} = 5.061 \times 10^{-41}$.

The paper’s per-bin total-uncertainty median is 6.86%; the ML noise is therefore $\approx 2.4\%$ of the paper’s quoted uncertainty and far subdominant to data-stat and systematics. Going from $n = 4$ to $n = 10$ trials moved the headline numbers within rounding, so the seedscan envelope is converged.

A.2.2 Statistical uncertainty: the Poisson bootstrap

What it estimates. The combined data-statistical and MC-statistical variance, propagated *coherently* through the full OmniFold pipeline. Coherent means: if some data event happens to be “up-weighted” in a replica, the same event flows through both the step-1 classifier training and the final reweighted-MC cross-section construction; if an MC event is up-weighted, the same up-weight is applied to its reco-tree and truth-tree copies, preserving the response matrix.

Procedure. For each replica $k = 1, \dots, N$:

1. Draw per-event Poisson(1) factors $b_i^{\text{data},(k)}$ for every data event and $b_i^{\text{MC},(k)}$ for every MC event, using two independent sub-RNGs (a data RNG seeded from k , and an MC RNG seeded from $k + 10^7$). The two sources are uncorrelated by construction.
2. Multiply data weights by b^{data} and MC weights (both reco and truth copies of each MC event) by the *same* b^{MC} . This keeps the migration matrix self-consistent.
3. Run OmniFold (5 iterations) with the GBDT seed *pinned*. Pinning the seed isolates the bootstrap signal from the ML noise of §A.2.1.
4. Save `hXSec2D` (and 1D projections).

Stage-1 used $N = 50$ replicas (rank-deficient on 205 bins); Stage-2 scaled to $N = 300 > 205$ for a full-rank statistical covariance.

Caveat (seed-varying replicas). The $N = 300$ covariance was produced with replicas that varied *both* `--bootstrap-seed N` and `--seed N`, so it carries a small ML-stochastic contribution on top of the pure Poisson statistics. The leak is numerically small relative to the systematic block (the ML floor is 0.166% vs the stat 0.564%); a purely Poisson block would fix `--seed` while varying only `--bootstrap-seed`.

Math. Flatten each replica’s 14×16 histogram into a 205-vector $\mathbf{X}^{(k)}$ over reported bins (row-major over $(p_T, p_{\parallel})$). The bootstrap covariance is the unbiased sample covariance:

$$(\mathbf{C}^{\text{boot}})_{ij} = \frac{1}{N-1} \sum_{k=1}^N (X_i^{(k)} - \bar{X}_i)(X_j^{(k)} - \bar{X}_j), \quad \bar{X}_i = \frac{1}{N} \sum_k X_i^{(k)}. \quad (6)$$

This is exactly `np.cov(Xrep, rowvar=False, ddof=1)` in `uq/analyze_uq.py`. The rank-deficiency check is the Cholesky-plus-jitter pattern from §A.1.

Why Poisson(1) is the right primitive. A non-parametric bootstrap with replacement assigns event i a multinomial count n_i with $\mathbb{E}[n_i] = 1$, $\text{Var}[n_i] = (N-1)/N \rightarrow 1$, and $\mathbb{E}[\sum n_i] = N$. In the large- N limit the n_i become independent and each is Poisson(1) (Appendix A.9). For a weighted analysis you multiply $w_i \rightarrow w_i n_i$; since n_i may equal zero, some events are dropped from a given replica, exactly mimicking the “with replacement” resample.

Code pointer. Bootstrap weight draws and seed pinning: `unfold_2d_omnifold_unbinned.py` (`--bootstrap-seed`, `--seed`).

Covariance rollup: `uq/analyze_uq.py`.

Output: `uq/bootstrap_MEFHC_300/uq_covariance_boot300.root`.

Worked example ($N = 50$ vs $N = 300$, MEFHC lgblm).

Metric (205 bins)	$N = 50$	$N = 300$
Total σ relative std	0.068%	0.061%
Per-bin median relative spread	0.532%	0.564%
Per-bin p_{84}	1.235%	1.215%

The per-bin median is stable from $N = 50$ to $N = 300$ (0.53% $\rightarrow$ 0.56%); the $N = 300$ block is the one used in the headline because it is full rank ($N > 205$), whereas $N = 50$ was rank-deficient and its Cholesky needed jitter 1.6×10^{-94} . The bootstrap median is $\approx 3.4\times$ the seedscan median (0.166%), as expected for a sample of $\mathcal{O}(10^6)$ data events. Across datasets the per-bin median scales as $\sqrt{N_{\text{events}}}$: playlist 1A ($\approx N/12$ of MEFHC) gives 1.46%, consistent with $\sqrt{12} \times 0.56\% \approx 1.9\%$ at the same backend.

A.2.3 Systematic uncertainty: universes

What it estimates. Model uncertainty, by re-running the full unfold under each of several physically-motivated reweightings of the MC. Where the bootstrap shakes the *events*, the universes shake the *model*: a different flux $\rightarrow$ different per-event weights $\rightarrow$ different unfolded cross section.

Procedure. The C++ event loop (`runEventLoopOmniFold.cpp`) dumps, for each MC event, alternative weight branches $w_{B,u}^{\text{truth}}$ and $w_{B,u}^{\text{reco}}$ for every (band B , universe index u) pair. The Stage-2 sweep has **187 universes across 44 bands**. Five bands are genuinely kinematic: BeamAngleX/Y, MuonResolution, and Muon_Energy_MINERvA/MINOS. They require selection-complete MNV101_ACTIVE_UNIVERSE event loops when migration across the event selection matters. MinosEfficiency and GEANT are weight-only. The dominant ensemble is the 100-universe PPFX flux band; almost all other bands are $\pm 1\sigma$ pairs.

Band (examples)	Variation	N_u
Flux	PPFX hadron-production sampling	100
Muon_Energy_MINOS	Muon energy scale, $\pm 1\sigma$	2
Muon_Energy_MINERvA	Muon energy scale (lateral), $\pm 1\sigma$	2
MinosEfficiency	MINOS muon-match efficiency, $\pm 1\sigma$	2
MaRES, MvRES, MaCCQE	GENIE knobs, $\pm 1\sigma$	2 ea.
GEANT_{Neutron,Pion,Proton}	FSI response (lateral), $\pm 1\sigma$	2 ea.

For each (B, u) , `unfold_2d_omnifold_unbinned.py --universe B:u` reweights all MC events (truth and reco) by the matching branch and runs OmniFold with a fixed seed. A separate matched “CV” job runs without `--universe`, with the *same* omnifile, backend, seed, iteration count, and flux, so that ML noise cancels between CV and universe.

Math (MAT-conformant convention). For universe (B, u) , define the per-bin value $\mathbf{X}^{(B,u)}$, a 205-vector over reported bins. MINERvA’s MAT framework (`PlotUtils/MnvVertErrorBand::CalcCovMx`, see §A.2.4) builds the per-band covariance as the *mean-centered, biased* sample covariance, referenced to the **universe mean** (not the CV) and normalized by $1/N_u$:

$$\mathbf{C}^B = \frac{1}{N_u} \sum_{u=1}^{N_u} (\mathbf{X}^{(B,u)} - \langle \mathbf{X} \rangle_B) (\mathbf{X}^{(B,u)} - \langle \mathbf{X} \rangle_B)^\top, \quad \langle \mathbf{X} \rangle_B = \frac{1}{N_u} \sum_u \mathbf{X}^{(B,u)}. \quad (7)$$

This single formula is applied *uniformly* to all $N_u \geq 2$ — there is **no special-case $\pm 1\sigma$ pair formula**. For a $\pm 1\sigma$ pair ($N_u = 2$) it collapses to the rank-1 $\frac{1}{4}(\mathbf{X}_+ - \mathbf{X}_-)(\mathbf{X}_+ - \mathbf{X}_-)^\top$; mean-centering kills the symmetric component, so a one-sided or asymmetric pair correctly contributes near-zero variance rather than the rank-2 contribution a naive CV-centered formula would give (Appendix A.10).

Target-nucleon normalization (rank-1 add-on). MINERvA quotes a 1.4% overall target-nucleon normalization uncertainty (Aliaga NIM, arXiv:1305.5199) that lives outside the `MnvVertErrorBand` machinery. We add it as a fully-correlated rank-1 outer product on top of the band sum:

$$\mathbf{C}^{\text{norm}} = (\sigma_N \mathbf{X}^{\text{CV}})(\sigma_N \mathbf{X}^{\text{CV}})^\top, \quad \sigma_N = 0.014, \quad (8)$$

via `analyze_universes.py --add-norm 0.014`.

Total systematic covariance. Bands are assumed independent (different physics sources, different RNGs), so

$$\mathbf{C}^{\text{syst}} = \sum_B \mathbf{C}^B + \mathbf{C}^{\text{norm}}. \quad (9)$$

Independence is a modeling assumption with one known correlated exception — the Bashyal flux $\leftrightarrow$ Muon_Energy_MINOS joint block (§A.4), which we have since rederived ($\rho = 0.900$, `uq/flux_muonE_cross.root`); it is real but rank-preserving, so for the 2D budget the block sum above is retained as the headline and the cross term is carried as a diagnostic.

There is, however, a second and larger correction that Eq. (9) omits: the band sum is linear in the per-band covariances, whereas the OmniFold reweighter is *nonlinear* in the input perturbations, so jointly perturbing all bands and re-unfolding produces a cross term beyond any pairwise correlation. We measure this directly with a *unified throw* — N joint re-unfolds in which every reweightable band and the flux multisim are perturbed simultaneously per event ($w_{CV} \prod_B \rho_B^{g_B}$, $g_B \sim \mathcal{N}(0, 1)$) and the cross section re-extracted. The corrected implementation uses asymmetric $\pm 1\sigma$ endpoint interpolation, one fixed estimator seed, and covariance about the throw mean; the mean shift from CV is stored separately. It does not subtract a scalar run-to-run jitter floor (`nd-unfolding/unified_throw_cov.py`). The 5D result uses the strict joint ensemble and matched-endpoint comparator described in §4.4. No corrected 4D or extended-fiducial covariance is reported. A single-throw superposition probe is insufficient to test the nonlinear response. Any adoption must document the low-rank treatment, mean shift, and selection-support bound before transferring information onto a sweep block, $\mathbf{C}^{\text{syst}} \rightarrow (\mathbf{C}^{\text{syst}} - \mathbf{C}^{\text{vert}}) + \mathbf{G} \mathbf{C}^{\text{vert}} \mathbf{G}$ ($\mathbf{G} = \text{diag } g$), which is positive-semidefinite by construction (a direct rank- N swap is not) and never falls below the block baseline.

Code pointer. Event-loop dump: `runEventLoopOmniFold.cpp` (env `MNV101_DUMP_UNIVERSES`).
Per-universe unfold: `unfold_2d_omnifold_unbinned.py --universe BAND:IDX`.
Per-band rollup: `uq/analyze_universes.py (--add-norm 0.014)`.
Output: `uq/universe_stage2_MEFHC_full_matcorr/uq_universe_covariance_full_matcorr.root`.

Worked example (full matcorr sweep). Top contributing bands, per-bin median relative σ_i/X_i^{CV} over 205 reported bins:

Band	Per-bin median rel. σ
Muon_Energy_MINOS	2.305%
Muon_Energy_MINERvA	1.285%
MinosEfficiency	1.472%
Flux (PPFX, $N_u = 100$)	1.008%
MaRES	0.548%
MvRES	0.377%
MaCCQE	0.357%
Systematic total (44 bands + norm)	4.783% (median); $\sqrt{\text{Tr} \mathbf{C}} = 2.463 \times 10^{-39}$

Grouped by category the medians are: Muon reconstruction 3.366%, Models 2.034%, Flux 1.008%, Hadronic response 0.758%, Statistical 0.563%, Normalization 0.081%. Muon energy scale (MINOS+MINERvA) and the MINOS-efficiency band dominate; these are exactly the bands the Ruterbories paper flags as flux-correlated (§A.4).

A.2.4 How MINERvA builds its published covariance

MINERvA’s covariance construction is not spelled out in the paper; it is defined in the MAT C++ framework. The takeaway: **MAT samples per-band exactly as we do**, using the mean-centered $1/N$ sample covariance, and sums bands element-wise.

What the Ruterbories 2021 paper says (Section V). Three uncertainty categories — flux, detector response, neutrino-interaction model — evaluated by “re-extracting the cross section using modified simulations”, i.e. the universe method, but with *no covariance formula given*. Specifics quoted: flux below 4% across most of phase space thanks to the ν - e -scattering constraint (Valencia, arXiv:1906.00111); 1.4% target-nucleon normalization; in-situ test-beam hadron response with per-particle GEANT inelastic variations; muon energy scale 1% constrained by the low-recoil fit of Bashyal (JINST 16 P08068), with the explicit statement that “*the fit causes the flux and muon energy scale uncertainties to be correlated and that correlation is propagated to the final result*” — the only non-block-diagonal piece the paper admits. The only covariance that appears in equations is V in the model-test χ^2 (Eq. 1, “the measurement covariance matrix”), with 205 degrees of freedom (Table I), and a log-normal χ^2 variant motivated by the highly-correlated flux normalization (Peelle’s Pertinent Puzzle).

What the paper does *not* say. No per-band covariance construction; no $\pm 1\sigma$ pair formula; no band-independence/block-sum statement; nothing about rank, condition number, or regularization of V ; no bootstrap or coverage validation (the paper uses D’Agostini iterative unfolding). All of this is delegated to MAT or simply unstated.

Per-band covariance: `MnvVertErrorBand::CalcCovMx`. Reading the MAT source establishes three facts, encoded in Eq. (7):

- the reference is the **universe mean** $\langle x \rangle_u$, not the central value X_{CV} ;
- the normalization is $1/N$ (biased), not Bessel’s $1/(N-1)$ — a 1% offset for PPFX ($N=100$);
- there is **no special case for $N=2$** — the mean-centered N -universe form is applied uniformly. The “MINERvA-101 $\pm 1\sigma$ pair” formula $\frac{1}{2}(\Delta_+ \Delta_+^\top + \Delta_- \Delta_-^\top)$ is a folk re-derivation that agrees only for CV-symmetric shifts.

We verified our rollout against MAT byte-for-byte: populating a `PlotUtils::MnvH1D` with the same 44 bands / 188 universes and comparing `MnvH1D::GetTotalErrorMatrix` against our `--add-norm 0` rollout gives a max element-wise relative difference of 5.5×10^{-17} (machine ε), with $\sqrt{\text{Tr}}$ matching to seven digits (`uq/verify_matcorr_vs_mnvh1d.py`).

Total covariance: `MnvH1D::GetTotalErrorMatrix`. A straight element-wise sum across bands plus the statistical matrix. Band-independence is assumed; cross-band correlations exist only if encoded inside a single combined band (e.g. the Bashyal joint fit, handled outside this loop). MAT does *no* ridge / shrinkage / SVD truncation on the published total cov and no special handling of rank deficiency — the published V is full-rank because it includes the *statistical* block, not because it sums richer systematics (its systematic-only rank is $120 < 140$; see §A.4).

Upstream references that set each band’s $\pm 1\sigma$ amplitude.

Input	Reference
Flux / PPFX (100 universes)	Aliaga, PRD 94 092005 (2016), arXiv:1607.00704
ν - e flux constraint ($< 4\%$)	Valencia, PRD 100 092001 (2019), arXiv:1906.00111
Muon-E scale $\leftrightarrow$ flux fit	Bashyal, JINST 16 (2021) P08068
Test-beam / GEANT hadron response	Aliaga, NIM A 789 28 (2015), arXiv:1501.06431
GENIE reweight knobs	Andreopoulos, NIM A 614 87 (2010); GENIE manual arXiv:1510.05494
1.4% normalization (detector mass)	Aliaga, NIM A 743 130 (2014), arXiv:1305.5199

A.2.5 Block-sum combination

What it estimates. The total uncertainty covariance, combining the three statistically independent components above.

Independence assumption. The bootstrap samples *events* (with two distinct sub-RNGs for data and MC). The universes vary *model weights*, with the GBDT seed pinned. The seedscan varies the *GBDT seed*, with all weights fixed. The three Monte Carlos are independent by construction:

$$\mathbf{C}^{\text{tot}} = \mathbf{C}^{\text{ML}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{syst}}. \quad (10)$$

The combined-cov hook in `compare_to_paper_fullcov.py` (`--omnifold-cov ROOT:HIST, repeatable`) adds OmniFold- derived covariances on top of the paper’s `TotalCovariance` through exactly this rule. $\mathbf{C}^{\text{ML}}$ is more than an order of magnitude below the other two; the headline includes it but it is numerically negligible.

Worked example (UQ envelope). Component traces and per-bin median relative spreads:

Component	$\sqrt{\text{Tr}\mathbf{C}}$	Per-bin median rel. σ
ML noise (lgbm seedscan, $n = 10$)	5.061×10^{-41}	0.166%
Statistical (Poisson bootstrap, $N = 300$)	1.828×10^{-40}	0.564%
Systematic (matcorr sweep + norm)	2.463×10^{-39}	4.783%
Combined (block sum)	2.470×10^{-39}	4.822%
Paper <code>TotalCov</code> (reference)	2.676×10^{-39}	6.86%

The combined OmniFold envelope is $\approx 70\%$ of the paper’s per-bin median uncertainty, dominated by the systematic block. (The combined $\sqrt{\text{Tr}}$ is the quadrature sum of the three independent component traces; the per-bin median 4.822% is taken directly from the matcorr rollout summary.)

A.3 Comparing to the paper

A.3.1 The 205-bin selection

The paper reports $d^2\sigma/(dp_T dp_{\parallel})$ on a 14-bin p_T axis $\times$ 16-bin $p_{\parallel}$ axis = 224 cells (edges in `minerva_paper_anc/bin_mapping.txt`). Of those, 205 have $x^{\text{paper}} > 0$ in `data_result_ptpl_2D_minerva_inclusive_6GeV.txt`; the remaining 19 are paper-unreported cells that the paper’s analysis suppresses via an efficiency cut. We drop those 19 from both sides of every χ^2 and pull — that is the unambiguous “apples-to-apples” choice.

A secondary diagnostic mask, used only as a sensitivity check, is the strict-interior set: cells where $p_T^{\text{high}}/p_{\parallel}^{\text{low}} \leq \tan 20^\circ$. Some cells in the 205-bin set straddle the $\theta_\mu < 20^\circ$ kinematic cut and have wildly suppressed paper efficiencies that we do not replicate; pulls on those cells run to $\mathcal{O}(100)\sigma$ and dominate the χ^2 . The 185-bin interior cut removes them, and we quote it only as a sensitivity check that the headline tension is not an artifact of those few boundary cells. **The canonical metric is the 205-bin set** used everywhere below.

A.3.2 Covariance-aware χ^2/ndf

Math. With $\Delta \equiv \mathbf{x}^{\text{ours}} - \mathbf{x}^{\text{paper}}$ on the 205-bin set and $\mathbf{C}$ the chosen covariance,

$$\chi^2 = \Delta^\top \mathbf{C}^+ \Delta, \quad \text{ndf} = n_{\text{reported}} = 205, \quad \chi^2/\text{ndf} = \chi^2/205. \quad (11)$$

The implementation (`chi2_with_cov` in `compare_to_paper_fullcov.py`) builds the 205×205 block, inverts with `np.linalg.pinv` (§A.1), and forms the quadratic form.

Code pointer. `compare_to_paper_fullcov.py` (headline); `--omnifold-cov ROOT:HIST` (repeatable) adds each OmniFold covariance via Eq. (10); `--subtract-stat` removes the paper’s stat block to avoid double-counting the bootstrap (§A.4).

Worked example: comparison to paper (headline numbers).

Metric (205 bins)	Paper	OmniFold (combined)
Width-weighted total σ , $\sqrt{\mathbf{w}^\top \mathbf{C} \mathbf{w}}/(\mathbf{w}^\top \mathbf{x})$	4.61%	2.26%
Per-bin median relative σ	6.86%	4.822%
χ^2/ndf	3.661 (paper-only)	1.481 (paper+ours)
Log-normal χ^2/ndf	—	1.468

The $3.661 \rightarrow 1.481$ drop (-60%) is the central UQ result: once the OmniFold-side error budget is included, the summed-covariance distance to the paper falls to $\chi^2/\text{ndf} \approx 1.5$. (A perfectly correct model with correct uncertainties gives exactly 1; ≈ 1.5 tolerates small residual shape mismatch in high- $p_\parallel$ tails, see §A.3.4.) **This is an indicative distance, not a calibrated goodness-of-fit:** the published (IBU-era) result and this OmniFold result share the same MINERvA data and flux/detector/GENIE systematics, and without the (unavailable) OmniFold $\leftrightarrow$ paper cross-covariance the summed covariance both double-counts those shared systematics and omits their correlation. We therefore quote it as a distance and ratio-to-paper, not a p -value; see §A.4 for why it remains the headline and why we report per-bin pulls rather than a single unqualified ours-only χ^2 . The log-normal variant (matching the paper’s Table I treatment of the correlated flux normalization) gives 1.468, essentially identical.

Edge-bin sensitivity ($p_\parallel$ -min cut). `compare_to_paper_fullcov.py` ships a side-table that recomputes χ^2/ndf on progressively shrinking subsets, using the paper covariance only (no OmniFold add):

$p_\parallel \geq$ (GeV/c)	N_{bins}	χ^2	χ^2/ndf	median ratio
1.5	205	750.49	3.661	1.0064
2.0	196	685.36	3.497	1.0059
2.5	186	545.87	2.935	1.0059
3.0	175	474.84	2.713	1.0059
3.5	163	452.95	2.779	1.0073
4.0	151	428.72	2.839	1.0068

The residual is concentrated in the low- $p_\parallel$, near-edge cells; the high- $p_\parallel$ tails carry sub-2% shape disagreement. This is a robustness check, not a separate method.

A.3.3 Pull distribution

Math. For each of the 205 reported bins,

$$\text{pull}_i = \frac{x_i^{\text{ours}} - x_i^{\text{paper}}}{\sqrt{C_{ii}^{\text{tot}}}}, \quad \mathbb{E}[\text{pull}_i] = 0, \quad \text{Var}[\text{pull}_i] = 1 \text{ under the null.} \quad (12)$$

Summary diagnostics: pull mean, RMS, the 14×16 pull heatmap, and a pull histogram (in `compare_to_paper_fullcov.py`).

Worked example (headline numbers). With the combined covariance (paper TotalCov + matcorr universe + bootstrap-300 + ML):

- Pull mean: 0.069 (centered on zero; no global offset).
- Pull RMS: 0.466.

Pull RMS < 1 means the combined error band is *conservative* relative to the actual disagreement — the band is wider than it needs to be to absorb the observed scatter. No bins have $|\text{pull}| > 3$ in the reported set. Compare with paper-cov-only: mean 0.089, RMS 0.598 — similar mean (no bias), slightly wider spread before we include our own uncertainties.

A.3.4 Shape-only χ^2 with Jacobian propagation

What it estimates. Whether the *shape* of the $(p_T, p_{||})$ cross section matches the paper, independent of absolute normalization. Useful because the absolute normalization depends on the flux integral, target nucleon count, and POT (all of which can drift by $\mathcal{O}(\%)$), whereas the shape is a pure unfolding diagnostic.

Definition. With bin widths $w_j = (\Delta p_T)_j (\Delta p_{||})_j$ and total $T(\mathbf{x}) = \sum_{j \in \text{mask}} w_j x_j$, the unit-area shape is

$$s_i(\mathbf{x}) = \frac{x_i}{T(\mathbf{x})}. \quad (13)$$

By construction $\sum_i w_i s_i = 1$.

Jacobian. We need the covariance of $\mathbf{s}$ given the covariance of $\mathbf{x}$. Differentiating,

$$\frac{\partial s_i}{\partial x_j} = \frac{\delta_{ij}}{T} - \frac{x_i w_j}{T^2} = \frac{1}{T}(\delta_{ij} - s_i w_j). \quad (14)$$

In matrix form (in `normalize_xsec_shape.py`),

$$\mathbf{J} = \frac{1}{T}(\mathbf{I} - \mathbf{s} \mathbf{w}^\top), \quad \mathbf{C}^{\text{shape}} = \mathbf{J} \mathbf{C}^{\text{paper}} \mathbf{J}^\top. \quad (15)$$

Rank reduction. $\mathbf{J}$ is rank $n - 1$: the unit-area constraint $\sum_i w_i s_i = 1$ kills one degree of freedom. Algebraically, $\mathbf{w}^\top \mathbf{J} = \mathbf{w}^\top (\mathbf{I} - \mathbf{s} \mathbf{w}^\top)/T = (\mathbf{w}^\top - \mathbf{w}^\top)/T = \mathbf{0}^\top$, so the “all-bins-up” direction is the kernel of $\mathbf{J}$ and lives in the null space of $\mathbf{C}^{\text{shape}}$. Accordingly:

- $\mathbf{C}^{\text{shape}}$ has one zero eigenvalue and rank $n - 1$.
- $\chi_{\text{shape}}^2 = (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})^\top (\mathbf{C}^{\text{shape}})^+ (\mathbf{s}^{\text{ours}} - \mathbf{s}^{\text{paper}})$ (pseudo-inverse drops the constrained direction).
- $\text{ndf} = n_{\text{bins}} - 1 = 204$ on the 205-bin set.

Code pointer. `normalize_xsec_shape.py`: `shape_jacobian`, `shape_chi2`.

Worked example (headline numbers). Shape χ^2/ndf on 205 bins with the paper’s TotalCovariance (Jacobian-propagated): 3.596. This is essentially the absolute-normalization χ^2/ndf of 3.661 *minus* the contribution of overall scale. Total- σ ratio is $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.0111$ — 1% scale difference. Shape disagreement at the $\sim 1\%$ level (median bin ratio 1.0064, 94% of bins within 10% of the paper) accounts for the remaining shape residual.

A.4 Ill-conditioning and rank deficiency

The obstruction. For publication we would like an *ours-only* χ^2/ndf — the data-to-model agreement evaluated with our own covariance $\mathbf{C}^{\text{ours}} = \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}} + \mathbf{C}^{\text{ML}}$ rather than the paper’s V . That requires inverting $\mathbf{C}^{\text{ours}}$, and it is rank-deficient:

$$\text{rank}(\mathbf{C}_{\text{sample}}^{\text{syst}}) = 140/205, \quad \text{cond}(\mathbf{C}^{\text{syst}}) = 8.7 \times 10^8. \quad (16)$$

The deficiency is set by **band count, not universe count**: under the MAT formula Eq. (7) each $\pm 1\sigma$ pair is rank 1 and the 100-universe PPF band is rank ≤ 99 (and *effectively* rank-1: a single normalization mode holds 99.6% of its variance, §4.1), so the 44-band sum cannot fill the 205-dimensional bin space. *This is not a missing-bands problem.* The paper’s own systematic-only covariance (`TotalCov` – `StatOnlyCov`) is rank 120 — *lower* than our 140 — so we propagate at least as many independent systematic directions as the published analysis (we use the same `GetStandardSystematics`). The paper’s V is full-rank (rank = 204, cond = 1.5×10^{12}) purely because it **includes the statistical block** (`StatOnlyCov`, rank 205, near-diagonal), not because it sums richer systematics. Adding our 300-replica bootstrap does the same for us: $\text{rank}(\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}) = 201$, cond = 3.8×10^{14} .

The real obstruction is statistical, not rank. Even at rank 201 a defensible single ours-only χ^2 is out of reach — but for a *statistical-precision* reason, not a missing-band one. Our OmniFold bootstrap stat block is $2.5\times$ smaller by trace than the paper’s stat block ($\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.8 \times 10^{-40}$ vs $\sqrt{\text{Tr}\mathbf{StatOnlyCov}} = 4.6 \times 10^{-40}$), so it regularizes the small systematic modes $2.5\times$ less. The naive pseudo-inverse then gives $\chi^2/\text{ndf} = 252$ for $\mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{boot}}$, dominated by poorly-determined directions, against the paper’s 3.66 with its larger stat floor. The lever toward an ours-only χ^2 is therefore the *statistical* block — not the hunt for more systematic bands. The gap is *not* a missing MC-statistical term: our bootstrap already draws independent Poisson(1) multipliers on *both* data and MC per replica (§A.2.2; Practical Guide 2507.09582 §5.1), so data-stat and MC-stat contribute jointly. With that excluded, the smaller stat covariance most likely reflects genuinely higher unbinned-OmniFold statistical efficiency relative to the published binned D’Agostini unfold (whose iterative regularization amplifies statistical fluctuations), or a difference in what the paper’s `StatOnlyCov` encompasses — the split-bootstrap test below resolves this.

Resolution (data/MC split bootstrap). A 200+200-replica *split* bootstrap Poisson-fluctuates the data weights only (`boot_data`) or the MC weights only (`boot_mc`) with independent RNGs, so in expectation $\mathbf{C}^{\text{boot}} = \mathbf{C}_{\text{data}}^{\text{boot}} + \mathbf{C}_{\text{mc}}^{\text{boot}}$. Three findings settle the question. (i) *Closure holds*: $\sqrt{\text{Tr}(\mathbf{C}_{\text{data}} + \mathbf{C}_{\text{mc}})}/\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.07$, with the variance split 77% data / 23% MC. (ii) The *data stream alone* gives $\sqrt{\text{Tr}\mathbf{C}_{\text{data}}}/\sqrt{\text{Tr}\mathbf{StatOnly}} = 0.356$ ($\mathbf{C}^{\text{boot}}/\mathbf{StatOnly} = 0.392 = 1/2.55$): the $2.5\times$ deficit is present in the data-only stat error, so it is *not* an MC-statistics omission — it reflects genuinely higher unbinned-OmniFold statistical efficiency than the binned D’Agostini unfold. (iii) There is *also* a structure difference: our stat block is correlated (mean |off-diag ρ | = 0.108, leading-eigenvector overlap with `StatOnly` 0.46) whereas the paper’s is near-diagonal (0.009); scaling our block to the paper trace still leaves $\chi^2/\text{ndf} = 76.7$, so the gap is shape as well as magnitude. **Consequence:** the correct ours-only χ^2 uses our *own* $\mathbf{C}^{\text{ours}}$, not the paper’s stat floor. The truncation-dependence below remains (the inversion is still regularizer-set), so we lead with per-bin pulls and the scan; the ours-only goodness-of-fit is reported on $\mathbf{C}^{\text{ours}}$ via a truncated-spectral χ^2 (retain $\lambda > 10^{-10}\lambda_{\text{max}}$), the same convention carried into the 3D generator comparison (§A.4 is the 2D anchor; the 3D comparison applies it on all 1431 reported bins).

Regularization scans (all cutoff-dependent). On $\mathbf{C}^{\text{ours}}$ over the 205 reported bins, no regularizer gives a stable single number:

Method	Setting	Result
Eigenvalue truncation	$\lambda/\lambda_{\text{max}} > \{10^{-3}, 10^{-4}, 10^{-5}\}$	rank $\{19, 49, 82\}$; $\chi^2/205 = \{0.13, 1.12, 3.08\}$
Ridge $\mathbf{C} + \varepsilon\lambda_{\text{max}}\mathbf{I}$	$\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$	$\chi^2/205 = \{0.35, 1.38, 4.30\}$
Diagonal-target shrinkage	$\lambda = 0.05$	rank $178 \rightarrow 204$, cond 1.3×10^{11} , $\chi^2/205 = 3.29$

Each method spans roughly two decades of $\chi^2/205$ across a two-decade scan of its knob, with no defensible single cutoff. The useful conclusion is diagnostic, not a number: the naive inverse is dominated by tiny, poorly-determined modes, and in the retained high-variance modes the central-value difference is not large. We therefore report direct per-bin pulls (§A.3.3) and this scan rather than a single unqualified ours-only χ^2 .

Bootstrap/paper-stat double-count. The combined headline (§A.3.2) adds $\mathbf{C}^{\text{boot}}$ on top of the paper’s **TotalCov**, which already contains the paper’s statistical block — a double count. **--subtract-stat** replaces the paper baseline by **TotalCov** – **StatOnlyCov** before adding our covariances, driving χ^2/ndf $1.481 \rightarrow 11.56$. That *over*-corrects: our $\sqrt{\text{Tr}\mathbf{C}^{\text{boot}}} = 1.83 \times 10^{-40}$ is far smaller than the paper’s published stat block ($\sim$ the 2.7×10^{-39} scale of **TotalCov**), so subtracting the latter without a comparable replacement removes too much. The headline therefore keeps the double-counted 1.481 as the well-defined “what does OmniFold UQ add on top of the published V ” statement, with the de-double-counted number on file as a secondary diagnostic.

Rank and conditioning: summary. The 140/205 systematic rank is richer than the paper’s systematic-only 120 (not a missing-bands problem); the Bashyal flux $\leftrightarrow$ muon-energy joint block is rederived ($\rho = 0.900$) and carried as a diagnostic — adding it leaves the systematic rank unchanged, its directions already lying within the span of the universe covariance (§4.1); the PPFX band is effectively rank-1 (§4.1); and the $2.5\times$ smaller bootstrap is explained by the data/MC split as genuine unbinned-OmniFold efficiency, not a missing MC term. Two paper-independent refinements are possible: a fitted $(\Phi_{\text{norm}}, \Phi_{\text{shape}}, \text{MuonE}_{\text{scale}})$ posterior from MAT-MINERvA for a self-derived ρ , and published precedent for the truncated-spectral “ours-only” χ^2 convention.

Adopted conventions. The publication-grade headline uses the combined covariance for $\chi^2/\text{ndf} = 1.481$ with the double-count caveat documented; the per-band rollout is MAT-conformant and byte-for-byte verified (§A.2.4); the 1.4% normalization is included as a rank-1 band; and the ours-only cov is reported via the truncated-spectral χ^2 on $\mathbf{C}^{\text{ours}}$ (rank 201) with the regularization scan disclosed.

A.5 Anatomy of the paper-covariance tension

The paradox. The central values agree well: $\sigma_{\text{tot}}^{\text{ours}}/\sigma_{\text{tot}}^{\text{paper}} = 1.011$, median bin ratio 1.006, and the *per-bin pull RMS* against the paper covariance is only 0.598 (§A.3.3). A diagonal-only χ^2 — ignoring all off-diagonal correlations — is therefore

$$\chi_{\text{diag}}^2/\text{ndf} = \frac{1}{n} \sum_i \frac{\Delta_i^2}{C_{ii}} = \text{RMS}(\text{pull})^2 + \text{mean}(\text{pull})^2 \approx 0.366, \quad (17)$$

yet the full-covariance $\chi^2/\text{ndf} = 3.661$. The tension is a $10\times$ inflation that lives *entirely* in the off-diagonal / small-eigenvalue structure of the published **TotalCov** (`diagnose_tension.py`). The rest of this section dissects it.

Eigenmode decomposition. Diagonalize the reported-block paper covariance, $\mathbf{C} = \sum_k \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$ ($\lambda_1 \geq \dots \geq \lambda_n$), and project the residual Δ onto each eigenvector, $c_k = \mathbf{u}_k^\top \Delta$. Then

$$\chi^2 = \Delta^\top \mathbf{C}^+ \Delta = \sum_k \frac{c_k^2}{\lambda_k}, \quad \text{“eigen-pull”}_k \equiv \frac{c_k}{\sqrt{\lambda_k}}, \quad (18)$$

verified to reproduce the `pinv` $\chi^2 = 750.49$ to 3×10^{-9} . The decomposition shows the tension is **neither a few pathological null directions nor the absolute smallest eigenvalues**:

- The top contributing modes sit at $\lambda_k/\lambda_{\max} \sim 10^{-4}$ – 10^{-5} — the lower-middle of the spectrum by magnitude, but still ~ 7 – 8 decades above the cond-floor ($\lambda/\lambda_{\max} \sim 10^{-12}$) — with standardized-component magnitudes 5–6. These are descriptive coordinates in the published covariance basis, not calibrated significances for two independent results.
- The ten *smallest*- λ modes carry only 3% of the χ^2 ; about 90 modes are needed to reach 90%. The χ^2 is broadly distributed, not localized to the ill-conditioned tail.

Regularization robustness. A `pinv-rcond` scan confirms the tension is not a numerical artifact: retaining only the 73 best-determined directions ($\lambda_k/\lambda_{\max} > 10^{-4}$) still gives $\chi^2/\text{ndf} = 1.42$; 139 directions ($> 10^{-6}$) give 2.79. A rank-truncation scan keeping the r largest- λ modes rises smoothly $0.69 (r=50) \rightarrow 2.35 (100) \rightarrow 3.30 (180) \rightarrow 3.66 (205)$ — no late jump, i.e. the tension does not ride on the last few least-determined modes.

It is shape, not normalization. Splitting Δ into a component along the measured spectrum $\mathbf{x}^{\text{paper}}$ (an overall scale) and the orthogonal remainder, the scale piece contributes $\chi_{\text{norm}}^2 = 0.10$ and the shape piece $\chi_{\text{shape}}^2 = 750.1$. The +1.1% normalization offset is statistically irrelevant to the tension; it is a correlated *shape* difference. (Consistent with the Jacobian shape-only $\chi^2/\text{ndf} = 3.60$, §A.3.4.)

Where it lives. Mapping the dominant eigenvectors and the per-bin contribution $\Delta_i(\mathbf{C}^+ \Delta)_i$ back onto the 14×16 grid (`tension_mode_maps.png`, `tension_chi2_map.png`) localizes the tension to the **low- p_T cross-section peak ridge** ($p_T \lesssim 0.4$ GeV/c, p_T -bins 2/7/10 carry 16/11/12%) and the low- $p_{\parallel}$ near-edge cells ($p_{\parallel}$ -bins 0/1/8). The carrier modes are sign-alternating *shape* oscillations across the peak, not vertical p_T -columns (which a flux $1/\Phi(p_T)$ normalization error would produce) and not the $\theta_{\mu} < 20^\circ$ acceptance boundary. This is exactly where the flux and Muon_Energy bands — the two largest, and the only named cross-band correlation (Bashyal, §A.4) — dominate the covariance: tight bin-to-bin correlation there leaves little freedom to absorb a coherent ~ 1 – 2% shape difference, so a small OmniFold-vs-IBU shape difference becomes a large χ^2 .

A methodological component (estimator and iteration). Part of the tension is regularization-dependent — the same data, same normalization ($\Sigma\sigma$ within 0.1%), unfolded with a different GBDT backend moves the paper-cov χ^2 by ~ 1 unit while central values barely change (iteration count, by contrast, is negligible):

OmniFold configuration	paper-cov χ^2/ndf	pull RMS
exact-GBT, 5 iter (frozen production)	3.66	0.598
HistGBT, 5 iter (10-seed mean)	2.70 ± 0.04	0.56
LightGBM, 5 iter	2.65	0.563
LightGBM, 10 iter	2.66	0.565

Two things separate cleanly. (i) **Iteration count is negligible**: LightGBM 5 $\rightarrow$ 10 iters moves the χ^2 by +0.01 (2.645 $\rightarrow$ 2.657), so OmniFold is well-converged at 5 iters and the tension is *not* an under-iteration artifact. (ii) **The GBDT estimator carries a ~ 1 -unit band**: the frozen exact-GBT production (3.66) is the outlier, while both histogram-binned backends agree at 2.65–2.70 (HistGBT is a 10-seed mean, range 2.63–2.74; the ± 0.04 ML-seed jitter is $\sim 25\times$ smaller than the 0.96 exact $\rightarrow$ hist gap, so that gap is genuinely the estimator, not seed noise). The exact-gradient trees fit finer shape structure (less implicit smoothing) and so deviate more from the paper’s IBU-regularized result. This is direct evidence that ~ 1 χ^2 -unit of the tension is OmniFold’s implicit regularization rather than a physical data–MC disagreement (`tension_iter/`).

Conclusion. The tension is (i) a genuine, broadly-distributed, *correlated shape* difference — not a normalization offset and not a small- λ inversion artifact — concentrated in the low- p_T peak where the flux/Muon_Energy bands dominate, plus (ii) a ~ 1 - χ^2 -unit methodological band from the GBDT estimator/iteration regularization. Both point to the same structure: the Bashyal flux $\leftrightarrow$ Muon_Energy joint block (rederived at $\rho = 0.900$, §A.4) is exactly the covariance structure governing the region that carries the tension.

A.6 Pull diagnostics and the open 2D coverage test

What it estimates. The existing closure-toy ensemble supports a same-ensemble standardized-pull calculation that checks whether the toy distribution is approximately Gaussian. It does not support a valid 2D coverage claim: the MC bootstrap also fluctuates the stored `hTruthXSec2D`, so the all-toy truth mean is not an independent reference. A future disjoint calibration/evaluation test must use an independently stored unfluctuated truth (or a redesigned toy ensemble).

Procedure (`sbatch_coverage_toys_MEFHC_200.sh`). Generate $N_{\text{toy}} = 200$ closure unfolds, each one combining two existing modes of `unfold_2d_omnifold_unbinned.py`:

- `--closure`: pseudo-data are MC reco events (weighted); a truth marginal is written to `hTruthXSec2D` in every output file.
- `--bootstrap-seed k` (seeds chosen to avoid collision with the production replicas): per-event Poisson(1) draws, same as the production bootstrap.

Toy seeds are independent and bins within a toy are correlated. Because the MC bootstrap multiplies `w_truth`, the stored truth marginal also varies from toy to toy and cannot serve as the independent coverage target.

Math. Stack the toys into an $N_{\text{toy}} \times 14 \times 16$ array $\mathbf{U}^{(t)}$. Here $\bar{X}^{\text{truth}}$ is the all-toy mean used only to center this same-ensemble diagnostic. For each reported bin (where $\bar{X}_i^{\text{truth}} > 0$, i indexing the 205 reported cells):

$$\sigma_i = \text{std}(\mathbf{U}_i^{(t)}; \text{ddof} = 1) \quad (\text{spread across toys}), \quad (19)$$

$$r_i^{(t)} = \frac{U_i^{(t)} - \bar{X}_i^{\text{truth}}}{\sigma_i} \quad (\text{per-toy, per-bin standardized residual}), \quad (20)$$

$$c_i = \frac{1}{N_{\text{toy}}} \sum_{t=1}^{N_{\text{toy}}} \mathbf{1}\{|r_i^{(t)}| \leq 1\} \quad (\text{same-ensemble per-bin fraction}). \quad (21)$$

The reference value of c_i for Gaussian standardized residuals is the central- σ probability of a unit normal:

$$c^{\text{expected}} = \Phi(1) - \Phi(-1) = \text{erf}(1/\sqrt{2}) \approx 0.6827. \quad (22)$$

A second diagnostic is the mean absolute pull, $\langle|r|\rangle$, which for a unit normal is $\sqrt{2/\pi} \approx 0.798$.

Code pointer. `uq/coverage_toys.py (rollup); driver sbatch_coverage_toys_MEFHC_200.sh; outputs uq/coverage/2d_xsec_MEFHC_5iter_lgbm_coverage_toy*.root and uq/coverage/coverage_summary.txt.`

Worked example (same-ensemble pull diagnostic). With $N_{\text{toy}} = 200$ on the 205 reported bins:

- Fraction with $|r| \leq 1$: 68.71% (Gaussian reference 68.27%; $\Delta = +0.44\%$); median 68.50%.
- $\langle|r|\rangle = 0.794$ (target 0.798; within 0.5%).
- Signed residual mean: $+0.006 \pm 0.082 \sigma$ (no bias).
- Fraction of bins whose within-bin fraction exceeds 65%: 97.6% — only 5 of 205 bins fall below.

The agreement of these same-ensemble quantities with their Gaussian references shows that the standardized toy residuals are approximately normal. Because σ_i is estimated from the same toys being scored, this is not a frequentist coverage measurement and does not certify the bootstrap band. The attempted 100/100 split-sample reinterpretation is also withdrawn because it used the all-toy mean of the fluctuated truth histograms; no 2D containment number is quoted until an independent unfluctuated reference is supplied.

A.7 Closure tests: detecting bias

The variance methods of §A.2 answer the question “how much does the unfold wobble.” Closure tests answer a different question: “is the unfold biased — does it return a value that is consistently offset from the right answer when we inject a known truth?” Bias does not show up in the variance procedures; you have to look for it.

The three closure tests inject a known deformation into the MC, unfold, and check that the answer reproduces the deformation. Each test has its own “acceptance threshold,” set comfortably above the ML-noise floor of 0.166%.

A.7.1 Truth-reweight closure

Procedure. For a known shape $f(p_T, p_{\parallel})$, multiply both the MC reco weights *and* the MC truth weights by f at the event level. Unfold the reweighted pseudo-data against the unchanged (CV) response. The reference is the reweighted truth marginal — the unfolded answer should match it.

Two shapes tested.

- **gauss_pt:** $f = 1 + A \exp[-((p_T - p_T^0)/\sigma)^2]$ with $A = 0.2$, $p_T^0 = 0.4$ GeV/c, $\sigma = 0.1$ GeV/c.
- **tilt_pz:** $f = 1 + \alpha(p_{\parallel} - p_{\parallel}^{\text{ref}})$ with $\alpha = 0.1$, $p_{\parallel}^{\text{ref}} = 5$ GeV/c.

Acceptance. Median per-bin $|\text{residual}| \leq 1.5\%$.

Code pointer. `uq/closure/closure_truth_reweight.py.`

Worked example (1A lgbm 5-iter).

- **gauss_pt:** median $|\text{residual}| =$ 0.046% (PASS).
- **tilt_pz:** median $|\text{residual}| =$ 0.013% (PASS).

Both are $\sim 100\times$ tighter than the threshold; OmniFold recovers the injected truth shape essentially perfectly.

A.7.2 Hidden-variable closure

Procedure. This is the subtle one. The OmniFold feature set sees $(p_T, p_{\parallel})$ on reco and truth, but *not* the resolution variable $\Delta p_T \equiv p_T^{\text{reco}} - p_T^{\text{truth}}$. So we inject a bias on Δp_T only — a kinematic axis the network cannot see — and ask: how much of that bias leaks into the measured $(p_T, p_{\parallel})$ marginal?

The pseudo-data weight bump is

$$w^{\text{reco}} \rightarrow w^{\text{reco}} \cdot \left(1 + A \exp\left[-((\Delta p_T - c)/\sigma_{\Delta p_T})^2\right]\right), \quad (23)$$

with $c = +0.1$ GeV/c, $\sigma_{\Delta p_T} = 0.05$ GeV/c. Truth weights are *unchanged*, so the reference is the unaltered CV truth marginal. The amplitude is scanned: $A \in \{0.05, 0.10, 0.30\}$.

Leakage coefficient. Define the signed residual $\bar{r}(A)$ as the median of the per-bin signed residual at amplitude A . The leakage coefficient is the slope of a linear fit:

$$\bar{r}(A) \approx \kappa_{\text{leak}} A \quad \Rightarrow \quad \kappa_{\text{leak}} = \frac{\partial \bar{r}}{\partial A}. \quad (24)$$

κ_{leak} tells you what fraction of an unseen Δp_T -shaped bias contaminates the visible answer.

Acceptance. Median per-bin $|\text{residual}| \leq 3.0\%$ (looser than truth- reweight because Δp_T and p_T are weakly correlated, so the unfolding can never fully kill the leak).

Code pointer. `uq/closure/closure_hidden_var.py`.

Worked example (1A lgblm 5-iter, 205 reported bins).

- Per-amplitude median residuals: 0.57%/1.14%/3.43% for $A = 0.05/0.10/0.30$ — linear in A , as expected for small perturbations.
- Linear fit slope: $\kappa_{\text{leak}} = \boxed{0.113}$.

About 11% of a hidden Δp_T bias leaks into the measured $(p_T, p_{\parallel})$ marginal — under the 3% threshold for the largest realistic deformation ($A = 0.30$). Larger leakage would indicate that Δp_T ought to be added to the feature set; the present number is small enough to be safely ignored.

A.7.3 Alternative-model closure

Procedure. Train the OmniFold response on the CV weights. Build the pseudo-data from an alternative-universe reweighting (e.g. **MaCCQE:0** $+1\sigma$, or **Flux:50** PPFx index). The reference is the in-acceptance-extrapolated alt truth:

$$\tilde{X}_i^{\text{ref}} = X_i^{\text{CV,truth}} \cdot \frac{X_i^{\text{alt,in-acc}}}{X_i^{\text{CV,in-acc}}}. \quad (25)$$

This scales the CV truth marginal by the ratio of in-acceptance counts between alt and CV, isolating the unfolding's response to a model-shape change from acceptance/efficiency effects. **MaCCQE** probes an interaction-model shape change; **Flux** probes flux reweighting — two distinct kinds of model dependence.

Acceptance. Median per-bin $|\text{residual}| \leq 2.0\%$.

Code pointer. `uq/closure/closure_alt_model.py`.

Worked example (1A lgbm 5-iter).

- **MaCCQE:0** ($+1\sigma$): median $|\text{residual}| = \boxed{0.068\%}$ (PASS); in-acceptance bias -1.68% folded out by the extrapolation.
- **Flux:50** (PPFX idx 50): median $|\text{residual}| = \boxed{0.376\%}$ (PASS); in-acceptance bias -7.32% .

Both pass with $\gg 5\times$ headroom. The unfolding is robust to model variations on the order of $\pm 1\sigma$ of GENIE and full PPFX flux universes.

A.8 How the pieces fit together

The combined result, in one sentence: *across 205 paper-reported bins, the combined OmniFold + paper error budget gives $\chi^2/\text{ndf} = 1.481$ and pull mean/RMS 0.069/0.466. Because the paper and OmniFold results share data and their cross-covariance is unavailable, this is an indicative distance rather than a calibrated goodness-of-fit. The same-ensemble toy pulls are approximately Gaussian, and the 0.166% ML-noise floor is more than an order of magnitude below the systematic envelope.*

Audit trail of the headline

Number	Section	Procedure
$\chi^2/\text{ndf} = 1.481$	§A.3.2	Eq. (11) with $\mathbf{C}^{\text{paper}} + \mathbf{C}^{\text{syst}} + \mathbf{C}^{\text{stat}} + \mathbf{C}^{\text{ML}}$
$\mathbf{C}^{\text{syst}}$	§A.2.3	187 universes, MAT-conformant Eq. (7) + norm
$\mathbf{C}^{\text{stat}}$	§A.2.2	Poisson bootstrap, 300 replicas, Eq. (6)
$\mathbf{C}^{\text{ML}}$	§A.2.1	lgbm seedscan, $n = 10$ trials
$\mathbf{C}^{\text{paper}}$	§A.2.4	Paper’s published TotalCovariance
Pull RMS = 0.466	§A.3.3	Per-bin standardized residual on combined $\mathbf{C}$
Combined median rel. = 4.822%	§A.2.5	$\sqrt{\text{diag} \mathbf{C}^{\text{tot}} / \mathbf{X}^{\text{CV}}}$
Same-ensemble $ r \leq 1$: 68.71%	§A.6	Pull diagnostic, $N_{\text{toy}} = 200$

Standing assumptions

1. **Gaussian approximation.** Pulls and χ^2 treat each bin as Gaussian. Justified because per-bin entries are weighted sums over $\mathcal{O}(10^4)$ MC events; central limit theorem applies.
2. **Independence of error sources.** Eq. (10) requires no cross-correlations between ML, statistical, and systematic. The bootstrap pins the GBDT seed (so ML noise cancels at first order); universes pin the seed too; the seedscan pins the bootstrap. Cross-terms are higher-order. (The available $N = 300$ bootstrap block carries a small residual ML leak, §A.2.2.)
3. **Bootstrap diagnostic.** §A.6’s same-ensemble 68.71% fraction is a Gaussianity check, not a half-percent certification of $\mathbf{C}^{\text{stat}}$.
4. **Band independence in $\mathbf{C}^{\text{syst}}$.** Different physics sources are taken as uncorrelated; the one known exception, the Bashyal flux $\leftrightarrow$ muon-E joint block, is rederived and carried as a diagnostic — adding it leaves the systematic rank unchanged (§A.4).
5. **Combined- χ^2 double-count.** The headline shares systematics with the paper’s V and is the “what OmniFold UQ adds on top of V ” statement, not an independent ours-only test (§A.4).
6. **Closure passes are valid for the perturbations tested.** Nothing guarantees they pass for unphysical deformations far outside the training distribution.

A.9 Multinomial $\rightarrow$ independent Poisson bootstrap limit

A non-parametric bootstrap with replacement draws N events from the original N -event sample. Let n_i be the number of times event i is drawn. Then $(n_1, \dots, n_N) \sim \text{Multinomial}(N, \mathbf{p})$ with $p_i = 1/N$ uniform, $\sum_i n_i = N$.

The marginal $n_i \sim \text{Binomial}(N, 1/N)$, whose mean is 1 and variance $(N-1)/N \rightarrow 1$. For large N , the binomial converges to Poisson(1) (Poisson limit theorem: many trials, small success probability, fixed expected count).

The full multinomial has the constraint $\sum n_i = N$, which introduces $\mathcal{O}(1/N)$ correlations between the n_i . In the large- N limit those correlations vanish and the n_i become *independent* Poisson(1) variables. This is the Poissonization of the multinomial: the conditioned N -event multinomial is a Poisson process *conditioned* on hitting N counts, and the unconditional version is independent Poissons.

In practice, the independent-Poisson form is what we implement (one `np.random.default_rng().poisson(1.0)` draw per event), and the $\sum n_i = N$ constraint is dropped. For $N \gtrsim 10^6$ (our regime), the relative error from the dropped constraint is $\mathcal{O}(1/\sqrt{N})$ on the total event count — far smaller than the per-bin Poisson variance.

A.10 The $\pm 1\sigma$ pair covariance: mean-centered (MAT-conformant)

For a single universe band with two universes $u = 0, 1$ at $\pm 1\sigma$, write the two value vectors as $\mathbf{X}_0$ and $\mathbf{X}_1$, with universe mean $\langle \mathbf{X} \rangle = \frac{1}{2}(\mathbf{X}_0 + \mathbf{X}_1)$. The MAT-conformant estimator Eq. (7) at $N_u = 2$ is

$$\mathbf{C}_{\text{mean}}^B = \frac{1}{2} \sum_{u=0,1} (\mathbf{X}_u - \langle \mathbf{X} \rangle)(\mathbf{X}_u - \langle \mathbf{X} \rangle)^\top = \frac{1}{4}(\mathbf{X}_0 - \mathbf{X}_1)(\mathbf{X}_0 - \mathbf{X}_1)^\top, \quad (26)$$

a rank-1 outer product of the *half-difference*. It is mean-centered, so any common offset of the pair away from X_{CV} contributes nothing.

A naive alternative centers on the CV rather than the pair mean, with $\Delta_u = \mathbf{X}_u - \mathbf{X}_{\text{CV}}$ and decomposition $\Delta_0 = \mathbf{m} + \mathbf{a}$, $\Delta_1 = \mathbf{m} - \mathbf{a}$:

$$\mathbf{C}_{\text{CV-centered}}^B = \frac{1}{2}(\Delta_0 \Delta_0^\top + \Delta_1 \Delta_1^\top) = \mathbf{m} \mathbf{m}^\top + \mathbf{a} \mathbf{a}^\top. \quad (27)$$

The two agree *only* for a CV-symmetric pair ($\mathbf{m} = 0$), where both reduce to $\mathbf{a} \mathbf{a}^\top$. For an asymmetric or one-sided pair the CV-centered form adds a spurious rank-1 $\mathbf{m} \mathbf{m}^\top$ term — it treats the common offset as a second variance direction, which mean-centering correctly collapses to near zero. We mean-center throughout, matching MAT (§A.2.4); the bands where the distinction matters are the asymmetric ones (BeamAngle, MuonResolution, and the one-sided NC band).

B Unbinned background subtraction by negative-weight injection

The background subtraction of §3.1 is the one step in which the analysis binning enters before the final histogramming: each data event in reconstructed bin b is down-weighted by the purity $w_{\text{pur}}(b) = \max(0, N_{\text{data}}(b) - N_{\text{bkg}}(b))/N_{\text{data}}(b)$. This appendix documents the fully unbinned realization of the same correction — injecting the background simulation onto the measured side of the fit with negative weight — shows that it targets an identical estimand, and quantifies its agreement with the binned correction on the 2D measurement. It is the continuous version of the purity weight recommended for nontrivial backgrounds by Ref. [21], and the natural treatment for the extended-fiducial analysis, where the background is larger.

B.1 Estimator

Step 1 of OmniFold trains a classifier that separates the simulated reconstruction (class 0, weighted density S) from the measured sample (class 1, weighted density ρ_1), then reweights the simulation by the learned density ratio $r(x) = p(x)/(1 - p(x))$, whose population optimum is $r^*(x) = \rho_1(x)/S(x)$. The background treatment is entirely a choice of ρ_1 :

- the binned purity weight sets $\rho_1(x) = D(x) w_{\text{pur}}(x) = \max(0, D(x) - B(x))$, realized by a per-reco-bin scalar applied to each data event, with D the data density and B the POT-scaled background density;
- negative-weight injection appends the background-simulation reconstruction events to the measured side with weight $-w_{\text{bkg}} \times (\text{POT scale})$, leaving the data events at $+1$, so that $\rho_1(x) = D(x) - B(x)$ *continuously*, with no analysis binning.

Both give the same background-subtracted estimand $r^*(x) = (D(x) - B(x))/S(x)$; they differ only in whether the subtraction is performed per reco bin or event by event in the full reconstructed phase space. The injected background is restricted to the same reconstructed fiducial window as the data, so the two constructions act on identical support.

The floor. Where the background locally exceeds the data, $D - B < 0$, the binned method floors the correction at zero through the $\max(0, \cdot)$. In the unbinned case the classifier output $p \in (0, 1)$ forces $r = p/(1 - p) \geq 0$, so a negative reweight cannot arise; the floor is therefore structural rather than imposed. The fit is not well-posed in such a region, however — the class-1 weighted mass is locally negative — so the principled treatment there is the “Stay Positive” refinement of Ref. [42], which resamples the signed-weight sample into an equivalent positive-weight one before training. Whether the plain injection and the refinement coincide in practice depends on the classifier. With the histogrammed gradient-boosted density estimator used for the production result they agree to the median of 1.000 reported below, at the sub-percent genuine-background scale of this measurement. With an *exact* (tree-boosted) classifier the raw signed weights instead destabilize training — a full low- p_T row of the cross section inflates by up to a factor $\sim 5 \times 10^4$ where the local background is large — and the Stay-Positive resampling removes the pathology, recovering agreement with the binned purity to a median of 0.999 (141/148 bins within 5%). The refinement is therefore what makes the unbinned subtraction robust across estimators, and it matters most for the larger, more structured backgrounds of the extended-fiducial analysis.

B.2 Closed-form toy

The estimator is verified on a one-dimensional problem with a known answer: signal simulation drawn from $\mathcal{N}(0, 1)$, a data signal from $\mathcal{N}(0.3, 1.1)$ with equal normalization, and a background $\mathcal{N}(2, 0.7)$ contributing 16.7% of the pseudo-data, so that the target reweight $(D - B)/S = \phi(x; 0.3, 1.1)/\phi(x; 0, 1)$ is a smooth positive function. A gradient-boosted classifier trained with the signed-weight construction recovers this ratio to a median 6.8% relative error — matching the binned purity construction’s 6.7% on the identical problem, and reproducing it bin-for-bin — while omitting the subtraction leaves a 10.9% bias from the residual background. The result is stable across classifier seeds ($\pm 0.3\%$), and the negative sample weights introduce no numerical instability. In a stress configuration with the background moved into the sparse signal tail and deliberately over-subtracted, the learned reweight floors cleanly at zero over the region where $B > D$, as anticipated above.

Table 3: Binned purity versus unbinned negative-weight background subtraction on the 2D ME FHC measurement (common estimator and seed). The per-bin figures are the negative-weight/purity ratio over the populated cross-section bins.

	binned purity	negative-weight
total σ (cm ² /nucleon)	$3.0727e - 38$	$3.0687e - 38$
total ratio (neg/pur)	0.9987 (−0.13 %)	
median per-bin ratio	1.000	
per-bin RMS deviation	1.4 %	
largest single-bin deviation	−12.6 % (low p_T , high background)	

B.3 Comparison on the 2D measurement

Table 3 compares the two background treatments on the ME FHC data, unfolded with identical settings and a common classifier seed so that the difference isolates the background treatment. The two agree to −0.13 % on the total cross section — an order of magnitude inside the $\sim 3\%$ purity correction itself — with a median per-bin ratio of 1.000 and a 1.4 % RMS scatter across the plane (Fig. 38). The only bin differing by more than a few percent is a single low- p_T , high-background edge bin (−12.6 %), exactly where the background concentrates and the subtraction is largest. The unbinned subtraction thus reproduces the adopted binned correction wherever the correction is well determined, and departs from it only in the regime where the binned $\max(0, \cdot)$ floor and the structural floor of §B.1 are themselves the operative choice.

B.4 Propagation through systematics and statistics

A central-value match is necessary but not sufficient: to replace the binned correction the unbinned subtraction must also reproduce the *uncertainty* budget. The full 2D ME FHC covariance was therefore rebuilt end to end under negative-weight injection — the systematic ensemble of 187 universes, with the genuine background varied consistently within each universe rather than frozen at nominal, and a statistical bootstrap — and compared to the binned-purity construction with identical settings. The two agree at the level the $(D - B)/S$ identity predicts: the systematic covariance matches in overall magnitude to a $\sqrt{\text{tr}}$ ratio of 0.986 (negative-weight/purity), and the statistical covariance, compared on matched bootstrap seeds, to 0.982. The residual few-per-mille differences are bootstrap and classifier-seed sampling noise, not a method bias. Negative-weight injection thus reproduces not only the central cross section but its full covariance, per bin and in aggregate.

The equivalence at the 2D scale — in the central value, the closed-form estimator recovery, and now the full systematic and statistical covariance — together with the Stay-Positive refinement that keeps the estimator robust on the larger backgrounds of the extended-fiducial analysis, establishes negative-weight injection as a fully validated unbinned replacement for the binned purity correction. The binned purity is retained as the default for the reproduction measurement of this note, with negative-weight injection carried as the validated unbinned cross-check and as the baseline subtraction for the extended-fiducial analysis (§10), where its binning-independence and continuous floor are essential.

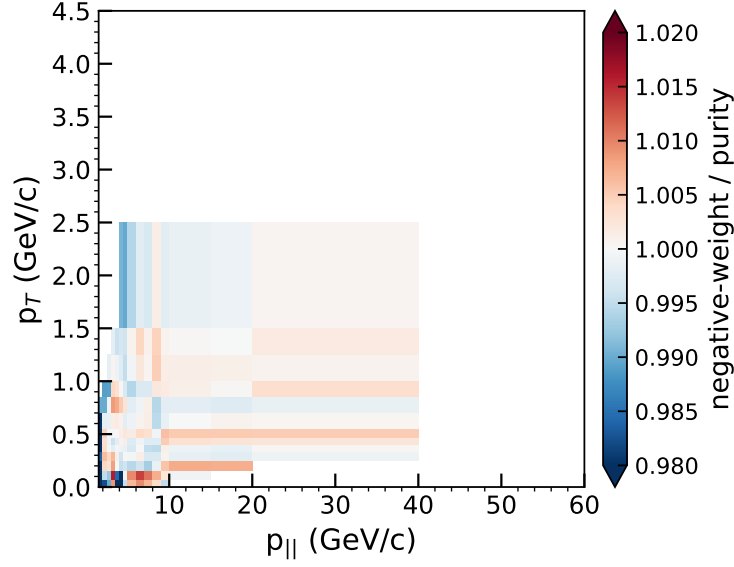


Figure 38: Per-bin ratio of the negative-weight to the binned-purity cross section on the 2D ME FHC measurement (common estimator and seed), over the populated bins. The two agree to a median of 1.000 across the plane (near-white); the only departure beyond a few percent is the single low- p_T , high-background edge bin, where the background is largest and the $\max(0, \cdot)$ /structural floor of §B.1 is the operative choice.

C The MINERvA measurement-dimensionality survey

The full classification behind Fig. 20: every MINERvA differential cross-section measurement, the number of kinematic variables its headline result unfolds simultaneously, and the reference.

Table 4: Full MINERvA differential cross-section survey: the number of kinematic variables unfolded simultaneously (Dim.), classified from each analysis’s headline differential result. Channels: CC-inclusive (incl.), CCQE/QE-like incl. transverse kinematic imbalance (QE), pion production CC and coherent (π), deep-inelastic / kaon (DIS), electron-neutrino (ν_e), nuclear-target ratio (ratio).

Year	Channel; observables	Dim.	Ref.
2013	QE; ν_μ CCQE $p_T, p_{ }$	2	[43]
2013	QE; $\bar{\nu}_\mu$ CCQE $p_T, p_{ }$	2	[44]
2014	ratio; CC on C/Fe/Pb to CH	1	[45]
2014	π ; coherent $\pi^\pm$ (total & diff.)	1	[46]
2015	π ; CC $1\pi^+$ (T_π, θ_π)	1	[47]
2015	QE; μ +p final states	1	[48]
2015	π ; $\bar{\nu}_\mu$ single π^0	1	[49]
2016	ν_e ; CCQE-like on CH	2	[50]
2016	π ; $\nu_\mu/\bar{\nu}_\mu$ pion prod.	2	[51]
2016	DIS; partonic nuclear effects	2	[52]
2016	DIS; CC K^+ production	1	[53]

continued...

Year	Channel; observables	Dim.	Ref.
2017	π ; CC single π^0	2	[54]
2017	QE; nuclear-dependence (multi-target)	1	[55]
2018	QE; $\bar{\nu}_\mu$ QE-like $p_T, p_\parallel$	2	[56]
2018	QE; μ -p mesonless TKI	2	[57]
2018	π ; coherent $\pi^\pm$ on C	2	[58]
2019	QE; quasielastic-like ν_μ	2	[59]
2019	π ; $\bar{\nu}_\mu$ single π^-	2	[60]
2020	incl.; CC-incl $\langle E_\nu \rangle \sim 3.5$ GeV $p_T, p_\parallel$	2	[61]
2020	QE; nuclear binding & TKI	2	[62]
2020	QE; high-statistics QE-like	2	[63]
2020	π ; CC π^0 nuclear effects	1	[64]
2021	incl.; CC-incl $\langle E_\nu \rangle \sim 6$ GeV $p_T, p_\parallel$ (target)	2	[4]
2022	QE; QE-like muon & proton kinematics	2	[1]
2022	incl.; CC-incl low momentum transfer	1	[11]
2023	π ; CC $1\pi^+$ (multi-target)	2	[65]
2023	π ; coherent π^+ (multi-target)	1	[66]
2023	QE; $\bar{\nu}_\mu$ multi-neutron low- E_{avail}	1	[67]
2024	ν_e ; low- Q^2 (E_{avail}, p_T)	2	[41]
2025	QE; A -dependence TKI (multi-target)	2	[68]
2026	this work ; unbinned $p_T, p_\parallel$	2	—
2026	this work ; unbinned $p_T, p_\parallel, E_{\text{avail}}$	3	—
2026	this work ; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3$	4	—
2026	this work ; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3, W$	5	—

D Codebase layout and terminology

The analysis is a chain of four components; the italicized names are used throughout the note.

- *Event loop* (C++): `runEventLoopOmniFold`, built on the MINERvA Analysis Toolkit (MAT) and the MINERvA-101 framework. It reads the production AnaTuples, applies the selection and the systematic-universe machinery, and writes per-event ROOT trees.
- *Omnifiles*: the event-loop outputs — per-event records for data, truth/reco-paired simulated signal, simulated background, and the truth-level efficiency denominator, with POT bookkeeping; the `_universes_full` variants add per-universe weight and shifted-kinematics branches.
- *Driver* (Python): the per-analysis orchestration — `2d-unfolding/unfold_2d_omnifold_unbinned.py` and its N -dimensional generalization `nd-unfolding/unfold_nd_omnifold_unbinned.py`. The driver assembles training arrays from the omnifiles, applies the purity-weight background subtraction (§3.1), invokes the engine, and extracts the cross section.
- *Engine* (Python): `unbinned_unfolding/python/omnifold.py` — the OmniFold iteration itself, agnostic to the learner backend (LightGBM in production; sklearn `exact/hist` alternates; MLP and point-cloud-transformer cross-checks).

The recoil-point-cloud cross-checks of §9 replace the driver’s scalar classifier features with the non-muon cluster and truth-hadron inputs of `nd-unfolding/pet/`, reusing the same omnifile and cross-section machinery. Muon scalars are retained for selection and binning but are not classifier inputs in these legacy PET products.

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